AI & COMPUTING

Solid Geometry with Problems and Applications (Revised edition)

N. J. Lennes and H. E. Slaught

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Bonaventura Cavalieri(1598–1647) was one of the most influential mathematicians of his time. He was chiefly noted for his invention of the so-called “Principle of Indivisibles” by which he derived areas and volumes. See pages 143 and 214.

SOLID GEOMETRY WITH PROBLEMS AND APPLICATIONS REVISED EDITION BY H. E. SLAUGHT, Ph.D., Sc.D. PROFESSOR OF MATHEMATICS IN THE UNIVERSITY OF CHICAGO AND N. J. LENNES, Ph.D. PROFESSOR OF MATHEMATICS IN THE UNIVERSITY OF MONTANA ALLYN and BACON Bo<on New York Chicago

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PREFACE In re-writing the Solid Geometry the authors have consistently car- ried out the distinctive features described in the preface of the Plane Geometry. Mention is here made only of certain matters which are particularly emphasized in the Solid Geometry. Owing to the greater maturity of the pupils it has been possible to make the logical structure of the Solid Geometry more prominent than in the Plane Geometry. The axioms are stated and applied at the precise points where they are to be used. Theorems are no longer quoted in the proofs but are only referred to by paragraph numbers; while with increasing frequency the student is left to his own devices in supplying the reasons and even in filling in the logical steps of the argument. For convenience of reference the axioms and theorems of plane geometry which are used in the Solid Geometry are collected in the Introduction. In order to put the essential principles of solid geometry, together with a reasonable number of applications, within limited bounds (156 pages), certain topics have been placed in an Appendix. This was done in order to provide a minimum course in convenient form for class use and not because these topics, Similarity of Solids and Applications of Projection, are regarded as of minor importance. In fact, some of the examples under these topics are among the most interesting and concrete in the text. For example, see pages 180–183, 187–188, 194– 195. The exercises in the main body of the text are carefully graded as to difficulty and are not too numerous to be easily performed. The concepts of three-dimensional space are made clear and vivid by many simple

illustrations and questions under the suggestive headings “Sight

PREFACE Work.” This plan of giving many and varied simple exercises, so effective in the Plane Geometry, is still more valuable in the Solid Geometry where the visualizing of space relations is difficult for many pupils. The treatment of incommensurables throughout the body of this text, both Plane and Solid, is believed to be sane and sensible. In each case, a frank assumption is made as to the existence of the concept in question (length of a curve, area of a surface, volume of a solid) and of its realization for all practical purposes by the approximation process. Then, for theoretical completeness, rigorous proofs of these theorems are given in Appendix III, where the theory of limits is presented in far simpler terminology than is found in current text-books and in such a way as to leave nothing to be unlearned or compromised in later mathematical work. Acknowledgment is due to Professor David Eugene Smith for the use of portraits from his collection of portraits of famous mathematicians. H. E. SLAUGHT N. J. LENNES Chicago and Missoula, May, 1919.

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SOLID GEOMETRY

SOLID GEOMETRY INTRODUCTION 1. Two-Dimensional Figures. In plane geometry each figure is restricted so that all of its parts lie in the same plane. Such figures are called two-dimensional figures. A figure, all parts of which lie in one straight line, is aone-dimensional figure, while a point is of zero dimensions. 2. Three-Dimensional Figures. A figure, not all parts of which lie in the same plane, is athree-dimensional figure. Thus, a figure consisting of a plane and a line not in the plane is a three-dimensional figure because the whole figure does not lie in one plane. 3. Solid Geometry treats of the properties of three-dimensional figures. 4. Representation of a Plane.While a plane is endless in extent in all its directions, it is represented by a

parallelogram, or some other limited plane figure. A plane is designated by a single letter in it, by two letters at opposite corners of the parallelogram representing it, or by any three letters in it but not in the same straight line. Thus, we saythe planeM, the planePQ, or the planeABC.

2 SOLID GEOMETRY 5. Figures in Plane and Solid Geometry.In describing a figure in plane geometry, it is assumed, usually without special mention, that all parts of the figure lie in the same plane, while in solid geometry it is assumed that the whole figure need not lie in any one plane. Thus, in plane geometry we have the theorem: “Through a fixed point on a line one and only one perpendicular can be drawn to the line.” If all parts of the figure are not required to lie in one plane, the theorem just quoted is far from true. As can be seen from the figure, an unlimited number of lines can be drawn perpendicular to a line at a point in it. Thus, all the spokes of a wheel may be perpendicular to the axle. 6. Loci in Plane and Solid Geometry.In plane geometry, “the locus of all points at a given distance from a given point” is a circle, while in solid geometry this locus is a sphere. In plane geometry, “the locus of all points at a given distance from a given line” consists of two lines, each parallel to the given line and at the given distance from it, while in solid geometry this locus is a cylindrical surface whose radius is the given distance.

INTRODUCTION 3 7. Parallel Lines. Skew Lines. In plane geometry, twolineswhichdonotmeetareparallel, while in solid geometry, two lines which do not meet need not be parallel. That is, they may not be in the same plane. Lines which are not parallel and do not meet are calledskew lines. In solid geometry, as in plane geometry, the definition of parallel lines implies that the lines lie in the same plane. That is, if two lines are parallel, there is always some plane in which both lie. Thus, in the figure,l1 and l2 are parallel, as are alsol1 and l3, whilel3 and l4 are skew. sight work Note. In exercises 1–4 give the required loci for both plane and solid geometry. No proofs are required. 1. The locus of all points six inches distant from a given point. 2. The locus of all points ten inches distant from a given point. 3. The locus of all points at a perpendicular distance of four inches from a given straight line. 4. The locus of all points at a perpendicular distance of nine inches from a given straight line. 5. Find the locus of all points one foot from a given plane. Is this a problem in plane or in solid geometry? 6. Find the locus of all points equidistant from two parallel lines and in the same plane with them. Is this a problem in plane or in solid geometry? 7. Find the locus of all points equidistant from two given parallel planes. Is this a problem in plane or in solid geometry? 8. The side walls of your schoolroom meet each other in four vertical lines. Are any two of these parallel? Are any three of them parallel? Do any three of them lie in the same plane? 9. The side walls of your schoolroom meet the floor and the ceiling in straight lines. Which of these lines are parallel to each other? Do any of these lines lie in the same plane?

4 SOLID GEOMETRY 8. Representation of Solid Figures on a Plane Surface.To represent a figure on a plane surface when at least part of the figure does not lie in that surface requires special devices. Thus, in the parallelogramABCD used to represent a plane, the edges AB and BC are made heavier than the other two. Thisindicatesthatthelowerandright-hand sides are nearer the observer than the other

edges. Hence, the plane represented does not lie in the plane of the paper, but the lower part of it stands out toward the observer. The figure ABCD represents a triangular pyramid. The corner marked B is nearest the observer and this is indicated by the heavy lines. The triangleACD lies behind the pyramid and is thus farther from the observer. The lineAC is dotted to indicate that it is seen through the figure. In the closed boxAG, the linesAD, DC, andDH lie behind the figure and are dotted, while the others are in full view and are solid. If the box were open at the top, part of the lineDH would be in full view and would be represented by a solid line. 9. Representation of Lines.The following plan for representing lines is generally adhered to in this book: (1) A line of the main figure which is not ob- scured by any other part of the figure is represented by a solid line. (2) An auxiliary line, which is drawn inciden- tally in making a proof or constructing a figure, is marked in long dashes if it is in full view. (3) Any line whatever which is behind a part of the figure is marked in short dashes or dots, or sometimes is not shown at all. (4) Where a figure is shaded it is usually regarded as opaque and the lines behind it cannot be seen at all. (5) In some cases a shaded surface is regarded as translucent and the lines behind it are seen dimly. Such lines are marked in short dashes.

INTRODUCTION 5 The following Axioms and Theorems from plane geometry are re- ferred to in the solid geometry. The special axioms of solid geometry will be given as they arise in the text. axioms 10. Things equal to the same things are equal to each other. 11. If equals are added to equals, the sums are equal. 12. If equals are subtracted from equals, the remainders are equal. 13. If equals are multiplied by equals, the products are equal. 14. If equals are divided by equals, the quotients are equal. 15. If equals are added to unequals, the sums are unequal and in the same order. 16. If unequals are added to unequals, in the same order, then the sums are unequal and in that order. 17. If equals are subtracted from unequals, the remainders are un- equal and in the same order. 18. If unequals are subtracted from equals, the remainders are un- equal and in the opposite order. 19. If a is less thanb and b less thanc, thena is less thanc. 20. If a and b are quantities of the same kind, then eithera>b , or a =b, ora<b . 21. Through a point not on a given line only one straight line can be drawn parallel to that line. 22. A straight line-segment is the shortest distance between two points. 23. Corresponding parts of equal figures are equal.

6 SOLID GEOMETRY theorems 24. If two lines intersect, the vertical angles are equal. 25. Two triangles are equal if two sides and the included angle of one are equal respectively to two sides and the included angle of the other. 26. Two triangles are equal if two angles and the included side of one are equal respectively to two angles and the included side of the other. 27. Two triangles are equal if three sides of one are equal respectively to three sides of the other. 28. Two points each equidistant from the extremities of a line-segment determine the perpendicular bisector of the segment. 29. One and only one perpendicular can be drawn to a line through a point whether that point is on the line or not. 30. The sum of all consecutive angles about a point in a plane is four right angles. 31. The sum of all consecutive angles about a point and on one side of a straight line is two right angles. 32. If two adjacent angles are supplementary, their exterior sides lie in the same straight line. 33. If in two triangles two sides of one are equal respectively to two sides of the other, but the

third side of the first is greater than the third side of the second, then the included angle of the first is greater than the included angle of the second. 34. Two lines which are perpendicular to the same line are parallel. 35. If a line is perpendicular to one of two parallel lines, it is per- pendicular to the other also. 36. If two given lines are perpendicular respectively to each of two intersecting lines, then the given lines are not parallel. 37. In a right triangle there are two acute angles.

INTRODUCTION 7 38. From a point in a perpendicular to a straight line, oblique seg- ments are drawn to the line. Then, (1) If the distances cut off from the foot of the perpendic- ular are unequal, the oblique segments are unequal, that one being the greater which cuts off the greater distance; and (2) Conversely, if the oblique segments are unequal, the distances cut off are unequal, the greater segment cutting off the greater distance. 39. Two angles whose sides are perpendicular, each to each, are equal or supplementary. 40. Two right triangles are equal if the hypotenuse and a side of one are equal respectively to the hypotenuse and a side of the other. 41. Two right triangles are equal if a side and an acute angle of one are equal respectively to the corresponding side and acute angle of the other. 42. Two right triangles are equal if the hypotenuse and an acute angle of one are equal respectively to the hypotenuse and an acute angle of the other. 43. A quadrilateral is a parallelogram (1) if both pairs of opposite sides are equal; or (2) if two opposite sides are equal and parallel. 44. Opposite sides of a parallelogram are equal. 45. Two parallelograms are equal if an angle and the two adjacent sides of one are equal respectively to an angle and the two adjacent sides of the other. 46. The segment connecting the middle points of the two non-parallel sides of a trapezoid is parallel to the bases and equal to one half their sum. 47. The locus of all points equidistant from the extremities of a line-segment is the perpendicular bisector of the segment.

8 SOLID GEOMETRY 48. In the same circle or in equal circles equal chords subtend equal arcs. 49. A line perpendicular to a radius at its extremity is tangent to the circle. 50. If a line is tangent to a circle, it is perpendicular to the radius drawn to the point of contact. 51. If in a proportion the antecedents are equal, then the consequents are equal and conversely. 52. In a series of equal ratios the sum of any two or more antecedents is to the sum of the corresponding consequents as any antecedent is to its consequent. 53. If a line cuts two sides of a triangle and is parallel to the third side, then any two pairs of corresponding segments form a proportion. 54. If two sides of a triangle are cut by a line parallel to the third side, a triangle is formed which is similar to the given triangle. 55. In two similar triangles corresponding altitudes are proportional to any two corresponding sides. 56. Two triangles are similar if an angle of one is equal to an angle of the other and the pairs of adjacent sides are proportional. 57. Two triangles are similar if their pairs of corresponding sides are proportional. 58. The area of a parallelogram is equal to the product of its base and altitude. 59. Two parallelograms have equal areas if they have equal bases and equal altitudes. 60. The area of a triangle is equal to one half the product of its base and altitude. 61. If a is a side of a triangle andh the altitude on it andb another side andk the altitude on it, thenah =bk. 62. The area of a trapezoid is equal to one half the product of its altitude and the sum of its bases.

INTRODUCTION 9 63. The area of a circle is one half the circumference times the ra- dius, or in symbols: a = 1 2· 2πr·r =πr2.

BOOK I PROPER TIES OF THE PLANE 64. Relations of Points, Lines, and Planes.If a line or a plane contains a point, the point is said to beon the line orin the plane and the line or plane is said to passthrough the point. If a plane contains a line, the line is said to bein the plane and the plane is said to pass through the line. 65. Axiom 1. If two points of a straight line lie in a plane then the whole line lies in the plane. Since a line is endless, it follows from this axiom that a plane is endless in all its directions. 66. Axiom 2. Through three non-collinear points one and only one plane can be passed. 67. Axiom 3. Two distinct planes cannot meet in one point only. 68. Determination of a Plane.A plane is said to bedetermined bycertainelements(linesorpoints)ifthisplanecontainstheseelements while no other plane does contain them. While two points determine a straight line it is obvious that two points do not determine a plane. The figure shows three planes,L,M,N, all passing through the two pointsAandB. But only a certain one of these planes contains a given pointC which is not in the lineAB. We, therefore, say that three non-collinear points determine a plane, while any number of collinear points fail to determine a plane.

PROPERTIES OF THE PLANE 11 line common to two planes 69. Theorem I. Two intersecting planes meet in a straight line. Given two intersecting planesM and N. To prove that they meet in a straight lineAB. Proof: If two planes intersect they meet in at least two points, as A and B. Ax. 3, § 67 But A and B determine a line which lies wholly inM and also wholly inN. Ax. 1, § 65 Hence the planes have the straight lineAB in common. A pointC not inAB cannot lie in bothM and N, for in that case the planes would have three non-collinear points in common and hence would coincide. Ax. 2, § 66 Hence the planesM and N meet in the straight lineAB. Q. E. D. 70. Foot of a Line Meeting a Plane. The point in which a straight line meets a plane is called thefoot of the line. 71. Line and Plane Perpendicular to Each Other.A line is said to beperpendicular to a planeif it is perpendicular to every line in the plane passing through its foot. In this case the plane is also said to be perpendicular to the line. 72. Line and Plane Oblique to Each Other. A line which meets a plane and is not perpendicular to it is said to beoblique to the plane. The plane is also said to be oblique to the line. In the figure,PA is perpendicular to the planeM andQA is oblique to it.

12 SOLID GEOMETRY: BOOK I elements which determine a plane 73. Theorem II. A plane is determined by(1) a line and a point not on it,(2) two intersecting lines, and (3) two parallel lines. Given (1) a linel and a pointP not on it; (2) two inter- secting linesl1 and l2; (3) two parallel linesl1 and l2. To prove that in each case a plane is determined. Proof: (1) LetA and B be two points onl. Then one and only one planeM can be passed through l and P because one and only one plane can be passed throughA, B, andP. Ax. 2, § 66 (2) Let A be the intersection point of l1 and l2, and B and C any other points, one onl1 and the other onl2. Then A, B, andC determine the plane N in which liel1 and l2. Axs. 2, 1. §§ 66, 65 (3) By definitionl1 andl2 lie in a planeR. They lie in only one such plane since the pointsA andB on l1 and C on l2 lie in

only one plane.a Q. E. D. 74. Corollary 1. Through a line there is more than one plane. Suggestion. Let M be a plane through the given line l, andC a point not inM. Thenl and C deter- mine a planeN through l different fromM. 75. Corollary 2. At a point on a line there is more than one perpendicular to the line. aSee Transcriber's Notes.

PROPERTIES OF THE PLANE 13 Suggestion.Let M andN be planes each passing through the given linel. Then in each plane there is a line⊥l at any pointA on it. line perpendicular to the plane of two given lines 76. Theorem III. If a line is perpendicular to each of two lines at their point of intersection, it is perpendicular to the plane of these lines. Given a linel perpendicular to each of the linesl1 and l2 at the pointP. To prove thatl is perpendicular to the plane ofl1 and l2. Proof: LetM be the plane ofl1 andl2, and letl3 be any line inM throughP. Draw a line meetingl1,l2, andl3 in the pointsB,C, andD respectively. LetE and F be points onl, on opposite sides ofP, and such thatEP =FP . DrawEB, ED, EC, FB, FD, FC. Then prove: (1)△EBP =△FBP ; (2)△ECP =△FCP ; (3)△EBC =△FBC ; (4)△EBD =△FBD ; (5)△EPD =△FPD ; (6) ∠EPD = ∠FPD. □ EP is perpendicular tol3. Why? But l3 is any line inM through P. □ line l⊥ plane M. § 71 Q. E. D.

14 SOLID GEOMETRY: BOOK I 77. Corollary. If each of two lines is perpendicular to a third line at the same point, then the plane of the two lines is perpendicular to the third line. sight work The diagram on this page represents a three-dimensional figure in the shape of an ordinary rectangular box. In this figure the pointsA,K, andB, for instance, do not determine a plane, since they all lie on the same straight line, whileA, B, andC do not lie in a straight line, and hence determine a plane. 1. In this figure pick out several lines which lie in one of the surfaces and are not obscured by the figure. 2. Pick out several lines which are obscured by the figure; also some which lie within the figure. 3. Pick out four sets of three points each which do not determine planes, and also four sets which do determine planes. 4. Is the lineAB perpendicular to the planeBCG? Why? Is AB per- pendicular to the planeAEH? 5. Pick out six planes in the figure, each determined by parallel lines. 6. Do the pointsC, Z, E determine a plane? the pointsC, Z, G? the points B, F, Z? 7. Using the schoolroom, or a room at home, locate planes corresponding to the planesAEG, KLM, NOP , andEFG, in the above figure.

PROPERTIES OF THE PLANE 15 8. Point out in some room planes determined by points corresponding to D, E, B; D, F, B; D, C, F; A, B, H in the above figure. plane perpendicular to a line 78. Theorem IV. Through a point there is one and only one plane perpendicular to a line. Given a linel and a pointP. To prove that throughP there is one and only one plane⊥l. Proof: (1) When the pointP is on the linel. Fig. 1. Through P draw linesPQ and PQ′ both⊥ l. Then the planeM, determined byPQ and PQ′, is⊥l. § 77 To prove thatM is the only plane throughP which is⊥l, suppose that a planeM′ through P is also⊥ l. Let R be a plane throughl meeting M and M′ in two lines. Then these lines would both lie inR and be⊥l, which is impossible. § 29 (2) When the pointP is not on the linel. Fig. 2. Let PQ be a line⊥l and letN be a plane⊥l at Q. Then the planeN passes throughP and is the plane required. For ifP does not lie inN, then a planeR′ determined byl andPQ cuts N in a linel′ which is⊥ l (§ 71), andPQ and l′ are each⊥ l at the pointQ in planeR′, which is impossible. § 29 Suppose, now,

that there are two planes through P each⊥ l. These planes cannot meet l in the same point (Case 1). Let them meet l in Q and Q′. Then PQ and PQ′ are each⊥l, which is impossible. Q. E. D. 79. Corollary. All lines perpendicular to a line at the same point lie in the plane perpendicular to the line at this point.

16 SOLID GEOMETRY: BOOK I line perpendicular to a plane 80. Theorem V. Through a point there is one and only one line perpendicular to a plane. Given a planeM and a pointP. To prove that throughP there is one and only one line⊥M. Proof: (1) When P is in the planeM (first figure). Letl1 be any line inM through P, and letN be a plane⊥l1 at P and meetingM in the linel2. Let PA be a line inN and⊥l2. Then PA is also⊥l1. § 71 ∴PA⊥M. § 76. (2) WhenP is not in the planeM (second figure). Letl be any line in the planeM. Through P pass a planeN′⊥l at A and meetingM in the lineAK. FromP in planeN′ draw a linePO⊥AK and extend it toP′ so thatOP′ =PO. Let B be any point inl different fromA. DrawPB, PA, P′B, P′A, andOB. Then prove (1)△POA =△P′OA; (2) △PAB =△P′AB; (3)△POB =△P′OB; (4) ∠POB = ∠P′OB; (5) PO⊥OB. ∴PO⊥M. § 76 If in either case (1) or case (2) there were two lines PA and PB each⊥ M, then the plane R of these lines would cutM in a line l. Hence PA and PB would both lie in R and be⊥l, which also lies inR. But this is impossible by § 29. Hence PA is the only line throughP which is⊥M. Q. E. D.

PROPERTIES OF THE PLANE 17 sight work 1. Does a stool with three legs always stand firmly on a flat floor? Why? 2. Does a table with four legs always stand firmly on a flat floor? Why? On what conditions will such a table stand firmly on a flat floor? 3. If the pointC does not lie in the plane ABD, how many different planes are determined by the pointsA, B, C, D? 4. How many planes are determined by any four points which do not all lie in one plane? 5. How many planes are determined by the pointsA, B, C, D, E, ifA, B, C lie in a straight line andC, D, E lie in another straight line? 6. How many planes are determined by five points, no four of which lie in the same plane? 7. How many planes are determined by three linesl1, l2,l3 all passing through the same point but not all lying in the same plane? 8. How many planes are determined by four lines which all meet in a point, but no three of which lie in the same plane? 9. How many planes are determined by three lines all parallel to each other, and not all lying in the same plane? l1 l2 l3 10. How many planes are determined by four lines all parallel to each other, and no three lying in the same plane? 11. A line cannot be perpendicular to each of two intersecting planes. Why? Suggestions. (1) If a linel is per- pendicular to the planesM and N at thepoints AandB, andC isapointin their intersection, then△ABC would contain two right angles. (2) Ifl is perpendicular toM and N at a pointP in their intersection, pass a plane throughl, meetingM and N in l1 and l2. Then in this plane l1 and l2 are both⊥l.

18 SOLID GEOMETRY: BOOK I oblique lines from a point to a plane 81. Theorem VI. Oblique lines from a point to a plane meeting the plane at equal distances from the foot of the perpendicular are equal; and conversely, two equal oblique lines from a point to a plane meet the plane at equal distances from the foot of the perpendicular. (1) GivenPC⊥M, andAC =BC. To prove thatPA =PB. Suggestion. Suggestion. Prove△PCA =△PCB . (2) GivenPC⊥M, andPA =PB. To prove thatAC =BC. Suggestion. Suggestion. Prove△PCA =△PCB . 82. Corollary. The perpendicular is the

shortest distance from a point to a plane. Hence the distancefrom a point to a plane means the perpendicular distance. sight work Without giving proofs describe the following loci: 1. All points equidistant from the points on a circle. 2. All points equidistant from the vertices of a triangle. 3. All points in a plane which are at a given distance from a given point outside the plane. If a perpendicular be drawn to the plane from this outside point, how is its foot related to this locus?

PROPERTIES OF THE PLANE 19 exercises 1. Show how a carpenter could use the theorem of § 76 to stand a post perpendicular to the floor, if he has at hand two ordinary steel squares. 2. Show how a back-stop on a ball field can be made perpendicu- lar to the line through second base and the home plate. What theo- rems of solid geometry are used? 3. If a plane is perpendicular to a line-segmentPP′ at its middle point, prove: (1) Every point in the plane is equally distant fromP andP′; (2) every point equally distant fromP andP′ lies in this plane. What is the locus of all points in space equidistant fromP and P′? Compare § 47. 4. Given the pointsA and B not in a planeM. Find the locus of all points inM equidistant fromA and B. Suggestion. All such points must lie in the planeM and also in the plane which is the perpendicular bisector of the segmentAB. 5. Find the locus of all points equidistant from two given pointsA and B, and also equidistant from two pointsC and D. Discuss. 6. State and prove a theorem of solid geometry corresponding to the theorem of plane geometry given in § 38. 7. If in the figurePD⊥ plane M, andDC⊥ AB, a line of the planeM, prove thatPC⊥AB. Suggestion. Lay offCA = CB, and compare tri- angles. 8. If in the same figurePD ⊥ M, andPC ⊥ AB, a line of the plane, prove thatDC⊥AB.

20 SOLID GEOMETRY: BOOK I p arallel lines perpendicular to the same plane 83. Theorem VII. Two lines perpendicular to the same plane are parallel; and Conversely, if one of two parallel lines is perpendicular to a plane, the other is also. Given (1)AB and CD each⊥ the planeM. Fig. 1. To prove thatAB∥CD. Proof: DrawBD and makeDE⊥DB. Take pointsA and E so thatBA = DE, and drawAD, AE, and BE. Now prove: (1)△ABD =△BDE and □AD =BE; (2)△ADE =△ABE and □ ∠ADE = ∠ABE = Rt. ∠. □DC, DA, DB, andBA all lie in the same plane. § 79 AB∥CD. § 34 Given (2)AB∥CD and AB⊥M. Fig. 2. To prove thatCD⊥M. Proof: IfCD is not⊥M let C′D be⊥M. Then C′D∥AB by case (1), andC′D coincides withCD § 21 and □CD⊥M. 84. Corollary 1. If each of two lines is parallel to a third line they are parallel to each other. 85. Corollary 2. If a plane is perpendicular to one of two parallel lines, it is perpendicular to the other.

PROPERTIES OF THE PLANE 21 p arallel planes and lines 86. Parallel Planes. Two planes which do not meet are said to be parallel. 87. Line Parallel to a Plane.A straight line and a plane which do not meet are said to be parallel. 88. Intercepted Segments. If a straight linel2 meets two planes inA and B, then the segmentAB is said to bein- tercepted by the planes. Any line, as l1, in either of two paral- lel planes,M and N, is parallel to the other plane. The segment AB on the linel2 is in- tercepted by the planes. line p arallel to a plane 89. Theorem VIII. If a straight line is parallel to a given plane, it is parallel to the intersection of any plane through it with the given plane. Suggestion for proof. If l1 is the given line,M the given plane, andl2 the intersection of a planeN through l1 with M, show thatl1 and l2 lie in plane N and

cannot meet. 90. Corollary 1. If a line outside a plane is parallel to some line in the plane, then the first line is parallel to the plane.

22 SOLID GEOMETRY: BOOK I 91. Corollary 2. If a line is parallel to a plane, then through any point in the plane there is a line in the plane parallel to the given line. 92. Corollary 3. The intersections of a plane with two parallel planes are parallel lines. planes perpendicular to a line are p arallel 93. Theorem IX. If each of two planes is perpendicular to the same line, they are parallel; and Conversely, if one of two parallel planes is perpendicular to a line, the other is also. Given (1) planeM⊥AB and planeN⊥AB. Fig. 1. To prove thatM∥N. Proof: Suppose M and N to meet in some pointP. DrawAP in M and BP in N. Then AB⊥ AP and AB⊥ BP (§ 71), which is impossible. § 29 Given (2)M∥N and M∥AB. Fig. 2. To prove thatN∥AB. Proof: Through AB pass a plane cuttingM and N in AC and BD, and a second plane cuttingM and N in AE and BF. Then, AC∥BD and AB∥BF. § 92

PROPERTIES OF THE PLANE 23 Now prove (1)AB⊥BD, (2)AB⊥BF. § 35 AB⊥N. § 76 94. Corollary 1. Parallel line-segments included between parallel planes are equal. 95. Corollary 2. If a line is perpendicular to one of two parallel planes, it is perpendicular to the other also. 96. Corollary 3. Two planes each parallel to a third plane are parallel to each other. p arallel planes 97. Theorem X. If a plane is parallel to each of two intersecting lines, it is parallel to the plane of these lines. Given a planeM parallel to the intersecting linesl1 and l2. To prove that the planeM is∥ the planeN of l1 and l2. Proof: IfM is not∥N, these planes meet in a linel3. § 69 Then neitherl1 nor l2 can meetl3, since they are∥M. Hence l1∥l3, andl2∥l3, which is impossible. § 21 □M and N cannot meet and are parallel. Q. E. D. 98. Theorem XI. Through a point not in a plane there is one and only one plane parallel to this plane.

24 SOLID GEOMETRY: BOOK I Given a planeM and a pointP not in M. To prove that throughP there is one and only one plane∥M. Proof: Letl be a line throughP and⊥M. Through P drawl1 and l2 each⊥l. § 75 The planeN of l1 and l2 is⊥l, and henceN∥M. §§ 76, 93 If throughP there were another planeR∥M, thenR would be⊥l at P. § 93 But N and R cannot both be⊥l at P. § 78 Hence N is the only plane throughP∥M. Q. E. D. planes p arallel to given lines 99. Theorem XII. Through one of two skew lines there is one and only one plane parallel to the other line. Given two skew linesl1 and l2. See § 7. To prove that throughl1 there is one and only one plane∥l2. Proof: ThroughP, a point inl1, draw a linel3∥l2. Then l1 and l3 determine a planeM∥l2. § 90 Any other planeN through l1 would meet the plane ofl3 and l2 in a line throughP not∥l2 and henceN would meetl2. Q. E. D. 100. Theorem XIII. Through a point outside of each of two non-parallel lines there is one and only one plane parallel to both of these lines.

PROPERTIES OF THE PLANE 25 Given a pointP outside of the non-parallel linesl1 and l2. To prove that there is one and only one planeM throughP parallel to l1 and l2. Proof: ThroughP pass l3∥l1 and l4∥l2. Then the planeM of l3 and l4 is parallel tol1 and l2. § 90 In any other planeN through P∥ l1 and l2, there are linesl′ 3 and l′ 4 through P such thatl′ 3∥l1 and l′ 4∥l2. § 91 In that casel′ 3 is identical withl3 and l′ 4 with l4. § 21 Hence any such planeN is identical withM. Q. E. D. exercises 1. Given a planeM and a pointP not inM. Find the locus of the middle points of all segments connectingP with

points inM. Suggestion. Use the fact that a line parallel to the base of a triangle and bisecting one side bisects the other side also. 2. Show that a plane containing one only of two parallel lines is parallel to the other. 3. If in two intersecting planes a line of one is parallel to a line of the other, then each of these lines is parallel to the line of intersection of the planes. 4. Show that three lines which do not meet in one point must all lie in the same plane if each intersects the other two.

26 SOLID GEOMETRY: BOOK I 5. Show that three planes, each of which intersects the other two, have a point in common unless their three lines of intersection are parallel. Suggestion.Suppose two of the intersection lines are not parallel, but meet in some pointO. Then show that the other line of intersection passes through O, and hence thatO is the point common to all three planes. 6. Given two intersecting planesM and N. Find the locus of all points inM at a given perpendicular distance fromN. 7. Given two non-intersecting linesl1 and l2. Find the locus of all lines meetingl1 and parallel tol2. 8. Prove that the middle points of the sides of any quadrilateral in space are the vertices of a parallelogram. Suggestion.Use the fact that a line bisecting two sides of a triangle is parallel to the third side. Note that the four vertices of a quadrilateral in space do not necessarily all lie in the same plane. State the corresponding theorem in plane geometry. 9. In erecting a flagpole on a level space, show how it can be made perpendicular by means of three ropes of equal length. See § 81. angles whose sides are p arallel 101. Theorem XIV. If two intersecting lines in one plane are parallel, respectively, to two intersecting lines in another plane, then the two planes are parallel, and the cor- responding angles formed by the lines are equal.

PROPERTIES OF THE PLANE 27 Given the planesM and N in whichAB∥ A′B′, and AC∥ A′C′. To prove thatM∥N and ∠1 = ∠2. Proof: (1) If M is not∥ N, then these planes meet in a linel. Why? Then neitherAB nor AC can meetl since each is∥N. □AB∥l and AC∥l which is impossible. Why? (2) To prove∠1 = ∠2, lay offAB = A′B′, AC = A′C′, and draw BC, B′C′, AA′, BB′, andCC′. Analysis: ∠1 = ∠2 if△ABC =△A′B′C′, which is true ifBC = B′C′. But BC =B′C′ if BB′C′C is a , which is so ifBB′ =CC′ and BB′∥CC′. This last is true ifAA′C′C and AA′B′B are S , for then BB′ =AA′ =CC′ and BB′∥AA′∥CC′. Hence we need to prove in order (1)AA′C′C and AA′B′B are S , (2) BB′C′C is a , (3) △ABC = △A′B′C′, and (4) ∠BAC = ∠B′A′C′. Hence it is proved that the planes are parallel and the angles are equal. Q. E. D. 102. Corollary. Two angles in space whose sides are parallel each to each are either equal or supplementary. p arallel planes intercept propor tional segments 103. Theorem XV. If two straight lines are cut by three parallel planes, the intercepted segments on one line are proportional to the corresponding segments on the other.

28 SOLID GEOMETRY: BOOK I Given the linesAB and CD cut by the planesM, N, andP. To prove thatAE EB = CG GD. Outline of Proof: DrawAD and let the plane determined byAD and CD cut the planesM and N in AC and FG respectively; and let the plane ofAB and AD cut N and P in EF and BD respectively. Then prove (1)FG∥AC and EF∥BD, (2) AE EB = AF FD and CG GD = AF FD, (3) AE EB = CG GD. 104. Corollary. Parallel planes which intercept equal segments on any transversal line intercept equal segments on every transversal line. sight work Review the theorems

of solid geometry up to this point by stating the corresponding theorems of plane geometry, in case such theorems exist. exercises 1. If the spaces between four parallel shelves are5, 8, and 10 inches respectively, and a slanting rod intersecting them has a7-inch segment between the first two shelves, find the other two segments of the rod. 2. If a line cuts three parallel planes,M,N,R, so that the segment intercepted betweenM and N is 7 and that betweenN and R is 21, and if another line cuts the same planes so that the segment between M andN is 11, find the segment on the second line betweenN andR.

PROPERTIES OF THE PLANE 29 3. Showthatline-segmentsincludedbetweenparallelplanesandper- pendicular to them are equal, and hence that parallel planes are every- where equally distant. How can a carpenter make use of this principle in placing two parallel shelves? How many distances must he measure? Why? 4. Show that through a point outside a plane any number of lines can be drawn parallel to the plane. How are all these parallels related? 5. Prove that if a plane bisects two sides of a triangle it is parallel to the third side. 6. The perpendicular distance from a pointP to a plane is 12 in. Find the radius of the circle which is the locus of all points in the plane at a distance of20 in. fromP. 7. Show that, if three line-segments not in the same plane are equal and parallel, the triangles formed by joining their extremities, as in the figure of § 101, are equal and their planes are parallel. 8. What is the relation of two lines if they are (a) parallel to a given line, (b) perpendicular to a given line, (c) parallel to a given plane, (d) perpendicular to a given plane? 9. What is the relation of two planes if they are both (a) parallel to a given plane, (b) parallel to a given line, (c) perpendicular to a given line? dihedral angles 105. Dihedral Angle. The part of a plane on one side of a line in it is called ahalf-plane. The line is called theedge of the half-plane. Two half-planes meeting in a common edge form adihedral angle. The common edge is theedge of the angle and the half-planes are itsfaces. 106. Plane Angle of a Dihedral Angle. Two lines in the respective faces of a dihedral angle and perpendicular to its edge at a common point form a plane angle, which is calledthe plane angle of the dihedral angle.

30 SOLID GEOMETRY: BOOK I In the figure, the half-planes M and N form the dihedral angleM-AB-N, read by naming the two faces and the edge. If CD in N is⊥ AB and ED in M is ⊥ AB, then ∠CDE is the plane angle of the dihedral angle M-AB-N. By § 101 all plane angles of a dihedral angle are equal to each other. 107. Generation of a Dihedral Angle. A dihedral angle may be thought of asgenerated by the rotation of a half-plane about its edge. The magni- tude of the angle depends solely upon theamount of rotation. 108. Equal Dihedral Angles. Two dihedral angles areequal when they can be so placed that they coincide. 109. Right Dihedral Angle.A right dihedral angle is one whoseplane angle is a right angle. 110. Perpendicular Planes. Two planes are said to bemutually perpendicu- lar if their dihedral angle is a right angle. Dihedral angles areacute or obtuse according as their plane angles are acute or obtuse. Two dihedral angles areadjacent, vertical, supplementary, orcomplemen- tary according as their plane angles possess these properties. dihedral angles and their plane angles 111. Theorem XVI. Two dihedral angles are equal if their plane angles are equal; and Conversely, if two dihedral angles are equal, their plane angles are equal.

PROPERTIES OF THE PLANE 31 Given (1) the dihedral anglesM−AB−N andM′−A′B′−N′ in which the plane∠sCDE and C′D′E′ are equal. To prove that∠M−AB−N = ∠M′−A′B′−N′. Proof: Place the equal∠sCDE and C′D′E′ in coincidence. ThenAB coincides withA′B′ since these lines are perpendicular to the planeCDE at the pointD. § 80 ThenM andM′ coincide as do alsoN andN′ since they are deter- mined by coincident lines. § 73 ∠ M−AB−N = ∠M′−A′B′−N′. § 108 Given (2) ∠M−AB−N = ∠M′−A′B′−N′. To prove that the plane∠sCDE and C′D′E′ are equal. Proof: Place the equal dihedral∠sM−AB−N andM′−A′B′−N in coincidence so that the pointsD and D′ coincide. Then CD and C′D′ coincide as do alsoED and E′D′. § 29 ∠ CDE = ∠C′D′E′. Q. E. D. 112. Corollary. All right dihedral angles are equal. 113. Measure of a Dihedral Angle.It follows from § 111 that the plane angle of a dihedral angle may be regarded as its measure.

32 SOLID GEOMETRY: BOOK I mutuall y perpendicular planes 114. Theorem XVII. If two planes are mutually perpen- dicular, and if a line is drawn from a point in one perpen- dicular to their intersection, then this line is perpendicular to the second plane. Given planeM⊥ plane N and a pointP in M. Let l1 be the intersection ofM and N and l a line inM through P⊥l1. To prove thatl⊥N. Proof: Let l meet l1 in O. Through O draw l2 in N and perpen- dicular tol1. Then the angle formed by the linesl andl2 is the plane angle of the dihedral angle between the planesM and N. § 106 ∠ l⊥l2 § 109 ∠ l⊥N. Why? Q. E. D. 115. Corollary. If two planes are mutually perpendicular and if a line is drawn from a point in one and perpendicular to the other, then this line lies in the first plane and is perpendicular to the intersection of the two planes. Proof: By the theorem there is a line throughP which lies in the plane M and is⊥N. But throughP there is only one line⊥N. § 80

PROPERTIES OF THE PLANE 33 plane perpendicular to each of two planes 116. Theorem XVIII. If a plane is perpendicular to each of two planes, it is perpendicular to their line of inter- section. GivenQ⊥M andQ⊥N andl the intersection ofM andN. To prove thatQ⊥l. Proof: From a pointP common toM and N draw a linel′⊥Q. Then l′ lies in bothM and N. § 115 Hence, l′ and l are the same line. Why? That is,l⊥Q, orQ⊥l. Q. E. D. sight work 1. State theorems on dihedral angles corresponding to those in §§ 24, 30, 31, 32, on plane angles. 2. Name all dihedral angles in the accompanying fig- ure. 3. Any plane⊥ the edge of a dihedral angle is⊥ each of its faces. Why? 4. If each of three lines is⊥ the other two at the same point, then each is⊥ the plane of the other two. Why? 5. Find the locus of all points at a given distance from a given plane and also at a given distance from a second plane. Discuss this locus for the various cases possible.

34 SOLID GEOMETRY: BOOK I perpendicular lines and planes 117. Theorem XIX. If a line is perpendicular to a plane, every plane containing this line is perpendicular to that plane. Given a linel⊥ plane N at P. To prove that a planeM containingl is⊥N. Proof: Let l1 be the intersection ofM and N. In N draw l2⊥ l1 at P. Then l⊥l1, l2⊥l1, andl⊥l2. Why? ∠ M⊥N. Why? Q. E. D. 118. Projection of a Figure on a Plane. Theprojectionofapointonaplaneisthefootof the perpendicular from the point to the plane. The projection of anyfigure on a plane is the locus of the projections of all points of the figure on the plane. Thus,B is the projection of the pointA on M, andl2 is

the projection of the linel1 on M. 119. Angle Between Line and Plane. The angle between a plane and a line oblique to it is understood to be theacute angleformed by the line and its projection upon the plane. 120. Bisector of a Dihedral Angle. A half-plane bisects a dihedral angle if it passes through the edge of the dihedral angle and bisects its plane angle.

PROPERTIES OF THE PLANE 35 projection of a straight line on a plane 121. Theorem XX. The projection of a straight line on a plane is a straight line in that plane. Given a planeM and a lineAB projected upon it. To prove that the projection is a straight line. Proof: Project any three pointsA, E, B of the lineAB into the points C, F, D in the planeM. Then AC and BD determine a plane⊥M. §§ 83, 117 Hence, by §§ 115, 69, any three points,C, F, D, of the projection lie in a straight line, that is, the projection is a straight line.Q. E. D. 122. Theorem XXI. The acute angle formed by a straight line with its own projection on a plane is the least angle which it makes with any line in that plane. Given CB the projection ofAB on the planeM, and any other lineBD in M through B. To prove that∠ABD >∠ABC. Proof: Draw AC⊥ M, make BD = BC, and drawAD. Then proveAD >AC and ∠ABD >∠ABC. § 33

36 SOLID GEOMETRY: BOOK I plane bisecting a dihedral angle 123. Theorem XXII. The locus of all points equally distant from the faces of a dihedral angle is the half-plane bisecting the angle. Given the half-planeP bisecting the dihedral∠M−AB−N. To prove(1) that any pointE inP is equally distant fromM andN, and (2) that any point which is equally distant fromM andN lies inP. Proof: (1) DrawEC⊥M and ED⊥N. Then (a) PlaneCED⊥M and also⊥N. Why? (b) PlaneCED⊥AB. Why? Now ∠COE is the plane∠ of M−AB−P, and ∠DOE is the plane∠ of N−AB−P. Why? ∠COE = ∠DOE by the hypothesis. Hence EC =ED. §§ 42, 23 (2) LetE be any point such thatEC = ED, these being perpen- dicular respectively toM and N. Then ∠COE = ∠DOE. §§ 40, 23 Hence, the planeP determined byABE is the bisector of the given dihedral angle, that is, the pointE lies inP. Q. E. D. sight work Define a locus. Why are two conditions necessary in determining a locus? Are these conditions fulfilled in the above theorem?

PROPERTIES OF THE PLANE 37 problems in construction In making constructions in solid geometry, it is assumed that 124. A line can be passed through any two points. 125. A plane can be passed through three non-collinear points. 126. Through a point not on a given line there can be drawn a line parallel to this line. 127. Through a point a line perpendicular to a given line can be drawn. 128. Corollary. A plane can be passed through(1) a line and a point not on it,(2) two intersecting lines,(3) two parallel lines. § 73 Q. E. F. constructing a plane perpendicular to a line 129. Problem Through a given point to construct a plane perpendicular to a given line. (1) Given a linel and a pointP on it. Fig. 1. To construct a planeM⊥l at P. Construction: Through P draw two lines PA and PB each perpendicular tol. §§ 127, 75 Then the planeM determined byPA and PB is⊥l. § 76 (2) Given a linel and a pointP not on it. Fig. 2. To construct a planeN throughP and⊥l. Construction: From P draw a linePQ⊥l. § 127

38 SOLID GEOMETRY: BOOK I AtQ construct a planeN⊥l, as in (1). Then, by § 79,PQ lies inN, and henceN is the plane required. Q. E. F. constructing a line

perpendicular to a plane 130. Problem Through a given point to construct a line perpendicular to a given plane. (1) Given a planeM and a pointP in it. To construct a line throughP and⊥M. Construction: Let l be any line inM not passing throughP. DrawPB⊥l. ThroughB drawBK⊥l but not inM. In the planePBK drawPC⊥PB and meetingBK in C. ProvePC⊥M. §§ 76, 117, 114 (2) Given a planeN and a pointP not in it. To construct a line throughP and⊥N. Construction: Let l be any line inN. Draw PB ⊥ l and inN drawBK⊥l. In the planePBK drawPC⊥BK. Prove, as in (1),PC⊥N. (See also § 80.) Q. E. F.

PROPERTIES OF THE PLANE 39 constructing a plane p arallel to a line 131. Problem Through one of two skew lines to pass a plane parallel to the other. Given two skew linesl1 and l2. To construct a plane throughl1∥l2. Construction: Through a pointP inl1 pass a linel3∥l2.(§ 126.) The plane determined byl1 and l3 is∥l2. § 90 Q. E. F. constructing a plane perpendicular to a plane 132. Problem Through a given line to pass a plane perpendicular to a given plane. Given a linel and a planeM. To pass a plane throughl and⊥M. Construction: Through a point P in l pass a linel1⊥M. § 130 The planeN determined byl and l1 passes throughl and is⊥M. § 117 Q. E. F. constructing a plane p arallel to each of two lines 133. Problem Through a point to pass a plane parallel to each of two given lines. Given a pointP and two linesl1 and l2. To construct a plane throughP parallel tol1 and l2. Construction: Through P pass linesl′ 1 and l′ 2 such thatl′ 1∥ l1 and l′ 2∥l2. § 126

40 SOLID GEOMETRY: BOOK I Then the plane determined byl′ 1 andl′ 2 is parallel to bothl1 andl2. § 90 Q. E. F. sight work 1. Study the special case of § 132 in which the given line is perpendicular to the given plane. 2. Is it possible that the plane constructed by the method of § 133 may contain one or both of the given lines? Study all possible cases. 3. Through a given point pass a plane⊥ each of two given planes. How many such can be constructed? Study the case in which the two given planes are parallel to each other. Suggestion.Consider the relation of the required plane to the intersection of the two given planes. common perpendicular to two skew lines 134. Problem To construct a common perpendicular to two skew lines. Given two skew linesl1 and l2. To construct a lineBC perpendicular to each of them. Construction: Through A, any point inl2, drawl3∥l1. Let M be the plane determined byl2 and l3. Through l1 pass a planeN⊥M and meetingM in l4. AtB the intersection ofl2 and l4 erect BC⊥M.

PROPERTIES OF THE PLANE 41 Then BC is the required line perpendicular tol1 and l2. Outline of Proof: Show each of the following: (1) BC lies in the planeN, and □ meets l1. (2) l1∥l4, (3)BC∥l4, and □BC⊥l1. 135. Corollary. There cannot be more than one common per- pendicular to two skew lines. Suggestion.Suppose a second common perpendicular drawn from a point P1 inl1 to a pointP2 inl2. Then show that fromP1 it would be possible to have two lines⊥ plane M. sight work 1. If a line is⊥ a plane, show that its projection is a point. 2. If a line-segment is∥ a plane, show that its projection is a segment equal to the given segment. 3. If a line-segment is oblique to a plane, show that its projection is less than the given segment. exercises 1. Show that the projections of two parallel lines on a plane are parallel. Is the converse true? Illustrate with pieces of cardboard. 2. If two parallel lines meet a plane, they make

equal angles with it. Why? Is the converse true? What is the corresponding theorem in plane geometry? Is its converse true? 3. If a line cuts two parallel planes, it makes equal angles with them. Why? Is the converse true? Discuss the corresponding theorems in plane geometry. 4. If two parallel line-segments are oblique to a plane, their projec- tions on the plane are in the same ratio as the given segments. 5. A line and its projection on a plane determine a plane perpendic- ular to the given plane.

42 SOLID GEOMETRY: BOOK I 6. If a line is parallel to one of two given planes and perpendicular to the other, then the two planes are perpendicular to each other. 7. Find the locus of all points equidistant from two pointsA andB and also at a given distance from a planeM. Discuss. 8. Find the locus of all points equidistant from two given planes and also equidistant from two given points. Discuss. 9. Find the locus of all points equidistant from two given planesM and N, and also equidistant from two other given planesQ and R. Discuss. Compare with the corresponding loci in plane geometry. 10. Prove that there is a line in space every point of which is equidis- tant from three pointsA,B,C, provided these points do not lie on one line. pol yhedral angles 136. Polyhedral Angle. Given a convex polygon and a pointP not in its plane. If a half-linel with its end point fixed atP moves so thatitalwaystouchesthepolygonandismadetotraverseitcompletely, it is said to generate aconvex polyhedral angle. 137. Vertex. Edges. The fixed point is the vertex of the polyhedral angle, and the rays through the vertices of the polygon are theedges of the polyhedral angle. Any two consecutive edges determine a plane, and the portion of such a plane included between these edges is called aface of the poly- hedral angle. 138. Face Angles. The plane angles at the vertex are called the face angles of the polyhedral angle. A polyhedral angle having three faces is called atrihedral angle. Thus in the figure,P is the vertex,PA, PB, etc., are the edges, and ∠APB, ∠BPC , etc., are the face angles. 139. Equal Polyhedral Angles.Two polyhedral angles areequal if they can be made to coincide.

PROPERTIES OF THE PLANE 43 A polyhedral angle is read by naming the vertex and one letter in each edge, asP−ABCDE, or by naming the vertex alone where no ambiguity would arise. 140. Order of Parts in Triangles. In the triangles ABC, and A′B′C′, AB = A′B′, BC = B′C′, CA = C′A′. However, the sides AB, BC, CA, are arranged incounter-clockwiseorder, and the sidesA′B′, B′C′, C′A′, are arranged in clockwise order. That is, the parts of the two triangles are arranged in opposite orders. 141. Order of Parts in Polyhedral Angles. In the trihedral anglesO and O′ the face angles AOB, BOC, COA are ar- ranged incounter-clockwise order as viewed from the vertex, while A′O′B′, B′O′C′, C′O′A′ are arranged inclockwise order. The trihedral anglesO andO′ cannot be made to coincide, even though their corresponding parts are equal. This can be illustrated by attempting to put a left glove on the right hand. conditions which make trihedral angles equal 142. Theorem XXIII. Two trihedral angles are equal if two face angles and the included dihedral angle of one are equal respectively to two face angles and the included dihedral angle of the other and arranged in the same order.

44 SOLID GEOMETRY: BOOK I Suggestion for Proof: This and the following theorem may be proved by superposition in the same manner as the corresponding

the- orems on the equality of triangles in plane geometry. 143. Theorem XXIV. Two trihedral angles are equal if a face angle and the two adjacent dihedral angles of one are equal respectively to a face angle and the adjacent dihedral angles of the other and arranged in the same order. equal dihedral angles opposite equal f ace angles 144. Theorem XXV. If two trihedral angles have the three face angles of the one equal respectively to the three face angles of the other, the dihedral angles opposite the equal face angles are equal. Given the trihedral anglesP and P′, in which ∠a = ∠a′, ∠b = ∠b′, ∠c = ∠c′. To prove that the corresponding dihedral angles are equal. Outline of Proof: Let P and P′ be cut by the planesABC and A′B′C′, makingPA =PB =PC =P′A′ =P′B′ =P′C′. On the edgesPA and P′A′ lay offAF =A′F′, and throughF and F′ pass planes⊥ to PA and P′A′ respectively. AF andA′F′ are to be taken short enough so that these planes shall cut the segmentsAB, AC and A′B′, A′C′ respectively.

PROPERTIES OF THE PLANE 45 In order to prove that∠B−AP−C = ∠B′−A′P′−C′, i.e. that ∠DFE = ∠D′F′E′, use the pairs of△sAPB, A′P′B′; BPC , B′P′C′; CPA, C′P′A′; ABC, A′B′C′. Then prove in order: (1) △ADF =△A′D′F′; (2) △AEF =△A′E′F′; (3) △ADE =△A′D′E′; (4) △DEF =△D′E′F′. □ ∠ DFE = ∠D′F′E′. 145. Corollary. Two trihedral angles are equal if the face angles of one are equal respectively to the face angles of the other and arranged in the same order. symmetrical trihedral angles 146. Symmetrical Trihedral Angles.Two tri- hedral angles aresymmetrical, one to the other, if the face angles and the dihedral angles of one are equal respectively to the face angles and the dihedral angles of the other, but arranged in theopposite order. 147. Vertical Trihedral Angles. Two trihe- dral angles are vertical if the edges of one are the prolongations of the edges of the other. 148. Corollary 1. Two vertical trihedral angles are symmetrical. The proof is evident from the figure above. 149. Corollary 2. Two trihedral angles are symmetrical if the face angles of one are equal to the face angles of the other and arranged in the opposite order. sight work 1. Is it possible that all three face angles of a trihedral angle shall be right angles? Is it possible that all three dihedral angles shall be right angles? Suggestion. Consider a corner of a rectangular box.

 2. If three planes meet in a point how many trihedral angles are formed? If all face angles of one of these trihedral angles are right angles, what about the face angles of the other trihedral angles? 3. If in the figure of § 144 the∠a = ∠b = ∠c = 60◦, what can be said about the trianglesPAB, PBC , PCA, andABC? 4. Show how to locate a point which is at a distance of2 feet from each of the three planes determined by the faces of a trihedral angle. How many such points are there? Suggestion. Pass planes parallel to each face of the trihedral angle, one on each side of it, and at a distance of2 feet from it. conditions which make trihedral angles symmetrical 150. Theorem XXVI. Two trihedral angles are sym- metrical (1) if two face angles and the included dihedral angle of one are equal respectively to two face angles and the included dihedral angle of the other, but arranged in the opposite order; or (2) if a face angle and the adjacent dihedral angles of one are equal respectively to a face angle and the adjacent dihedral angles of the other, but arranged in the opposite order. Proof: Lett1 and t2 be two trihedral angles having the properties specified under (1). Lett3 be a trihedral angle symmetrical tot1. Then t2 =t3. § 142 □ t1 and t2 are symmetrical.

PROPERTIES OF THE PLANE 47 The proof of the second part is left to the student. 151. Theorem XXVII. The sum of two face angles of a trihedral angle is greater than the third face angle. Outline of Proof: Connect pointsA and C on two sides. Suppose ∠ADC > ∠ADB. Con- struct DE making ∠ADE = ∠ADB. Suppose that point B is taken so that DB = DE, and that AB and BC are drawn. Then prove (1) AB =AE, (2) AC <AB +BC, (3) EC <BC, (4) ∠EDC <∠BDC, (5) ∠ADC <∠ADB + ∠BDC. sum of the f ace angles of a pol yhedral angle 152. Theorem XXVIII. The sum of the face angles of any convex polyhedral angle is less than four right angles. Proof: Let ABCDE be a plane section of the given polyhedral angle. The number of triangles thus formed havingP for a vertex is equal to the number of face angles of the polyhedral angle. Let O be any point in the base, and drawOA, OB, OC, etc. Then ∠PBA + ∠PBC > ∠ABC, ∠PCB + ∠PCD > ∠BCD, and so on. § 151

48 SOLID GEOMETRY: BOOK I Now the sum of the∠s of the△sOAB, OBC, etc., is equal to the sum of the∠s of the△sPAB, PBC , etc. Hence, ∠APB + ∠BPC +··· < ∠AOB + ∠BOC +··· . But the sum of the∠s about O is four right angles. Therefore, the sum of the face angles of the polyhedral angle is less than four right angles. Q. E. D. Note. If as in § 144 the three edges of a trihedral angle are cut by a plane, the intersection is a triangle. Then the sides and angles of such a triangle correspond to the face angles and dihedral angles respectively of the trihedral angle. sight work 1. What theorem in plane geometry corresponds to the theorem of § 151 in the sense of the above note? 2. Discusspropositionsofplanegeometrycorrespondingtothoseof§150. summar y of book i 1. State the axioms used in this Book. 2. Describe the various ways of determining a plane. 3. State the definitions on perpendicular lines and planes. 4. State a theorem on a line through a point perpendicular to a plane. 5. State a theorem on a plane through a point perpendicular to a line. 6. State the definitions on parallel lines and planes. 7. State propositions on (1) a plane through a point parallel to a given plane; (2) a plane through a line parallel to a given line; (3) a plane through a point parallel to two given lines. 8. State some facts about perpendiculars in the plane which do not hold in space.

PROPERTIES OF THE PLANE 49 9. Define a dihedral angle and the plane angle of a dihedral angle. 10. Whattheoremsonperpendicularplanesareprovedinconnection with dihedral angles? Notice that these theorems include all those on one plane perpendicular to another. Why should this be the case? 11. State the definitions and theorems on projections given in this Book. 12. State the definitions and theorems on polyhedral angles thus far given. 13. Define symmetrical polyhedral angles. Are symmetrical trihedral angles equal? 14. State theorems on equal and symmetrical trihedral angles. 15. Give examples of loci in plane geometry which differ from cor- responding loci in solid geometry. See, for instance, § 6. miscellaneous exercises on book i 1. A Christmas tree is made to stand on a cross-shaped base. What must be the relation of the tree to each piece of the cross in order that it may be perpendicular to the floor? 2. IfA,B, andC do not lie in the same line, and if their projections on a planeM do lie in a straight line, what is the relation of the planes M and ABC? 3. Is it possible to project a circle upon a plane so that the projection shallbeastraightline-segment? Ifso, howmustthecircleandtheplane be related? 4. If the projections of a set of points on each of two planes not par- allel to each other lie

in straight lines, show that the points themselves lie in a straight line. 5. Find the locus of points3 feet from one of two intersecting planes and 6 feet from the other. 6. It is required that a series of electric lights shall be7 feet above the floor of a room and 3 feet from the walls. Find the locus of all points at which such lights may be placed.

50 SOLID GEOMETRY: BOOK I 7. Find the locus of a point in space such that the difference of the squares of its distances from two fixed points,A and B, is constant. Suggestion. Through eitherA or B construct a plane⊥AB. 8. Given two pointsA and B on the same side of a planeM. De- termine a pointP in M such thatAP +PB shall be a minimum. Suggestion. Pass a plane throughA andB perpendicular toM, and proceed as in Ex. 10, page 280, Plane Geometry. 9. Show that if the edge of a dihedral angle is cut by two parallel planes, the sections which they make with the faces form equal angles. Is the converse proposition true? 10. Show that if all edges of a trihedral angle are cut by a series of parallel planes, the intersections form a series of similar triangles. Is the converse proposition true? 11. Find the locus of the intersection points of the medians of the triangles obtained in Ex. 10. Also of the altitudes. 12. How many planes may be made to pass through a given point parallel to a given line? Discuss the mutual relation of all such planes. 13. Through a pointP construct a line meeting each of two linesl1 and l2. Suggestion. Let M and N be the planes determined byP and l1, P and l2. Consider their intersection. Is this construction always possible? 14. A linel is parallel to a planeM, and linesl1 andl2 inM are not parallel tol. Show that the shortest distance betweenl and l1 is equal to the shortest distance betweenl and l2. 15. Find the locus of all points equidistant from the planes deter- mined by the faces of a trihedral angle.

PROPERTIES OF THE PLANE 51 16. Find the locus of all points equidistant from the edges of a trihedral angle. Suggestion.On the edges lay offPA =PB =PC. LetO be equidistant fromA,B, andC. Then any point Q in PO is equidistant from the edges. 17. Planes determined by the edges of a tri- hedral angle and the bisectors of the opposite face angles meet in a line. 18. If three planes are so related that the segments intercepted on any transversal line are in the same ratio as the segments in- tercepted on any other transversal then the planes are parallel. Suggestion. LetM,N,Q be the three planes. FromA any point inM draw three lines, not in the same plane, meetingN in A′, B′, C′, andQ in A″, B″, C″. Use the hypothesis to show thatA′B′∥ A″B″ and C′B′∥ C″B″. Hence Q∥N. Similarly show thatM∥N.

52 SOLID GEOMETRY: BOOK I Thales of Miletus(640–542 B.C.) was one of the Seven Wise Men of Greece. He learned astronomy and geometry in Egypt and was the first to introduce the scientific study of geometry in Greece. He measured the height of the Pyramids in Egypt by similar triangles and found a method of computing the distance of a ship at sea. He predicted the solar eclipse of 585 B.C.

BOOK II REGULAR POL YHEDRONS 153. Polyhedron. A polyhedronis a geometric solid whose bound- ary consists of plane polygons. 154. Convex Polyhedron. A polyhedron is convex if every section of it made by a plane is a convex polygon. 155. Faces, Vertices, Edges. The polygons which bound the polyhedron are itsfaces; the sides of these faces are theedges and their vertices are the vertices of the

polyhedron. 156. Surface of Polyhedron. The faces, edges, and vertices taken together form the surface of the polyhedron. 157. Polyhedrons Classified According to the Number of Faces. A polyhedron of four faces is atetrahedron, one of six faces is ahexahedron, one of eight faces anoctahedron, one of twelve faces a dodecahedron, and one of twenty faces anicosahedron. These names are all derived from the Greek and refer to the number of faces. Compare other words derived from the same Greek roots such as hexagon, octagon, dodecagon, etc. sight work 1. What is the smallest number of polygons which maybe used to inclose a portion of space? Suggestion. Remembering that a triangle is regarded as a polygon, con- sider a triangular pyramid. 2. How many edges and how many vertices has a tetrahedron? 3. How many faces, edges, and vertices has a hexahe- dron? Suggestion. Consider a cube.b bSee Transcriber's Notes.

54 SOLID GEOMETRY: BOOK II models of the regular pol yhedrons 158. Regular Polyhedrons. A polyhedron whose polyhedral angles are equal and whose faces are equal regular polygons is aregular polyhedron. The following are regular polyhedrons: 159. Models of the Regular Polyhedrons.Models of the regu- lar polyhedrons may be made with cardboard by cutting out patterns, as shown in the figures below, folding along the dotted lines, and fasten- ing the edges together by means of gummed paper strips corresponding to the dotted margins.

REGULAR POLYHEDRONS 55 sight work 1. What is the smallest possible number of face angles in a polyhedral angle? 2. How many faces meet in a vertex of a regular tetrahedron? What is the sum of the face angles forming one of its polyhedral angles? 3. How many faces meet in a vertex of a regular octahedron? in a regular icosahedron? 4. What is the sum of the face angles forming a polyhedral angle in a regular octahedron? in a regular icosahedron? 5. Why may not six or more equilateral triangles be used to form a polyhedral angle? 6. What is the sum of the face angles forming a poly- hedral angle of a cube (hexahedron)? 7. Why may not four or more squares be used to form a polyhedral angle? 8. How many faces meet in a vertex of a regular do- decahedron? What is the sum of the face angles forming one of its polyhedral angles? 9. Why may not more than three regular pentagons be used to form a polyhedral angle? 10. May regular hexagons be used to form a polyhedral angle? Why?

56 SOLID GEOMETRY: BOOK II 11. May any regular polygons of more than six sides be used to form polyhedral angles? Why? 12. Can we show that not more than three different regular polyhedrons may be formed having triangles as their faces? 13. How many regular polyhedrons may be formed having squares as their faces? How many having pentagons as their faces? the number of regular pol yhedrons 160. Theorem I. There are exactly five regular polyhe- drons. Proof: On page 54 it was shown how models of five different regular polyhedrons may be made. Hence there are at least five such polyhedrons. That there cannot be more than five regular polyhedrons follows from the two propositions: (a) Every polyhedral angle has at least three face angles. (b) The sum of the face angles of a polyhedral angle is less than360°. From these propositions it follows that each of the polyhedral angles of a regular polyhedron may be formed by three, four, or five (but not six) equilateral

triangles; by three (but not four) squares; or by three (but not four) regular pentagons. Regular polygons having more than five sides cannot form a polyhedral angle. The details of the proof are left to the student. The sight work on page 55 will furnish suggestions. 161. Construction of Regular Polyhedrons. The regular polyhedrons may be constructed by use of § 130. The construction for the tetrahedron and octahedron are given below. (1)The regular tetrahedron. At the centerE of an equilateral triangleABC erect a perpendicular to the plane of the triangle. On this take a pointD so thatAD =AC. Then the four triangles,ABC, ACD, ABD, BCD, are regular and equal, and the four trihedral angles are equal. Why?

REGULAR POLYHEDRONS 57 (2)The regular octahedron. Through the centerO of a squareABCD draw a perpendicular to the plane of the square and on it take pointsE and F such thatAF =AE =AB. JoinE and F to each of the four vertices,A, B, C, D. Then the eight faces are equal regular triangles, and the six polyhedral angles are equal. Why?

BOOK III PRISMS AND CYLINDERS 162. Closed Prismatic Surface. Given a convex polygon and a straight line not in its plane. If the line moves so as to remain parallel to itself while it touches the boundary of the polygon and is made to traverse it completely, the line is said to generate aclosed prismatic surface. 163.Generator. Element. Themovingline is thegenerator of the surface, and the generator in any one of its positions is anelement of the surface. 164. Prism. A polyhedron bounded by a prismatic surface and two parallel plane sections cutting all its elements is called aprism. 165. Bases. Lateral Surface. Edges. The two parallel cross-sections which bound a prism are its bases and the other faces form itslateral surface. The edges are the lines in which its lateral faces meet. 166. Right Section. Right Prism. A right section of a prism cuts all its edges at right angles. A right prism is one whose bases are right sections. 167. Corollary 1. The lateral faces of a prism are parallelograms. 168. Corollary 2. The lateral edges of a prism are equal and parallel. 169. Corollary 3. The lateral faces of a right prism are rectangles.

PRISMS AND CYLINDERS 59 classifica tion of prisms 170. Kinds of Prisms. Prisms are classified according to the form of their right sections, as triangular, quadrangular, pentagonal, hexagonal, etc. Aregular prismis one whose right section is a regular polygon. 171. Altitude. Area. Thealtitude of a prism is the perpendicular distance be- tween the planes of its bases. The altitude of a right prism is equal to its edge. The lateral area of a prism is the sum of the ar- eas of its lateral faces. The total area is the sum of its lateral area and the areas of its bases. 172. Truncated Prism. A polyhedron which is a part of a prism cut off by a plane meeting all the lateral edges, but not parallel to the base, is called a truncated prism. Two polyhedrons are said to beadded when they are placed so that a face of one coincides wholly or in part with a face of the other, but otherwise each lies outside the other. 173. Parallelopiped. A parallelop- iped is a prism whose bases, as well as lat- eral faces, are parallelograms. A rectangular parallelopipedhas its bases and all its faces rectangles. A cube is a parallelopiped whose bases and faces are all squares. p arallel cross-sections of a prism are equal 174. Theorem I. The cross-sections of a prism made by parallel planes are equal polygons.

60 SOLID GEOMETRY: BOOK III Given a prism cut by two parallel planes forming the poly- gons ABCDE and A′B′C′D′E′. To prove thatABCDE =A′B′C′D′E′. Outline of Proof: (1) Show thatAB = A′B′, BC = B′C′, etc., by proving thatABB′A′, BCC′B′, etc., are S . (2) Show that∠ABC = ∠A′B′C′, ∠BCD = ∠B′C′D′, etc. (3) Hence show that ABCDE and A′B′C′D′E′ can be made to coincide. 175. Corollary. Every cross-section of a prism parallel to the base is equal to the base. sight work 1. Can a parallelopiped be a right prism without being a rectangular parallelopiped? Illustrate. 2. Show that in a rectangular parallelopiped each edge is perpendicular to the other edges which meet it. 3. Show that any section of a prism made by a plane parallel to an edge is a parallelogram. Suggestion. Use in order §§ 89, 84, and 92. 4. Is it possible to construct a prism such that there is no plane per- pendicular to its edges and cutting all of them unless some of them are extended?

PRISMS AND CYLINDERS 61 the la teral area of a prism 176. Theorem II. The lateral area of a prism is equal to the product of a lateral edge and the perimeter of a right section. Suggestion. Show that the lateral edges are mutually equal and that the area of each face is the product of a lateral edge and one side of the right-section polygon. Complete the proof. Note. The form of statement in this theorem is the usual abbreviation for the moreprecise form: The lateral area of a prism is equal to the product of the numerical measuresof a lateral edge and the perimeter of a right section. Similar abbreviations are used throughout this text. 177. Corollary. The lateral area of a right prism is equal to the product of its altitude and the perimeter of its base. sight work 1. If the lateral edge of a prism is8 inches and the perimeter of its right section 30 inches, what is the lateral area?

62 SOLID GEOMETRY: BOOK III 2. Of all prisms having an altitude6 inches and the perimeter of a right section 24 inches, which one has the smallest lateral area? Suggestion. Which one has the shortest lateral edge? 3. Why is not the lateral area of an oblique prism determined by the perimeter of the base and a lateral edge? conditions making prisms equal 178. Theorem III. Two prisms are equal if three faces having a common vertex in the one are equal respectively to three faces having a common vertex in the other, and similarly placed. Given the three faces meeting atB in prism P equal re- spectively to the three faces meeting atB′ in prism P′, and similarly placed. To prove thatP can be made to coincide withP′. Outline of Proof: Trihedral ∠sB and B′ are equal. Why? Now apply the two prisms, makingB coincide withB′, and then show in detail that: (1) The lower bases coincide. (2) The lateral faces atB and B′ coincide. (3) The upper bases coincide.

PRISMS AND CYLINDERS 63 (4) All the lateral faces coincide. 179. Corollary 1. Two truncated prisms are equal under the conditions of § 178. 180. Corollary 2. Two right prisms are equal if they have equal bases and equal altitudes. opposite f aces of a p arallelopiped 181. Theorem IV. Opposite faces of a parallelopiped are equal and parallel. Suggestions. Consider the opposite facesABFE and DCGH. (1) Show that the sides of the anglesABF and DCG are parallel, and hence that the planes determined by them are parallel. (2) Show that these faces are equal. In like manner argue about any other pair of opposite faces. 182. Corollary. Any section of a parallelopiped made by a plane cutting four parallel edges is a parallelogram. Suggestion. In the figure

show that RQ‖ ST and QT‖RS.

64 SOLID GEOMETRY: BOOK III sight work 1. A section made by a plane passed through two diagonally opposite edges of a parallelopiped is a parallelogram, for example, the section through DH and BF in the figure of § 181. 2. Show that any two of the four diagonals of a parallelopiped bisect each other. Suggestion. Make use of example 1. 3. If the parallelopiped in § 182 is rectangular, is the section necessarily a rectangle? May it be so? 4. Show that the section of a cube made by a plane through two diago- nally opposite edges is a rectangle. Why can it not be a square? Can you construct a parallelopiped in which such a section would be a square? plane through opposite edges of a p arallelopiped 183. Theorem V. A plane through diagonally oppo- site edges of a parallelopiped divides it into two triangular prisms having equal right sections and equal lateral edges. Given the parallelopiped ABCD−F with opposite edges AE and CG. To prove that the plane throughAE and CG divides it into two triangular prisms having equal lateral edges and equal right sections.

PRISMS AND CYLINDERS 65 Proof: By § 182 the right sectionKLMO of the given prism is a parallelogram. Hence KM divides it into two equal triangles. Hence the right sections of the prismsABC−F and ACD−H are equal. The lateral edgesBF and DH are equal by § 168. sight work 1. If a right section of a prismatic space is a parallelogram, what kind of prism will be cut out by two right sections? 2. If a right section of a prismatic space is a rectangle, what kind of prism will be cut out by two right sections? 3. If a right section of a prismatic space is a regular hexagon, what kind of prism will be cut out by two right sections? 4. What kind of prism will be cut out by two right sections of any closed prismatic surface? exercises 1. The face angles of one trihedral angle of a parallelopiped are65◦, 70◦, and75◦ respectively. Whatarethefaceanglesoftheothertrihedral angles? 2. Show that two prisms cut from the same prismatic space and having equal lateral edges have equal lateral areas. 3. Show that the diagonals of a rectangular parallelopiped are all equal to each other. 4. Show that the square on the diagonal of a rectangular parallelopiped is equal to the sum of the squares on the sides meeting at a vertex from which it is drawn. E.g. in the figureAG2 =AC2 +CG2 =AB2 +BC 2 +CG2. 5. Find the ratio of the diagonal to one edge of a cube. 6. Find the edge of a cube whose diagonal is14 inches. Find the diagonal of a cube whose edge is16 inches.

66 SOLID GEOMETRY: BOOK III 7. Find a diagonal of a rectangular parallelopiped whose edges are 6, 8, and 10 inches respectively. 8. Are the diagonals of a cube perpendicular to each other? 9. In a rectangular parallelopiped the diagonal of the base is12 in. and the altitude is5 in. Find the diagonal of the parallelopiped. 10. If two equal right prisms whose bases are equilateral triangles are placed together so as to form one prism whose base is a parallelogram, compare the lateral area of the prism so formed with the sum of the lateral areas of the original prisms. 11. Arightprismwhosebasesareregularhexagonsisdividedintosix prisms whose bases are equilateral triangles. Compare the lateral area of the original prism with the sum of the lateral areas of the resulting prisms. volume of a rect angular p arallelopiped 184. Thus far certain properties of prisms have been studied, but no attempt has been made tomeasure the space occupiedby such a solid. For this purpose we consider first

a rectangular parallelopiped. 185. Numerical Measure. In case each edge of a rectangular parallelopiped is commensurable with a unit segment, the number of times which a unit cube is contained in it is thenumerical measureor the volume of the parallelopiped. 186. The Commensurable Case. In the commensurable case just described, the volume is easily computed. E.g. if in the figure one edgeAC is 4 units, and an adjoining edgeAB is 3 units, then a cube asAK, whose edge is one unit, may be laid off4 times along AC and a tier of3· 4 = 12 such cubes will adjoin the face AD. Since the edge BE is 5 units long, 5 such tiers will exactly fill the space within the solid. That is, 5· 3· 4 = 60 is the number of cubic units in the solid. Hence in this case Volume = Length×Width×Height.

PRISMS AND CYLINDERS 67 Again, if the given dimensions are3.4, 2.6, 4.5 decimeters respectively, then unit cubes, with edge onedecimeter, cannot be made to fill exactly the space inclosed by the figure, but cubes with edge each onecentimeter will do so, giving34, 26, and 45 respectively along the three edges of the figure. Therefore, the volume is 34· 26· 45 = 39, 780 cubic centimeters, or39.78 cubic decimeters. Hence in this case also Volume = Length×Width×Height. 187. Formula for Volume of a Rectangular Solid.We there- fore conclude that the volume of a rectangular parallelopiped with com- mensurable dimensions is given by the formula Volume =Length×Width×Height. volume of a rect angular solid 188. The Incommensurable Case. If a rectangular parallelop- iped is such that any two of its dimensions are incommensurable with each other, then there is no unit cube, however small, in terms of which the volume can be exactly expressed. But by choosing a unit cube sufficiently small we may determine the volume of a rectangular solid which differs from the given one by as little as we please. E.g., if the length is5 inches and the width is √ 3 inches, then the base cannot be exactly covered with equal cubes, however small. But since √ 3 = 1.732... , if we take as a unit of volume a cube one one- hundredth of an inch on a side, then the base of a rectangular solid whose length is5 inches and whose width is1.73 inches can be exactly covered with a layer of these cubes and the number of cubes in such a layer is 500× 173 = 86, 500. If the height is2 inches, then 200 layers will just reach the top, making 500× 173× 200 = 17, 300, 000 cubes. And since 1, 000, 000 such small cubes make a cubic inch, the volume of this solid is17.3 cubic inches.

68 SOLID GEOMETRY: BOOK III The portion of the given solid not thus filled is5 inches long, 2 inches high and less than.003 of an inch in thickness. Hence, its volume is less than5× 2×.003 = .03 of a cubic inch. By expressing √ 3 to further places of decimals and then using smaller and smaller units of volume, successive rectangular solids may be found which differ less and less from the given solid. The foregoing considerations constitute an informal proof of the following theorem: 189. Theorem VI. The volume of a rectangular parallelopiped is equal to the product of its length, width, and height. In symbols, V =l·w·h. The above argument shows that this theorem holds for all rectangular parallelopipeds used in the process of approximation, and hence it applies to all practical measurements of the volumes of such solids. 190. Corollary 1. The volume of a rectangular parallelopiped is equal to the product of the numerical measures of its base and altitude. 191. Corollary 2. If two rectangular parallelopipeds have two dimensions respectively equal to each

other, their volumes are in the same ratio as their third dimensions; and if they have one dimension the same in each, their volumes are in the same ratio as the products of their other two dimensions. For ifV andV′ are the volumes, anda,b,c, anda′,b′,c′ the dimensions, then V V′ = a·b·c a′·b′·c′ = a a′ if b =b′ and c =c′; or V V′ = a·b a′·b′ if c =c′. 192. Volumes of Prisms in General. From the formula for volumes of rectangular solids,V = l·w·h, we deduce the volumes of prisms in general by means of the principle: Two polyhedrons have the same volume if they can be made to coin- cide, or if they can be divided into parts which can be made to coincide in pairs.

PRISMS AND CYLINDERS 69 The sign = between two polyhedrons means that they are equal in all respects; that is, can be made to coincide. The wordequivalent is used to mean equal in volume. exercises 1. The dimensions in inches of a rectangular parallelopiped are5, 10, and 2 √ 2. Taking the approximate value of √ 2 as 1.4142, find the approximate volume. Show that this differs from the exact volume by less than.001 of a cubic inch, it being given that √ 2 lies between 1.41421 and 1.41422. 2. If the dimensions in inches of a rectangular parallelopiped are 5, 4 √ 2, and 5 √ 3, find how near an approximation to the volume is possible by taking √ 2 between 1.414 and 1.4143 and √ 3 between 1.732 and 1.7321. 3. If the dimensions in inches of a rectangular parallelopiped are 3 √ 2, 3 √ 3, 2 √ 5, find the values to be used for √ 2, √ 3, √ 5, to obtain the volume within.001 of a cubic inch. volume of an oblique prism 193. Theorem VII. The volume of an oblique prism is equal to that of a right prism having for its base a right section of the oblique prism and for its altitude a lateral edge of the oblique prism.

70 SOLID GEOMETRY: BOOK III Given the oblique prismAD′ with FGHJK a right section and F′G′H′J′K′ a right section of the prism extended so that AA′ =KK′. To prove that the oblique prismAD′ has the same volume as the right prismKH′. Proof: In the truncated prismsAH and A′H′, baseABCDE = base A′B′C′D′E′. § 175 In facesABFK and A′B′F′K′, we have AB =A′B′, ∠1 = ∠3, ∠2 = ∠4, Why? and AK =A′K′ and BF =B′F′. Why? □ABFK and A′B′F′K′ will coincide and are equal. Likewise, AKJE =A′K′J′E′. □ AH =A′H′. § 179 But the given prism AD′ =AH +KD′, and the right prism KH′ =A′H′ +KD′. □ AD′ =KH′. Q. E. D. 194. Corollary. Two prisms having equal lateral edges and equal right sections are equal in volume. volume of any p arallelopiped 195. Theorem VIII. The volume of any parallelopiped is equal to the product of its base and altitude.

PRISMS AND CYLINDERS 71 Given any parallelopipedP with area of baseb and alti- tude h. To prove that its volume is equal tob×h. Proof: Considering faceAK as the base of prismP, produce the four edges parallel toAB, and lay offA′B′ =AB. Through A′ and B′ erect planes⊥ A′B′, thus cutting off the right prism Q with A′L as one base. Now consideringCL as the base of prismQ, produce the four edges ∥CB′ and lay offC′D =CB′. At C′ and D erect planes⊥ C′D, cutting off the right prismR, which is a rectangular parallelopiped with baseb″. Now prove in detail: (1)h =h′ =h″, (2) b =b′ =b″, (3) Vol.P = Vol.Q = Vol.R, § 193 (4) Vol.R =b″h″, (5) Vol.P =bh. 196. Corollary. Two parallelopipeds are equal in volume if they have equal altitudes and bases of equal areas.

72 SOLID GEOMETRY: BOOK III volume of a triangular prism 197. Theorem IX. The volume of a triangular prism is equal to the product of its base and altitude.

Given the triangular prism whose base isABC. To prove that the volume is equal to the area of△ABC multiplied by the altitudeh. Proof: Complete the S ABCD and EFGH and drawDH. Now show thatCDHG andADHE are S , and hence thatBH is a parallelopiped. Use § 43 (2) Let KLMO be a right section ofBH, and letKM be the line in which the planeACGE cuts the planeKLMO. Then (1) prismABC−F = 1 2 prism BH, §§ 183, 194 (2) But prismBH =h×area of ABCD, § 196 (3) Hence, prismABC−F =h× 1 2area of ABCD, that is, prism ABC−F =h×area of ABC. Therefore the volume of prismABC−F is equal to the area of its base times its altitude. Q. E. D. sight work 1. Find the volume of a prism whose altitude is8 in. and whose base is a right triangle with legs5 in. and 6 in. respectively. 2. Find the volume of a right prism whose altitude is6 in. and whose base is a rectangle with sides3 in. and 5 in.

PRISMS AND CYLINDERS 73 volume of any prism 198. Theorem X. The volume of any prism is equal to the product of its base and altitude. Outline of Proof: Any prism, can be divided into triangular prisms by planes passing through one edge and each of the other non- adjacent edges. The altitudes of the triangular prisms are the same as that of the given prism, and the sum of their bases is equal to the base of the given prism. Now use § 197 and complete the proof. 199. Corollary 1. The product of the base and altitude of any prism is equal to the product of a lateral edge and the area of a right section. For each equals the volume of the prism. See §§ 198, 193. 200. Corollary 2. If two prisms have equivalent bases, their volumes are in the same ratio as their altitudes; and if they have equal altitudes, their volumes are in the same ratio as the areas of their bases. sight work 1. Find the volume of a prism whose altitude is8 in. and the area of whose base is42 sq. in. 2. The volume of a prism is264 cu. in. and its altitude is8 in. Find the area of its base.

74 SOLID GEOMETRY: BOOK III exercises 1. The theorem that the volume of any prism is equal to the product of its base and altitude is of great importance. What theorems of Book III have been used directly or indirectly in proving it? 2. What dimensions of a prism must be known in order to determine its lateral area by means of the preceding theorems of Book III? What dimensions must be known to determine its volume? 3. The edge of a cube ise. Find the total surface and the volume in terms ofe. 4. Find the edge of a cube if its total area is equal numerically to its volume, an inch being used as the unit. 5. Find the volume of a regular right triangular prism whose edges are all equal to6 inches. 6. Find the volume of a regular right hexagonal prism whose edges are all equal to10 inches. 7. A side of the base of a regular right hexagonal prism is3 inches. Find its altitude if its volume is54 √ 3 cubic inches. What is the total area of this prism? 8. The volume of a triangular prism is equal to the area of one lateral face multiplied by half the perpendicular distance of this face from the remaining edge. Prove. 9. The volume of a regular right prism is equal to the lateral area multiplied by half the apothem of the base. Prove. 10. Prove that the sum of the squares of the four diagonals of a rectangular parallelopiped is equal to the sum of the squares of the twelve edges of the parallelopiped. 11. A prismatic space is cut by two pairs of parallel planes which are the same distance apart. Does it follow that the volumes of the prisms thus formed are equal? The lateral areas?

PRISMS AND CYLINDERS 75 cylinders 201. Curved Surface. A surface no part of which is plane is called acurved surface. E.g. the surface of an eggshell or of a stovepipe is a curved surface. 202. A Closed Plane Curve. A curve which can be traced continuously by a point moving in a plane so as to return to its original position without crossing its path is aclosed plane curve. A closed plane curve isconvex if it can be cut by a straight line in not more than two points.c 203. A Cylindrical Surface. If a straight line moves so as to remain parallel to itself, while it always touches a closed convex plane curve and is made to traverse it completely, the line is said to generate a closed convex cylindrical surface. The moving line is the generator, and the generator in any one of its positions is anelement of the surface. 204. Cylinder. A solid bounded by a cylindrical surface and two parallel plane sections cutting all its elements is called acylinder. 205. Bases. Lateral Surface.The two parallel cross-sections which bound a cylinder are itsbases, and the curved surface is itslateral surface. 206. Element. Altitude.That part of the generator of a cylindri- cal surface which is included between the bases of a cylinder is called an element of the cylinder. Thealtitude of a cylinder is the perpendicular distance between its bases. circular cylinders 207. Right Section. A right section of a cylinder is made by a plane cutting each of its elements at right angles. 208. Circular Cylinders. A circular cylin- der is one whose right section is a circle. cSee Transcriber’s Notes.

76 SOLID GEOMETRY: BOOK III The radius of a circular cylinder is the radius of its right section. A cylinder whose elements are at right angles to its bases is called a right cylinder. Otherwise it is anoblique cylinder. A right cylinder whose bases are circles is aright circular cylinder. 209. Axis of a Cylinder. The line passing through the centers of two right sections of a circular cylinder is theaxis of the cylinder. A right circular cylinder may be generated by revolv- ing a rectangle about one of its sides as an axis. The side opposite the axis generates the lateral surface, and the sides adjacent to the axis generate the bases. sight work 1. What is the locus of all points in space which are at a perpendicular distance of 6 in. from a straight line10 in. long? 2. Show that any two elements of a cylinder determine a plane section of the cylinder which cuts the two bases in parallel lines. 3. If every plane determined by two elements of a cylinder is perpendic- ular to its bases, what kind of cylinder is it? 4. If one right section of a cylinder is a circle, what would appear to be true of all its right sections? For proof see § 213. 5. If a circular cylinder is oblique, are its bases circles? 6. If an oblique cylinder has a circular base, is its right section a circle? longitudinal sections of a cylinder 210. Theorem XI. If a plane contains an element of a cylinder and meets it in any other point, then it contains another element also, and the section is a parallelogram.

PRISMS AND CYLINDERS 77 Given the cylinderAC and a planeM containing the ele- ment AD and another point asP or P′. To prove that it contains another element and that the section con- taining these two elements is a parallelogram. Proof: (1) When the pointP is in the lateral surface. Let BC be the element throughP. Then BC∥AD and they determine a planeM. §§ 203, 73 Also AB∥DC and ABCD is a parallelogram. § 92 (2) When the pointP′ is in one of the bases. Draw AB through P′. Then by (1) the planeM contains the ele- mentBC and the section is a parallelogram.

Q. E. D. 211. Plane Tangent to a Cylinder. If a plane contains an element of a cylinder and no other point in it, the plane is said to betangent to the cylinder, and the element is called theline of contact. It may be noted that a cross-section of a cylin- der and its tangent plane consists of a plane curve and a line tangent to it. the bases of a cylinder are equal 212. Theorem XII. The bases of a cylinder are equal plane figures.

78 SOLID GEOMETRY: BOOK III Given a cylinder with the basesb and b′. To prove thatb =b′. Proof: Take any three pointsA,B,C in the rim of the baseb and draw elements through these points, meeting the baseb′ in D, E, F. Show that△ABC =△DEF. Use §§ 210, 27 Now, while the elementsAD andBE remain fixed, conceiveCF to generate the surface of the cylinder. Evidently△ABC =△DEF in every position ofCF. Hence, if baseb′ is applied to baseb with these triangles coinciding in one position, they will coincide in every position corresponding to the moving generator. That is,b′ coincides withb. Q. E. D. The proposition of § 212 may also be stated in the form of the following corollary: 213. Corollary 1. Parallel plane sections of a cylinder are equal, if they cut all the elements. 214. Corollary 2. The axis of a circular cylinder passes through the center of all its right sections.

PRISMS AND CYLINDERS 79 measuring the surf ace and volume of a cylinder 215. Area of a Curved Surface.Thus far in geometry the word area has been used in connection with plane figures only. In some cases the computation of an area has been made by an approximation process only, as in the case of some rectangles and of the circle. Indeed, in these cases approximate measurement only is possible, since no square unit exists in terms of which such areas can beexactly measured. In the case of any curved surface it is evident that approximate measurement is theonly kind possible in terms of a plane area unit, since no such unit, however small, can be made to coincide with a part of such a surface.d 216. Volume of a Solid Having a Curved Surface.Since the unit of measure for solids is the cube, it is evident that a solid having a curved surface cannot contain such a unit an integral number of times. Hence approximate measurement only of the volumes of such solids is possible. To measure approximately the area and volume of a cylinder use is made of inscribed and circumscribed prisms. 217. Inscribed Prisms. A prism is said to be inscribed in a cylinderif its lateral edges are elements of the cylinder, and their bases lie in the same planes. 218. Circumscribed Prisms. A prism is said to becircumscribed about a cylinderif its lateral faces are all tangent to the cylinder, and their bases lie in the same planes. Itis evident thatby increasingthe numberof lateralfaces oftheinscribed or circumscribed prisms the surface of the latter may be made to lie as near the surface of the cylinder as we please. The lateral edges of the prisms will remain equal to an element of the cylinder, while the right sections of the prisms can be made to differ as little as we please from the right section of the cylinder. dSee Transcriber’s Notes.

80 SOLID GEOMETRY: BOOK III la teral area of a cylinder 219. Fundamental Assumption on the Area and Volume of a Cylinder. We now assume that A cylinder has a definite lateral area and a definite volume which may be approximated as nearly as we please by taking the lateral areas and the volumes of successive inscribed or circumscribed prisms. 220. Lateral Area of a Cylinder. Since the lateral

area of a prism is equal to the product of a lateral edge and the perimeter of a right section (§ 176), it follows that this theorem holds for every inscribed or circumscribed prism used in approximating the lateral area of a cylinder. The foregoing considerations constitute an informal proof of the following theorem: 221. Theorem XIII. The lateral area of a cylinder is equal to the product of an element of the cylinder and the perimeter of a right section. 222. Corollary 1. If r is the radius of a right section of a circular cylinder, ande is an element, the lateral surface isS = 2πre (§ 176). 223. Corollary 2. If r is the radius andh is the altitude of a right circular cylinder, then the lateral surface isS = 2πrh (§ 177).

PRISMS AND CYLINDERS 81 volume of a cylinder 224. Since the volume of a prism is equal to the product of its altitude and the area of its base, it follows that this theorem holds for every inscribed or circumscribed prism used in approximating the volume of a cylinder. All practical computationsof the surfaces and the volumes of cylinders are approximations based upon the propositions stated in §§ 220, 224. The foregoing constitutes an informal proof of the following theo- rem: 225. Theorem XIV. The volume of a cylinder is equal to the product of its altitude and the area of its base. 226. Corollary 1. If r1 is the radius of the right section of a circular cylinder ande is an element, then the volume is(§ 199)V =πr2 1e. 227. Corollary 2. If a cylinder of altitudeh has a circular base whose radius isr2, then the volume is V =πr2 2h. 228. Corollary 3. Ifh is the altitude andr the radius of the base of a right circular cylinder, then the volume is V =πr2h. In this caser = r1 = r2, and e = h, and the formulas of corollaries 1 and 2 become identical. Note. The theorems of §§ 221, 225 were stated (and are true) for any cylinders whatever. However, the theorem of § 225 is available for the com- putation of the volume of a cylinder only in case the area of the base or of the right section can be computed. And this is possible byelementary methods only in case these are circles. The lateral area of a cylinder can be computed only in case the perimeter of a right section can be found, and this is possible by elementary methods only in case the right section is a circle.

82 SOLID GEOMETRY: BOOK III sight work 1. If a prism is inscribed in a cylinder, is its lateral area greater or less than that of the cylinder? Is its volume greater or less than that of the cylinder? 2. If a prism of say1000 faces is inscribed in a cylinder, would the the- orems of § 221 and § 225 apply directly to this prism? Would it be easy to detect experimentally the difference between this prism and the cylinder? 3. How would the right section of the prism and the cylinder of example 2 be related? 4. Discuss the questions in examples 1, 2, 3 as related to prisms circum- scribed about a cylinder. 5. What is the lateral area of a cylinder whose element is8 inches and the perimeter of whose right section is24 inches? 6. What is the volume of a cylinder whose altitude is10 inches and the area of whose base is50 square inches? 7. What is the volume of a cylinder whose edge is4 inches, and the area of whose right section is24 square inches? exercises 1. Find the volume of a right circular cylinder with radius5 and altitude 8. 2. Find the volume of a circular cylinder if the radius of a right section is 6 and the length of an element is10. 3. Find the lateral area of a cylinder if the perimeter of a right section is 39 and the length of an element is8. 4. Find the lateral area of a right cylinder if the perimeter of the base is 30 and the altitude is5. 5. Find the total surface area of a right circular cylinder of radius3 and altitude 5. 6.

Find the volume of an oblique cylinder whose base is a circle of radius 4 and whose altitude is8.

PRISMS AND CYLINDERS 83 summar y of book iii 1. Define prismatic surface and prismatic space. 2. Define prism, base of prism, lateral surface, right section, right prism, parallelopiped, truncated prism. 3. Give the rule for finding the lateral area of an oblique prism. How may this rule be modified in the case of a right prism? 4. What can be said about opposite faces of a parallelopiped as to shape and size? 5. Give two rules for finding the volume of an oblique prism. See §§ 193, 198. 6. In proving the theorem on the volume of a prism, we considered triangular prisms, general parallelopipeds, rectangular parallelopipeds, and general prisms. In what order were these taken? Why? 7. Show how you would find the approximate volume of a rectangu- lar parallelopiped whose dimensions are √ 2, √ 3, √ 5. Find this volume accurately to two places of decimals. Discuss the problem of finding this volume correct to four places of decimals. See exercises on page 69. 8. Define closed plane curve, convex plane curve, cylindrical sur- face, cylindrical space, cylinder, right cylinder, circular cylinder, tan- gent plane to a cylinder. 9. Give the rule for finding the lateral surface of any cylinder, of a right circular cylinder. 10. Give two rules for finding the volume of an oblique cylinder. Show that these both hold true for a right circular cylinder. 11. Describe what is meant by the statement “the area and the vol- ume of a cylinder may be approximated by taking the area and the volume of inscribed or circumscribed prisms.”

84 SOLID GEOMETRY: BOOK III miscellaneous exercises on book iii 1. If the lateral surface of a cylinder and the length of an element are known, can the perimeter of a right section be found? If the lateral area is 400π, and an element15, find the perimeter of a right section. 2. The diameter of a right circular cylinder is8, and the diagonal of the largest rectangle made by a vertical section is16. Find its altitude. 3. The volume of a right circular cylinder is128π cubic inches and its altitude is equal to its diameter. Find the altitude and the diameter. 4. A rectangle whose sides area andb is turned about the sidea as an axis and then about the sideb. Find the ratio of the volumes of the two cylinders thus developed. Also find the ratio of the total surfaces of these cylinders. 5. Find the diameter of a right circular cylinder if its lateral area is equal numerically to its volume. Does the result depend upon the altitude of the cylinder? 6. If the altitude of a right circular cylinder is equal to its diameter, find the ratio of the numerical values of its total area and its volume. Does this depend on the radius? 7. A regular hexagonal prism is inscribed in a right circular cylinder whose altitude is equal to the diameter. Find the difference between the volumes of the cylinder and the prism, if a side of the hexagon is 4 inches. 8. A cylindrical tank8 feet in diameter, partly filled with water, is lying on its side. If the greatest depth of the water is 6 feet, what fraction of the volume of the tank is filled with water? 9. In the preceding problem find the fraction of the volume occupied by water if the width of the top of the water along a right cross-section of the tank is4 feet.

BOOK IV PYRAMIDS AND CONES 229. Pyramidal Surface. Given a convex polygon and a fixed point P not in its plane. If a line through the fixed point moves so as to

touch the boundary of the polygon and is made to traverse it completely, the line is said to generate a convex pyramidal surface. The moving line is thegenerator of the surface, and in any of its positions it is an element of the surface. The fixed point is the vertex of the pyramidal surface. 230. Nappes. A pyramidal surface has two parts, one on each side of the vertex, which are called nappes. 231. Pyramid. The solid bounded by a pyramidal surface and a plane section cutting all the elements, and not passing through the vertex, is called apyramid. sight work If a plane cuts both nappes of a pyramidal surface, is it possible that it should cut every element of the surface? Suggestion. If a planeM cuts both nappes, then a plane through the vertex ∥M meets the surface in two elements which are∥M and hence are not cut byM.

86 SOLID GEOMETRY: BOOK IV classifica tion of pyramids 232. Faces. Edges. The lateral surface of a pyramid is composed oftriangular faces having a common vertex at the vertex of the pyramid, and having as bases the sides of the polygon forming the base. The sides common to two such triangles are the edges of the pyramid. Pyramids are classified, according to the shape of the base, as triangular, quadrangular, pentagonal, etc. 233. Tetrahedron. A pyramid having a triangular base has in all four faces, and is called atetrahedron. In this case every face is a triangle, and any one of them may be taken as the base. The altitude of a pyramid is the perpendicular distance from the vertex to the plane of the base. 234. Regular Pyramid. A regular pyramidis one whose base is a regular polygon such that the perpendicular from the vertex upon it meets it at the center. 235. Properties of a Regular Pyramid. (1) The edges are equal to each other. For they cut off equal distances from the foot of the perpendicular. (2) The faces are equal isosceles triangles. (3) The altitudes of the triangular faces are equal to each other. 236. Slant Height. The altitude of any one of the triangular faces of a regular pyramid is called itsslant height. sight work 1. If the slant height of a regular pyramid and the apothem of its base are given, how may its altitude be computed? If the altitude and apothem are given, how may the slant height be found? 2. Can the expression “slant height” be applied to a pyramid if the alti- tudes of its faces are not all equal? Discuss fully.

PYRAMIDS AND CONES 87 la teral area of a regular pyramid 237. Theorem I. The lateral area of a regular pyramid is equal to one half the product of its slant height and the perimeter of the base. Suggestion for Proof: Let l = PK represent the slant height, S the lateral area, andp the perimeter of the base. We are to show that S = 1 2l·p. Give all the steps. 238. Pyramids Cut by a Plane. The part of a pyramid between its base and a plane cutting all its edges is called afrustum if the cutting plane is parallel to the base, and atruncated pyramid if the cutting plane is not parallel to the base. 239. The Parts of a Frustum.The two parallel faces of a frustum are itsupper and lower bases, the perpendicular distance between the bases is itsalti- tude, the trapezoidal faces are itslateral surface, and the common altitude of these faces is theslant height of the frustum. 240. Corollary. The lateral area of the frustum of a regular pyramid is equal to half the sum of the perimeters of the bases multiplied by the slant height. Suggestion. Use § 62 and the figure in § 238.

88 SOLID GEOMETRY: BOOK IV section p arallel to the base of a pyramid 241. Theorem II. If a pyramid is cut by a plane parallel to the base, then (1) the edges and

altitude are divided proportionally; (2) the polygonal section is similar to the base; (3) the areas of this section and of the base are in the same ratio as the squares of their perpendicular distances from the vertex. Outline of Proof: LetA′B′C′D′E′ be a section∥ABCDE. (1) To prove thatPK′ PK = PA′ PA = PB′ PB , etc., pass another plane through P parallel to the base and then use § 103. (2)Toprovethat ABCDE∼A′B′C′D′E′, weshowthat ∠A = ∠A′, ∠B = ∠B′, etc., and alsoA′B′ AB = B′C′ BC , etc. (3) Calling the area of the sectionb′ and that of the baseb, we are to prove thatb′ b = PK′2 PK 2 , and for this we need to show that A′B′ AB = PA′ PA = PK′ PK . Give all the steps in detail.

PYRAMIDS AND CONES 89 242. Corollary. If two pyramids have equal altitudes and bases of equal areas lying in the same plane, the sections made by a plane parallel to the plane of the bases have equal areas. Suggestion. Ift andt′ are the areas of the sections andb andb′ those of the bases, show thatt b = QL′2 QL2 , t′ b′ = PK′2 PK 2 , and hencet b = t′ b′, from which it follows thatt =t′ if b =b′. sight work 1. Find the lateral area of a regular pyramid if its slant height is10 in. and the perimeter of its base is25 in. 2. What is the perimeter of the base of a regular pyramid if its lateral area is 72 sq. in. and its slant height is8 in.? 3. What is the slant height of a regular pyramid if its lateral area is 160 sq. in. and the perimeter of its base is20 in.? 4. What is the perimeter of the base of a regular pyramid if its lateral area is 250 sq. in. and its slant height is20 in.? 5. The area of the base of a pyramid is36 sq. in. and its altitude is8 in. What is the area of a section of this pyramid made by a plane parallel to the base and4 in. from the vertex? 6. The area of the base of a pyramid is64 sq. in. and its altitude is8 in. Find the distance from the vertex to a plane parallel to the base if the area of its section is16 sq. in.

90 SOLID GEOMETRY: BOOK IV 7. If the area of one section of a pyramid is double that of another, both sections being parallel to the base, find the ratio of their distances from the vertex. Find this ratio if the area of one section is three times that of the other; also if it is four times. inscribed and circumscribed prisms 243. Construction. A triangular pyramid is cut by a series of planes parallel to the base, including one through the vertex and also the one in which the base lies. Through the lines of intersec- tion of these planes with one of the faces, planes are constructed parallel to the opposite edge, thus forming a set of prisms all lying within the pyramid, asa′,b′,c′ in pyramid P′, or a set lying partly outside the pyramid, asa, b, c, d in pyramidP. The inner prisms thus constructed are called a set of inscribed prisms, and the outer prisms are calleda set of circumscribed prisms. For our purpose equidistant planes are used. This process may be repeated by doubling the number of planes drawn parallel to the base and thus doubling the number of inscribed or circum- scribed prisms. By continuing in this way either set of prisms may be made to coincide as nearly as we please with the pyramid. 244. Fundamental Assumption on the Volume of a Pyra- mid. We now assume that A pyramid has a definite volume which is less than the combined volume of any set of circumscribed prisms and greater than that of any set of inscribed prisms.

PYRAMIDS AND CONES 91 sight work 1. Show that in the above figure prismsb and a′ are equal in volume if PM = P′M′ and △sABC and A′B′C′ have equal areas. See §§ 188, 242. Also prove that volumec = volumeb′, etc. 2. Show that the sum of the

volumes of the circumscribed prisms exceeds the sum of the volumes of the inscribed prisms by the volume ofa. pyramids ha ving equal volumes 245. Theorem III. If two triangular pyramids have equal altitudes and bases of equal areas, their volumes are equal. Given the pyramidsP and P′ in which altitudesPM and P′M′ are equal, and the bases ABC and A′B′C′ have equal areas. To prove thatP and P′ have equal volumes. Proof: If P differs at all in volume fromP′, letP be the greater, and let the difference be some fixed numberK, so that Vol. P− Vol.P′ =K. (1) Construct a set of inscribed and circumscribed prisms as in § 243. Then a′ =b, b′ =c, c′ =d. Why?

92 SOLID GEOMETRY: BOOK IV Denote a +b +c +d byV and a′ +b′ +c′ byV′. Then V−V′ =a. (2) We have Vol.P <V , and Vol.P′ >V . § 244 □ Vol.P−Vol. P′ <V −V′. Why? Hence from (2) Vol. P−Vol.P′ <a . Now take the divisions onPM small enough to makea<K . § 243 Hence, Vol. P−Vol. P′ <K . (3) Thus (3) contradicts (1), and hence the supposition thatP and P′ differ in volume is impossible. Q. E. D. volume of a triangular pyramid 246. Theorem IV. The volume of a triangular pyramid is one third of the product of its base and altitude. Given the triangular pyramidE−ABC. Leth, b, andV be the numerical measures respectively of the altitudeEM, the area of the baseABC, and the volume of the pyramid. To prove that V = 1 3bh. Proof: On the baseABC construct a triangular prism with alti- tude h and lateral edgeEB. This prism may be cut into three pyramids by the plane sections through DEC and AEC, as shown in the figure on the right.

PYRAMIDS AND CONES 93 Thepyramids E−ABC andC−DEF haveequalvolumessincethey have equal bases,ABC andDEF (§ 175), and the same altitude,EM. Likewise volumeE−ACD = volumeE−CFD. But C−DEF and E−CFD are the same pyramid. Hence, Vol.E−ABC = Vol.C−DEF = Vol.E−ACD. That is, Vol.E−ABC is one third of the volume of the prism. But volume of prism =bh. § 198 Hence, V = volume of pyramid= 1 3bh. Q. E. D. volume of any pyramid 247. Theorem V. The volume of any pyramid is one third of the product of its base and altitude. Given the pyramidP−ABCDE. Let V, b, and h be the numerical measures respectively of the volume, base, and al- titude. To prove that V = 1 3bh. Proof: By means of diagonal planes such asPAC and PAD, di- vide the given pyramid into triangular pyramids. Call the bases of the triangular pyramidsb1,b2,b3, etc., and their common altitudeh. Then by § 246, the volumes of the triangular pyramids are1 3b1h, 1 3b2h, 1 3b3h. Hence the total volume is1 3 (b1 +b2 +b3)h = 1 3bh. Q. E. D. 248. Corollary 1. The volumes of any two pyramids having equal altitudes are proportional to the areas of their bases.

94 SOLID GEOMETRY: BOOK IV 249. Corollary 2. The volumes of any two pyramids having bases of equal areas are proportional to their altitudes. volume of a frustum of a pyramid 250. Theorem VI. The volume of a frustum of a pyra- mid is equal to the combined volumes of three pyramids whose common altitude is the same as that of the frustum, and the areas of whose bases are those of the upper and lower bases of the frustum and the mean proportional be- tween these areas. Given the frustumAC′ with area of lower baseb, area of up- per baseb′, and altitudeh. Then √ bb′ is the mean proportional between b and b′. To prove that the volume ofAC′ is V = 1 3hb + 1 3hb′ + 1 3h √ bb′ = 1 3h[b +b′ + √ bb′].

PYRAMIDS AND CONES 95 Proof: Let h′ be the altitudePK of the complete pyramidP− ABCDE. Then the altitude of the pyramidP−A′B′C′D′E′ ish′−h. Hence, b b′ = h′2 (h′−h)2 § 241 (3) from which √ b√ b′ = h′ h′−h. Clearing of fractions and solving forh′, h′ = h √ b√ b− √ b′. (1) Now V is the difference between the pyramids whose altitudes are h′ and h′−h. Hence, V = 1 3bh′− 1 3b′(h′−h), § 247 or, rearranging, V = 1 3b′h + 1 3h′(b−b′). (2) Substituting the value ofh′ from (1) in (2) we have V = 1 3b′h + 1 3 h √ b√ b− √ b′ (b−b′) = 1 3b′h + 1 3 h √ b√ b− √ b′ (√ b− √ b′)(√ b + √ b′) = 1 3b′h + 1 3hb + 1 3h √ bb′ = 1 3h[b′ +b + √ bb′]. Q. E. D. sight work 1. A flower bed is in the form of a regular right pyramid, with a square base 5 ft. on a side. The altitude is3 ft. Find the number of cubic feet of soil used in its construction. 2. The altitude of a certain pyramid is15 in. and its volume is380 cu. in. Find the area of its base. 3. The area of the base of a pyramid is48 sq. ft. and its volume160 cu. ft. Find its altitude.

96 SOLID GEOMETRY: BOOK IV 4. Find the locus of the vertices of pyramids having the same base and equal volumes. 5. Two monuments having bases of equal areas are pyramidal in shape, one being 15 ft. high and the other18 ft. Find the ratio of their volumes. 6. Two pyramids with equal altitudes have bases whose areas are7 sq. ft. and 13 sq. ft. Find the ratio of the volumes of the pyramids. 7. Find the volume of a frustum of a pyramid the areas of whose bases are 36 sq. in. and144 sq. in., and whose altitude is10 in. 8. The volume of a frustum of a pyramid is332 cu. in. and the areas of its bases are9 sq. in. and36 sq. in. Find its altitude. exercises invol ving numerical comput a tion 1. The base of a regular pyramid is a square whose sides are16 ft. Find the slant height of the pyramid if its altitude is6 ft. Also find the lateral area. 2. The lateral area of a regular hexagonal pyramid is72 sq. ft. and the slant height is12 ft. Find the perimeter of the base, the apothem of the base, and the altitude of the pyramid. 3. Find the lateral area of the frustum of a pyramid if the perimeters of its bases are27 and 54 in. respectively and its slant height is12 in. Does the result depend upon the number of sides of the frustum? 4. The bases of a frustum of a pyramid are squares whose sides are 8in.and 2in.respectively. Findthevolumeofthefrustumifitsaltitude is 6 in. 5. Findthevolumeofaregulartriangularpyramidthesidesofwhose base are 5 in. and whose altitude is6 in. Suggestion. The area of a regular triangle with sidea is a2 4 √ 3. 6. Findthevolumeofaregularhexagonalpyramidthesidesofwhose base are 8 in. and whose altitude is10 in. Suggestion. The area of a regular hexagon with sidea is 3a2 2 √ 3.

PYRAMIDS AND CONES 97 7. Two marble ornaments of equal altitudes are pyramidal in form. One has a square base2 in. on a side and the other a regular hexagonal base 1 in. on a side. Find the ratio of their volumes. 8. A pyramid has for its base a right triangle with hypotenuse10 and shortest side6. Find the volume of the pyramid if its altitude is9. 9. The slant height of a frustum of a regular pyramid is10 in. and the apothems of its bases are 8 in. and 6 in. respectively. Find its altitude. exercises invol ving algebraic comput a tion 1. A tent is to be made in the form of a right pyramid with a regular hexagonal base. If the altitude is fixed at12 ft., what must be the side of the base in order that the tent may inclose400 cu. ft. of space? 2. A pyramid with altitude8 in. and a base whose area is36 sq. in. is cut by a plane parallel to the base so that the area of the section is 18 sq. in. Find the distance from the base to the cutting plane. 3. A frustum of a pyramid is cut from a pyramid the

perimeter of whose base is60 in. and whose altitude is15 in. What is the altitude of the frustum if the perimeter of its upper base is20 in.? Does the result depend upon the number of sides of the pyramid? 4. Solve the preceding problem if the perimeter of the upper base of the frustum is onenth that of the lower base. 5. The area of the base of a pyramid is180 sq. in. and its altitude is 20 in. Cut from it a frustum the area of whose upper base is45 sq. in.; also one the area of whose upper base is onenth of180 sq. in. Do these results depend upon the number of sides of the pyramid? 6. If the altitude of a pyramid ish, how far from the base must a plane parallel to it be drawn so that the area of its cross-section shall be half that of the base of the pyramid? 7. In a regular right pyramid a plane parallel to the base cuts it so as to make a section whose area is one half that of the base. Find the ratio between the lateral area of the pyramid and that of the small pyramid cut off by the plane.

98 SOLID GEOMETRY: BOOK IV 8. Find the volume and the total surface of a regular tetrahedron whose edges are9 in. 9. Find the total surface and the volume of a regular hexagonal pyramid the sides of whose base are eacha and whose altitude isa. cones 251.ConicalSurface. Givenaclosedconvex plane curve and a fixed pointP not in its plane. If a line throughP moves so as always to touch the curve and is made to traverse it completely, it is said to generate aconvex conical surface. Themovinglineisthe generator ofthesurface, and in any of its positions it is an element of the surface. The fixed point is thevertex. 252. Nappes. A conical surface has two parts, one on each side of the vertex, which are called nappes. 253. Cone. The solid bounded by a conical surfaceandaplanesectioncuttingallitselements, and not passing through the vertex, is called a cone. 254. Base. Lateral Surface. Altitude. The plane part of the surface of a cone is itsbase and the curved part is thelateral surface. The altitude of a cone is the perpendicular distance from the vertex to the plane of the base. 255. Circular Cone. A cone which has a circular cross-section such that the perpendicular upon it from the vertex meets it at the center is called acircular cone. If the base is such a circle, the cone is then aright circular cone. Otherwise, it is anoblique circular cone. The axis of a circular cone is the line from the vertex through the center of a circular section. sight work If a plane cuts both nappes of a conical surface, show that it cannot cut all the elements of the surface. Seesuggestion on page 85.

PYRAMIDS AND CONES 99 256. Generating a Right Circular Cone.A right circular cone may be generated by rotating a right trianglePMB about one of its legs,PM, as an axis. The hypotenusePB generates the lateral surface, and the other leg,MB, generates the base. 257. Slant Height.The generator of the convex surface of a right circular cone in any of its positions is called theslant height of the cone. 258. Frustum of a Cone.The part of a cone included between the base and a plane section parallel to the base is called a frustum of a cone. The base of the cone and the parallel section are thebases of the frustum. 259. Slant Height of a Frustum. A frustum cut from a right circular cone has two circular bases. The seg- ments of all the elements of the cone in- terceptedbetweenthesebasesareequal and their common length is theslant height of the frustum. sight work 1. Compare the definitions of pyramidal and conical surfaces. 2. Compare the definitions of a pyramid and a cone, and of a frustum of each. 3. What kind of cone corresponds to a regular pyramid? 4. Can the expression

“slant height” be applied to any other cone than a right circular cone? Discuss this question for the frustum of a cone. 5. In a right circular cone, if any two of the three quantities, altitude, slant height, radius of the base, are given, show that the third may be found.

100 SOLID GEOMETRY: BOOK IV section of a cone through an element 260. Theorem VII. If a plane contains an element of a cone and meets its surface in any other point, then it contains another element also, and the section is a triangle. Let a plane contain the elementPB of the coneP−ABC, and also one other pointK or K′. To prove that this plane contains another elementPD, and that the section is a trianglePBD. Outline of Proof: (1) When the pointK is in the lateral surface. Draw the elementPD throughK, and let the plane determined by PB and PD meet the base inBD. Then△PBD is the section made by the plane containingPB and the pointK. (2) When the pointK′ is in the base. DrawBD through K′ and also the elementPD. Then the plane determined byBD and PD con- tains PB and K′ and cuts the cone in the△PBD. 261. Plane Tangent to a Cone. If a plane contains an element of a cone and no other point of the cone, the plane istangent to the cone, and the element is called theelement of contact.

PYRAMIDS AND CONES 101 section of a cone p arallel to the base 262. Theorem VIII. If the base of a cone is circular, every plane section parallel to the base is also circular. Given a cone with a circular baseAD. To prove that the∥ sectionEH is also circular. Proof: Draw the straight line fromP to the centerM of the base, and let it meet the sectionEH in the pointO. Let F and G be any two points on the perimeter of the sectionEH. Pass planes containingPM through the pointsF and G, and let them cut the base inMB and MC respectively. Now in△sPMB and PMC , OF∥MB and OG∥MC. § 92 □ OF MB = OP MP = OG MC. But MB =MC. □ OF =OG. Why? Hence, since F and G, any two points on the perimeter of this section, are equally distant from O, this shows that EH is a circle whose center isO. Q. E. D. 263. Corollary. If a cone has a circular base, the areas of two cross-sections parallel to it are in the same ratio as the squares of their perpendicular distances from the vertex and also as the squares of the distances of their centers from the vertex.

102 SOLID GEOMETRY: BOOK IV inscribed and circumscribed pyramids 264. The lateral surface of a regular pyramid inscribed in a right circular cone may be computed and is equal to one half the product of the slant height and the perimeter of the base (§ 237). If the number of faces of the inscribed pyramid is doubled, the lateral surface of the resulting pyramid may again be computed in terms of its slant height and the perimeter of its base. In a similar manner, the lateral surface of a regular pyramid circumscribed about a right cir- cular cone may be computed in terms of the slant height of the cone and the perimeter of the poly- gon circumscribed about the base. The number of faces may be doubled and the lateral surface again computed, and so on. Evidently either of these processes may be repeated indefinitely and the surfaces of the inscribed or circumscribed pyramids may be made to lie as close to the surface of the cone as we please. The circumscribed pyramids all have the same slant height as that of the cone, and in case of the inscribed pyramids, the slant height may be made to differ by as little as we please from that of the cone by

making the number of faces great enough. 265. Fundamental Assumption on the Lateral Area of a Right Circular Cone. We now assume that A right circular cone has a definite lateral area which can be approx- imated as nearly as we please by taking the lateral area of the successive inscribed or circumscribed pyramids. Since, by the theorem of § 237, the lateral area of any regular pyramid is half the product of the perimeter of its base and its slant height, it follows that this theorem holds for all pyramids used in approximating the lateral area of a right circular cone and that all practical measurements of such lateral areas are based on this theorem.

PYRAMIDS AND CONES 103 la teral area of a right circular cone The foregoing considerations constitute an informal proof of the following theorem: 266. Theorem IX. The area of the lateral surface of a right circular cone is equal to one half the product of its slant height and the circumference of its base. The argument used on page 102 shows that a theorem of this kind holds for every inscribed or circumscribed pyramid used in the approx- imation process, and hence this theorem for the cone is established for all purposes of practical measurement. 267. Corollary. If l is the slant height of a right circular cone andr is the radius of its base, the area of the lateral surface is S = 1 2· 2πr·l =πrl. Note. In the case of a cone which is not a right circular cone the slant height varies from point to point and the process of computation of § 266 fails. Finding the lateral surface of such a cone depends on methods first introduced in the calculus and is a much more difficult problem than those solved in elementary plane and solid geometry. sight work 1. If a pyramid is inscribed in a cone, is its lateral area greater or less than that of the cone? 2. Does the theorem of § 237 apply to an irregular pyramid or to an oblique pyramid? Discuss fully. 3. If a regular pyramid of1000 faces is inscribed in a right circular cone, would the theorem of § 237 apply to this pyramid? Would it be easy to detect experimentally the difference between this pyramid and the cone? Would the size of the cone affect the answer to this question? Discuss similar questions about a circumscribed pyramid of1000 faces.

104 SOLID GEOMETRY: BOOK IV 4. Find the lateral surface of a right circular cone whose altitude is7 in. and the radius of whose base is2 in. la teral area of a frustum 268. Theorem X. The lateral area of a frustum of a right circular cone is equal to one half the sum of the circumferences of the bases multiplied by the slant height. Given the frustumABCD, with slant heightl and radii of bases r and r′. Let S represent its lateral area. To prove thatS = 1 2 (2πr + 2πr′)l =πl(r +r′). Proof: Complete the cone, and letPC =l′. Then S = 1 2 î 2πr(l +l′)− 2πr′l′ó § 267 =πrl +πl′(r−r′). (1) But r r′ = l +l′ l′ , from whichl′ = r′l r−r′. (2) Substituting l′ from (2) in (1), S =πrl +πr′l =πl(r +r′). Q. E. D.

PYRAMIDS AND CONES 105 269. Corollary. The lateral area of a frustum of a right circular cone is equal to the circumference of a section midway between the bases multiplied by the slant height. Suggestion. In the formula of the theorem we have S =πl(r +r′) = 2π (r +r′) 2 l. Now show thatr +r′ 2 is the radius of the section midway between the two bases. § 46 sight work 1. If a triangle which is not a right triangle revolves about one of its sides, does it generate a cone? Show that the figure generated by revolving any triangle about its longest side may be divided into two

cones. 2. Find the lateral area of a right circular cone with radius of base6 in. and slant height 10 in. In this and the succeeding exercises, express the results in terms ofπ, e.g., 60π sq. in. 3. Find the lateral area of a right circular cone with radius of base8 in. and altitude 6 in. 4. Find the lateral area of a frustum of a right circular cone, the radii of whose bases are8 in. and 4 in. and whose slant height is6 in. 5. A right circular cone having a base with radius6 ft. and altitude8 ft. is cut by a plane parallel to its base and at a distance of4 ft. from the vertex. Find the radius of this section. Also find its area. exercises 1. Thelateralareaofthesurfaceofarightcircularconeis 120π sq.in., and its radius is4 in. Find its slant height. 2. A circular chimney 100 ft. high is in the form of a frustum of a right circular cone whose lower base is10 ft. in diameter and upper base 8 ft. Find the lateral surface.

106 SOLID GEOMETRY: BOOK IV 3. The lateral area of a frustum of a right circular cone is60π sq. in.; the radii of the two bases are6 in. and 4 in. Find the slant height of the frustum. 4. The lateral area of a right circular cone isS, and the slant height is l. Express the radius of the base and also the altitude in terms ofS and l. 5. If the radius of the base of a right circular cone isr, and the lateral area isS, express the slant height in terms ofr and S. 6. If the slant height of a right circular cone isl, and the lateral area is S, express the circumference of the base in terms ofl and S. volume of a cone 270. Consider inscribed and circumscribed pyramids similar to those used in § 264, except that they need not be regular since the cone is not now required to be a right circular cone. By repeatedly increasing the number of faces of either the inscribed or circumscribed pyramids, they may be made to approach coincidence with the cone as nearly as we please. 271. Fundamental Assumption on the Volume of a Cone. We now assume thata cone has a definite volume which can be approximated as nearly as we please by taking the volume of successive inscribed or circumscribed prisms. Recalling that the volume of a pyramid is equal to one third of the product of its base and altitude, we see that the above considerations constitute an informal proof of the following theorem:

PYRAMIDS AND CONES 107 272. Theorem XI. The volume of a cone is equal to one third of the product of its base and altitude. 273. Corollary. If the base of a cone is a circle with radiusr and if the altitude ish, then the volume of the cone is V = 1 3·πr2·h = 1 3πr2h. sight work 1. Find the volume of an oblique cone with altitude8 in. and a circular base whose radius is6 in. 2. Find the volume of a right circular cone with slant height10 in. and radius of base6 in. 3. The area of the base of a cone is50 sq. in. and its volume is600 cu. in. Is the altitude the same whether the cone is right or oblique? 4. Show that if two cones have bases of equal areas their volumes are proportional to their altitudes. volume of a frustum of a cone 274. Theorem XII. The volume of the frustum of a cone is equal to the combined volumes of three cones whose common altitude is the altitude of the frustum and whose bases are the upper and lower bases of the frustum and a mean proportional between these bases.

108 SOLID GEOMETRY: BOOK IV Suggestion. The proof is exactly like that of § 250, making use of § 272, instead of § 247. The result in symbols isV = 1 3h(b +b′ + √ bb′). 275. Corollary. The volume of a frustum of a right circular cone isV = 1 3πh(r2

+r′2 +rr′), whereh is the altitude andr and r′ are the radii of the bases. For b =πr2, b′ =πr′2. ∴ V = 1 3h(πr2 +πr′2 + √ πr2·πr′2) = 1 3πh(r2 +r′2 +rr′). Note. The theorem of § 272 holds for any cone whatever, whether right or oblique. However, when the base is not a circle we have no means in elementary mathematics forcomputing its area. Hence the volume of such a cone cannot be found at this stage even though § 272 does apply. Similarly § 274 applies to a frustum of any cone whatever, but we are able tocompute its volume by elementary methods only in case the bases are circles. sight work The radius of the base of a cone is5 in. and its altitude is10 in. Find the volume of a frustum formed by a plane parallel to the base and6 in. from it. Find the total surface of this frustum in case the cone is a right circular cone.

PYRAMIDS AND CONES 109 summar y of book iv 1. Define pyramidal surface and conical surface. In what respects do they differ? 2. Define pyramid, cone, regular pyramid, circular cone, and right circular cone. 3. For what kind of cones may the lateral area be found by means given in this Book? What is the corresponding kind of pyramid? 4. For what kind of frustum of a cone may the lateral area be found by means given in this Book? What is the corresponding kind of frus- tum of a pyramid? 5. What assumption about the volume of a pyramid is made in Book IV? In the proof of what theorems is this assumption used? 6. Beginning with the theorem of § 245 state in order the theorems which lead to the rule for finding the volume of any pyramid. 7. What assumption about the lateral area of a cone is made in this Book? Compare this assumption with the one in § 219. 8. What assumption is made about the volume of a cone? Compare this with the assumption in § 219. 9. What theorems on cylinders have no corresponding theorems for cones? 10. Show that a frustum of a cone becomes more and more nearly identical with a cylinder if the vertex of the cone is removed farther and farther from the base. 11. If a cone were regarded as a pyramid with a very large num- ber of very narrow lateral faces, what pages in this Book would be superfluous? 12. Describe the difficulty in finding the lateral area of an oblique cone or of a cone whose base is not circular. Does the same difficulty exist in finding the volume of such a cone?

110 SOLID GEOMETRY: BOOK IV exercises on book iv 1. If several planes are tangent to the same cone, find one point common to them all. 2. Find the locus of all lines which make a given angle with a given line at a given point in it. 3. Findthelocusofalllineswhichmakeagiven angle with a given plane at a given point. 4. If the middle points of four edges of a tetra- hedron, nothreeofwhichmeetatthesamevertex, are joined, prove that a parallelogram is formed. 5. Show how to pass a plane through a tetrahe- dron so that the section shall be a parallelogram. Suggestion. Pass a plane parallel to each of two opposite edges. See § 133. 6. A mound of earth of the shape shown in the figure has a rectangular base 16yards long and8yards wide. Its perpen- dicular height is5 yards, and the length on top is 8 yards. Find the number of cubic yards of earth in the mound. Suggestion.If from each end a pyramid with a base8 yd. by4 yd. is removed, the remaining part is a triangular prism. 7. Given a figure in general shape the same as the preceding, with a rectangular base of length24 ft. and width6 ft. Find its volume and lateral area if the dihedral angles around the base are each45◦. 8. Find the area and volume of the figure developed by an equilateral triangle with sides a if it is

revolved about one of its sides. Suggestion. The figure may be divided into two cones.

PYRAMIDS AND CONES 111 9. Find the volume and area of the figure formed by revolving an equilateral triangle with sides s about an altitude. 10. Find the area and volume of the figure devel- oped by revolving a square whose side is a about one of its diagonals. 11. Through one vertex of an equilateral triangle with sides adraw a linel perpendicularto the altitude upon the opposite side. Find the volume and area of the figure developed by revolving the triangle about the linel. Suggestion. The volume may be obtained by subtracting the volumes of two cones from the volume of a cylinder. 12. Through a vertex of a square with sidesa draw a linel perpendicular to the diagonal at that point. Find the area and volume of the figure developed by turning the square aroundl. Suggestion. Notice that two frustums and two cones are developed. 13. In a regular hexagon with sidesa draw a linel bisecting two opposite sides. Find the area and volume of the figure developed by turning the hexagon aboutl as an axis. 14. One angle of a right triangle is30◦. Find the ratios of the surfaces and also of the volumes of the solids developed by revolving this triangle around each of its three sides in succession. Suggestion. The sides of the△ are a, 2a, a √ 3. 15. If through any pointP in a diag- onal of a parallelopiped planesKN and RM are drawn parallel to two faces, show that the parallelopipedsDQ and LN thus formed have equal volumes.

112 SOLID GEOMETRY: BOOK IV Archimedes (287–212 B.C.) was without doubt the greatest mathe- matician of his time. He is known not through any extended treatise, like that of Euclid, but through a series of monographs on practically every mathematical subject then known, including physics, mechan- ics, astronomy, and many phases of geometry. His work on the circle, cone, cylinder, and sphere is reflected in our treatment of surfaces and volumes at the present day.

BOOK V THE SPHERE 276. Sphere. Center. A sphere is a solid bounded by a surface all points of which are equally distant from a point within called the center. 277. Diameter. Radius. A line-segment connecting two points on the surface of a sphere and passing through its center is adiameter. A segment joining the center to a point on the surface is aradius. 278. Notation for a Sphere. A sphere may be designated by a single let- ter at the center or by two letters giving a radius. Thus the sphereC means one whose cen- ter isC, and the sphereCA is one whose cen- ter isC and whose radius isCA. 279. Generating a Sphere. The surface of a sphere may be generated by revolving a semicircle about its diameter as an axis. Thus the surface of the sphereCA may be generated by revolving the semicircleMNA about the diameterMN. 280. Corollary 1. If two spheres have equal radii they may be made to coincide and hence are equal. 281. Corollary 2. All radii of the same sphere or of equal spheres are equal. 282. Corollary 3. All diameters of the same sphere or of equal spheres are equal.

114 SOLID GEOMETRY: BOOK V plane section of a sphere 283. Theorem I. A section of a spherical surface made by a plane is a circle. Given a sphere with centerC cut by a planeM. To prove that the points common to the surface of the sphere and the plane form a circle. Proof: From the centerC drawCA perpendicular to the planeM. LetB andD be any two points common to the plane and the surface of the

sphere. Complete the figure and proveAB =AD. Hence any two points common to the surface of the sphere and the plane M are equidistant from the pointA. That is, these points lie on a circle. How must this proof be modified in case the planeM passes through the center of the sphere? 284. Corollary 1. Through three points on a spherical surface there is one and only one circle. Suggestion. How many planes pass through these points? 285. Corollary 2. A radius of a sphere through the center of a circle on its surface is perpendicular to the plane of the circle; and conversely, a radius of a sphere perpendicular to the plane of a circle on its surface passes through the center of the circle.

THE SPHERE 115 proper ties of circles on a sphere 286. Figures on a Sphere. Any figure drawn on the surface of a sphere is said tolie on the sphere. 287. Axis and Poles. The line per- pendicular to the plane of a circle at its center is called theaxis of the circle. The points in which the axis of a circle on a sphere meets the surface of the sphere are called thepoles of the circle. 288. Great and Small Circles on a Sphere.If the plane of a circle on a sphere passes through the center of the sphere, it is called a great circleof the sphere, and if not, it is called asmall circle. Thus, in the figure,AB is a small circle,PP′ is its axis, andP and P′ are its poles. The circle passing throughP and P′ and containing the centerC is a great circle. 289. Inside and Outside of a Sphere.A point is inside, outside, or on a sphere according as its distance from the center is less than, greater than, or equal to the radius of the sphere. 290. Corollary 1. The center of a great circle on a sphere is the center of the sphere. 291. Corollary 2. All great circles on a sphere are equal and any two such circles bisect each other. 292. Corollary 3. The axis of any circle on a sphere passes through the center of the sphere. 293. Corollary 4. Through two given points on a sphere there is one and only one great circle unless these points are at opposite ends of a diameter. 294. Corollary 5. Circles on a sphere formed by parallel planes have the same axis and the same poles.

116 SOLID GEOMETRY: BOOK V dist ances from points on a circle to its pole 295. Distance on a Sphere. The spherical distance, or simply the distance between two points on a sphere is the distance measured between these points along the minor arc of the great circle through them. 296. Theorem II. All points of a circle on a sphere are equidistant from either pole of the circle. Given P a pole of the circle whose center isA, withB, C, and D any points on this circle. To prove that the great circle arcsPB, PC, andPD are equal. Suggestion. Prove chordPB = chord PC = chord PD, and hence arc PB = arc PC = arc PD. § 48 297. Polar Distance of a Circle. The spherical distance from the points of a circle on a sphere to its nearest pole is called thepolar distance of the circle. 298. A Quadrant. One fourth of a great circle is aquadrant. 299. Corollary 1. The polar distance of a great circle is a quadrant. 300. Corollary 2. If a point on a sphere is at a quadrant’s distance from each of two points not at the extremities of the same diameter, it is the pole of the great circle through these points.

THE SPHERE 117 use of a spherical blackboard 301. Corollary 3. Polar distances of equal circles on a sphere are equal. 302. Corollary 4. Straight line- segments joining points of a circle on a sphere, to one of its poles are equal. It follows from the

preceding theorem and corollaries that, if a spherical blackboard is at hand, circles may be constructed on it by means of crayon and string the same as on a plane blackboard. Likewise, curve-legged compasses may be used. sight work 1. If two points are at the extremities of the same diameter of a sphere, how many great circles can be passed through these points? 2. What great circles on the earth's surface pass through both poles? Where are the poles of these circles located? 3. If A and B are at opposite ends of a diameter, can a small circle be passed through them? 4. If two circles on a sphere have the same poles, prove that their planes are parallel. See § 294. 5. What is the locus of all points on a sphere at a quadrant's distance from a given point? 6. What is the locus of all points on a sphere at any fixed distance from a given point on the sphere? What is the greatest such distance possible? Discuss fully. 7. If two planes cutting a sphere are parallel, what can be said of the centers of the circles thus formed? What can be said of the poles of these circles? Are these statements true of three or more such circles? 8. Find the locus of the centers of a set of circles on a sphere formed by a set of parallel planes cutting it.

118 SOLID GEOMETRY: BOOK V 9. AB is a fixed diameter of a sphere. A plane containingAB is made to revolve about it as an axis. Find the locus of the poles of the great circles on the sphere made by this revolving plane. How are the pointsA and B related to this locus? circles equidist ant from the center 303. Theorem III. If the planes of two circles on a sphere are equidistant from the center, the circles are equal; and conversely, if two circles on a sphere are equal, their planes are equidistant from the center. Suggestion. In the first figure show that (1) ifCA = CA′, thenAB = A′B′, and (2) ifAB =A′B′, thenCA =CA′. circles unequall y dist ant from the center 304. Theorem IV. If the planes of two circles on a sphere are unequally distant from the center, the circles are unequal, the one nearer the center being the greater; and conversely, if two circles on a sphere are unequal, the plane of the greater circle is nearer the center.

THE SPHERE 119 Suggestion.In Fig. 2 above, show (1) that ifCA<CA ′ thenAB >A′B′; and (2) ifAB >A′B′ then CA<CA ′. 305. Plane Tangent to a Sphere.A plane which meets a sphere in only one point istangent to the sphere. Two spheres aretangent to each otherif they are both tangent to the same plane at the same point. A line is tangent to a sphereif it contains one and only one point of the sphere. plane t angent to a sphere 306. Theorem V. A plane tangent to a sphere is per- pendicular to the radius from the point of tangency; and conversely, a plane perpendicular to a radius at its extrem- ity is tangent to the sphere. Given a sphereC with planeM tangent to it atA. To prove that the radiusCA is perpendicular to the planeM. Proof: (1) It is necessary to prove thatCA is perpendicular to every line inM through A. (Why?) Draw any such line asAB. The planeBAC cuts the sphere in a circle. ThenAB is tangent to this circle, and hence perpendicular toAC. Since CA is⊥ to every line inM through A, it is⊥ to M.

120 SOLID GEOMETRY: BOOK V (2) To prove the converse, note thatCA is the shortest distance from C to the planeM. § 82 Hence, every point ofM except A is exterior to the sphere. That is,M is a tangent plane. (§ 305.) Q. E. D. 307. Sphere Inscribed in a Polyhedron. A sphere is said to be inscribed in a polyhedronif every

face of the polyhedron is tangent to the sphere. The polyhedron is then said to becircumscribed about the sphere. 308. Polyhedron Inscribed in a Sphere. A polyhedron is inscribed in a sphereif all its vertices lie in the surface of the sphere. The sphere is thencircumscribed about the polyhedron. sphere inscribed in a tetrahedron 309. Problem To inscribe a sphere in a given tetrahe- dron. Given a tetrahedronD−ABC. To construct a sphere tangent to each of its four faces. Construction: Construct planes bisecting the dihedral angles whose edges are AB, BC, and CA. These three planes meet in a pointP, which is equally distant from the four faces of the tetrahedron. With the pointP as a center and a radiusPE equal to the distance fromP to one of the faces construct a sphere. This sphere is inscribed in the tetrahedron. Proof: Every point in the planePAB is equidistant from the faces ABD and ABC. § 123

THE SPHERE 121 Similarly every point in the plane PBC is equidistant from the planesABC andDBC, andeverypointintheplane PAC isequidistant from the planesABC and DAC. Hence, the pointP, common to all three planes, is equidistant from each of the four faces of the tetrahedron. each plane is tangent to the sphere, and the sphere is inscribed in the tetrahedron. (§ 307.) Q. E. F. sight work Discuss the problem of finding a point equally distant from three planes two of which are parallel. How many such points are there? sphere circumscribed about a tetrahedron 310. Problem To find a point equally distant from the four vertices of a tetrahedron. Given the tetrahedronP−ABC. To find a pointO equidistant fromP, A, B, C. Construction: At D, the middle point ofBC, construct a plane perpendicular toBC. This plane containsE, the center of the circle circumscribed about △PBC andalso F, thecenterofthecirclecircumscribedabout △ABC. Why?

122 SOLID GEOMETRY: BOOK V In the planeFDE drawEG⊥ED and FH⊥FD. Then EG and FH cannot be parallel and hence meet in some point O. ThenO is the point required. Proof: Since BC is⊥ to the planeFDE, it follows that each of the planesPBC and ABC is⊥ to planeFDE. § 117 Hence OE is⊥ plane PBC and OF is⊥ plane ABC. § 114 ThenO is equidistant fromP,B, andC, and also fromA,B, andC. § 81 Hence OA =OB =OC =OP. Q. E. F. 311. Corollary 1. A sphere may be passed through any four points, not all of which lie in the same plane. 312. Corollary 2. A sphere may be circumscribed about any tetrahedron. to find the diameter of a sphere 313. Problem To find the diameter of a given material sphere. With any pointP of the sphere as a pole, construct a circle, and on this circle select any three pointsA, B, C. Using a pair of compasses, measure the straight line-segmentsAB, BC, CA, and construct the triangleA′B′C′ equal toABC. Let B′D′ be the radius of the circle circumscribed aboutA′B′C′. If PP′ is the axis of the circleABC on the sphere andBD the radius of this circle, thenBD =B′D′.

THE SPHERE 123 Measure PB by means of the compasses. Then PBP′ is a right triangle, withBD perpendicular to its hy- potenuse PP′. PB andBD being known, we may nowcomputePD from the right trianglePBD, and then computePP′ from the similar trianglesPBD and PP′B, for the latter using the relationPD : PB = PB : PP′ or PD×PP′ =PB 2 . The segment PP′ may also be found by geometric construction; namely, by drawing a triangle equal toP′BP. Show how to do this whenBP and BD are

known. intersecting spherical surf aces 314. Theorem VI. The intersection of two spherical surfaces is a circle. Proof: Two intersecting spherical surfaces may be developed by two intersecting circles rotating about the line connecting their centers C and C′. LetA and B be the two points common to the two circles. Then BA is⊥CC′. (CC′ is the⊥ bisector ofBA.) § 28 As the figure rotates about the lineCC′, AB remains⊥ CC′ and therefore all its positions lie in a plane. § 79 Also DB and DA remain fixed in length. Therefore the pointsA and B trace out a circle. Q. E. D.

124 SOLID GEOMETRY: BOOK V sight work 1. Find the locus of the centers of all spheres tangent to a given plane at a given point. 2. Find the locus of the centers of all spheres of given radius tangent to a given line at a given point. 3. Find the locus of the centers of all spheres of given radius tangent to a fixed plane. 4. Find the locus of the centers of all spheres of given radius tangent to a fixed line. 5. Find the locus of the centers of all spheres tangent to two given inter- secting planes. exercises 1. The four lines perpendicular to the faces of a tetrahedron at their circumcenters meet in a point. 2. The six planes perpendicular to the edges of a tetrahedron at their middle points all meet in a point. 3. The planes bisecting the six dihedral angles of a tetrahedron all meet in a point. 4. Show that a sphere may be inscribed in a cube. 5. Show that a sphere may be circumscribed about a cube. 6. Can a sphere be circumscribed about a rectangular parallelopiped which is not a cube? Can a sphere be inscribed in it? Prove. 7. If a planeM is tangent to a sphere at a pointA, show that the plane of every great circle of the sphere throughA is perpendicular to M. 8. Show that the line of centers of two intersecting spheres meets the spherical surfaces in the poles of their common circle.

THE SPHERE 125 9. Show that two spheres are tangent if they meet on their line of centers. Distinguish two cases. State and prove the converse of this proposition. 10. If a sphere is tangent to a given planeM at a given pointA, how many additional points on the sphere are required to determine it? Suggestion. Suppose one pointP given. Pass a plane throughP⊥ to plane M at A. Is there only one such plane? Discuss fully. 11. Describe the set of all lines in space whose distances from the center of a sphere are all equal to the radius of the sphere. 12. Describe the set of all planes whose distances from the center of a sphere are all equal to the radius of the sphere. 13. Describe the set of all spheres of given radius which are tangent to a given sphere of greater radius. spherical angles 315. Spherical Angles.Two planes through the center of a sphere cut the surface of the sphere in two great circles which intersect in two points and form fourspherical anglesabout each of these points. Any two of these angles areadjacent or vertical as in the case of angles formed by straight lines. A spherical angle is measured by the angle between the tangents to its sides (arcs) at their common point. Only angles formed by great circles are considered in this book.

126 SOLID GEOMETRY: BOOK V measuring a spherical angle 316. Theorem VII. A spherical angle is measured by an arc of the great circle whose pole is the vertex of the angle and which is intercepted by the sides of the angle. Suggestion. Show that ¯AB measures the dihedral angle formed by the planes PAC and PBC and that ∠BCA = ∠TPR . 317. Corollary 1. The sum of the consecutive spherical angles about a point is four right angles. 318. Corollary 2. A spherical angle is equal to the dihedral angle

formed by the planes of its arcs. spherical pol ygons 319. Polygons and Triangles on a Sphere.The portion of a spherical surface contained within a polyhedral angle whose vertex is at the center of a sphere is called aspherical polygon. It follows that for every spherical polygon there is a corresponding polyhedral angle at the center of the sphere made by drawing radii to the vertices of the polygon. The face angles of the polyhedral angle correspond to the sides of the spherical polygon and the dihedral angles of the polyhedral angle to the angles of the spherical polygon. Since a plane through the center of a sphere intersects the surface

THE SPHERE 127 in a great circle, it follows that the sides of a spherical polygon arearcs of great circles. Since a plane may be passed through the vertex of a polyhedral angle such that the poly- hedral angle lies entirely on one side of it, it follows that a spherical polygonlies within one hemisphere. A spherical polygon of three sides is aspherical triangle. 320. Relation between the Parts of a Spherical Polygon and the Corresponding Polyhedral Angle. (1) The face angles of the polyhedral angle are measured by the arcs forming the sides of the spherical polygon. Why? (2) The dihedral angles of the polyhedral angle are equal in measure to the angles of the spherical polygon. § 318 sight work Why is the vertex of a polyhedral angle which is used in defining a spherical polygon placed at the center of the sphere? corollaries on spherical pol ygons The following propositions are now obvious corollaries of the pre- ceding definitions and discussions. 321. The sum of two sides of a spherical triangle is greater than the third side. This is a direct consequence of § 151. 322. The sum of the sides of a spherical polygon is less than360◦. This is a direct consequence of § 152. 323. Each side of a spherical polygon is less than180◦.

128 SOLID GEOMETRY: BOOK V sight work 1. What is the angle between a meridian on the earth’s surface and the equator? 2. If two meridians are drawn meeting the equator10◦ apart, what is the angle between these meridians? 3. If two meridians meet the equator90◦ apart, what can be said of the three angles of the spherical triangle thus formed? 4. If in two spherical triangles the three sides of one are equal respectively to the three sides of the other, what can be said of the face angles of the corresponding trihedral angles? 5. If in two spherical triangles the three angles of one are equal respec- tively to the three angles of the other, what parts are equal in the corre- sponding trihedral angles? 6. If in two spherical polygons the angles of one are equal respectively to the angles of the other, what parts are equal in the corresponding polyhedral angles? 7. If in two spherical polygons the sides of one are equal respectively to the sides of the other, what parts of the corresponding polyhedral angles are equal? 8. Prove that if the sides of two spherical triangles are equal, then the angles of the triangles are equal. Suggestion. Use § 144. shor test spherical dist ance between two points 324. Theorem VIII. The shortest distance on a sphere between two of its points is measured along the minor arc of a great circle passing through these points.

THE SPHERE 129 Proof: Let A and B be any two points on a sphere,AB the minor arc of a great circle through them, andADCB any other curve on the sphere connecting A and B. Let D and C be any two points on the curveADCB taken in order fromA to B. Draw the great circle arcsAD, AC, DC, andCB. Then by § 321 ¯AC + ¯CB

> ¯AB and ¯AD + ¯DC >¯AC. Hence, ¯AD + ¯DC + ¯CB > ¯AB. If in like manner we subdivideAD, DC, CB, and continue this process, weobtainasuccessionofpaths, eachlongerthanthepreceding, and thus we get closer and closer to the length of the curveADCB. Hence, the curveADCB must be greater than¯AB. Q. E. D. 325. Symmetrical Spherical Triangles. Two spherical triangles are symmetrical if the sides and angles of one are equal respectively to the sides and angles of the other, but arranged in the opposite order. Compare § 146. sight work 1. Is it possible to move the spherical triangleABC in the above figure so as to make it coincide with triangleA′B′C′? 2. If in two plane trianglesABC andA′B′C′ the corresponding parts are equal, is it always possible to make the triangles coincide? Is this always possible without inverting one of the triangles? Discuss the difference in this respect between plane and spherical triangles. trihedral angles and spherical triangles 326. Theorem IX. Two equal or symmetrical trihedral angles with vertices at the center of a sphere intercept equal or symmetrical triangles, respectively, on the sphere.

130 SOLID GEOMETRY: BOOK V Suggestions for Proof. In case the trihedral angles are equal they can be made to coincide, whereby the spherical triangles will also be made to coincide. The triangles are therefore equal. In case the trihedral angles are symmetrical it follows directly from §§ 146, 325 that the spherical triangles are symmetrical. 327. Theorem X. If the radii drawn from the vertices of a spherical triangle are extended, they meet the sphere in the vertices of a triangle symmetrical to the given triangle. The proof is left for the student. 328. Theorem XI. Two triangles on the same sphere, or on equal spheres, are equal or symmetrical if three sides of one are equal respectively to three sides of the other. This is a direct corollary of §§ 145, 149. See the figure under § 326. triangles equal or symmetrical 329. Theorem XII. Two triangles on the same sphere, or on equal spheres, are equal or symmetrical if two sides and the included angle of one are equal respectively to two sides and the included angle of the other.

THE SPHERE 131 Proof: This is a direct corollary of the theorems §§ 142, 150. 330. Vertical Spherical Angles.Spherical angles arevertical if they have the same vertex and if the sides of one are extensions of the sides of the other. 331. Right Spherical Triangle. A right spherical triangle has one of its angles a right angle. 332. Isosceles Spherical Triangle. An isosceles spherical tri- angle has two equal sides. sight work 1. Show that vertical spherical angles are equal. 2. What kind of spherical triangle is formed by two meridians on the earth’s surface and the arc of the equator which they intercept? 3. Show that every spherical triangle formed as described in Example 2 contains more than two right angles. 4. Show how to form a spherical triangle as in Example 2 in which each angle shall be a right angle. 5. Show how to form a spherical triangle as in Example 2 in which one angle shall be 179◦ and each of the other two 90◦.

132 SOLID GEOMETRY: BOOK V equal angles opposite equal sides 333. Theorem XIII. The angles oppo- site the equal sides of an isosceles spheri- cal triangle are equal. Suggestion. Let AC and BC be the equal sides. Draw ¯CD to the middle point of¯AB. Then use § 328. 334. Corollary. If two isosceles spherical triangles are symmet- rical, they are equal, and conversely. exercises 1. Compare fully the

theorems on the equality of plane triangles and of trihedral angles. Is there any theorem in either case for which there is no corresponding theorem in the other? 2. Compare in the same manner the theorems on the equality of plane triangles and of spherical triangles. 3. Compare in the same manner the theorems on the equality of trihedral angles and of spherical triangles. 4. If two face angles of a trihedral angle are equal, the opposite dihedral angles are equal. 5. If two face angles of a trihedral angle are equal, it is equal to its symmetrical trihedral angle. Suggestion. Compare the sides and then the angles of the corresponding spherical triangles. Use §§ 333, 329. 6. Show how to find a pole of the circle through three given points on a sphere. Suggestion. Let the given points beA,B,C. By § 296 a pole of the circle is equidistant fromA, B, andC. ConnectA andB by an arc of a great circle and construct another arc of a great circle bisecting ¯AB perpendicularly.

THE SPHERE 133 Similarly construct a perpendicular bisector of ¯BC. The points in which these two arcs meet will be the poles of the circle throughA, B, andC. polar triangles 335. Definition. If with the vertices of a given spherical triangle as poles arcs of great cir- cles are constructed, another spherical triangle is formed which is called the polar triangle of the first. Thusinthefigure, Aisapoleofthearc B′C′,B is a pole of the arcC′A′, andC is a pole of the arcA′B′. HenceA′B′C′ is the polar triangle of triangleABC. 336. Polar Triangle. How Selected. If with the verticesA, B, C of a spherical triangle as poles three complete great circles are constructed, each of these circles meets each of the others in two points, thus forming eight spherical triangles, as shown in the figure, namely,A′B′C′, A′B′F, B′C′D, C′A′E, A′EF, B′DF, C′DE, andDEF. There is one and only one of these, namely,A′B′C′, such thatA and A′ are on the same side of circleB′C′,B andB′ on the same side of circleA′C′, and C and C′ on the same side of circleA′B′. Thetriangle A′B′C′ asthusdescribed is the polar triangleof ABC. sight work In the above figure the parts of the great circles which are supposed to be on the front side of the figure are given in solid lines while the parts on the back side are dotted. Study the figure with care and state which triangles are entirely on the front side, which are entirely on the back side of the sphere, and which are partly on the front side and partly on the back side of the sphere.

134 SOLID GEOMETRY: BOOK V triangles polar to each other 337. Theorem XIV. If A′B′C′ is the polar triangle of ABC, thenABC is the polar triangle ofA′B′C′. Given△A′B′C′, the polar triangle ofABC. To prove thatABC is the polar triangle ofA′B′C′. Proof: (1)A′ is at a quadrant's distance fromB because B is the pole of ˘A′C′. A′ is also at a quadrant's distance fromC because C is the pole of ˘A′B′. Hence, A′ is the pole of¯BC. § 300 Similarly,B′ is the pole of¯AC and C′ the pole of ¯AB. (2) To show thatA andA′ lie on the same side of the circleBC, we note that sinceA is the pole of the circleB′C′ and A lies on the same side of this circle withA′, thenA and A′ are at less than a quadrant's distance. Hence, it follows that ifA′ is at a quadrant's distance from BC, A and A′ must be on the same side ofBC. In like manner we show thatB and B′ lie on the same side ofAC, and C and C′ on the same side ofAB. 338. Corresponding Parts of Polar Triangles. If ABC and A′B′C′ are polar triangles, and ifA is a pole of ˘B′C′, then ∠A and ˘B′C′ are said to becorresponding parts.

THE SPHERE 135 measure of p ar ts in polar triangles 339. Theorem XV. The sum of the measures of an angle of a spherical triangle and the corresponding arc of its polar triangle is180◦. Given the polar triangles ABC and A′B′C′. Denote the measures in degrees of the angles byA, B, C, A′, B′, C′, and of the corresponding sides bya′, b′, c′, a, b, c. To prove that A +a′ = 180◦ A′ +a = 180◦ B +b′ = 180◦ B′ +b = 180◦ C +c′ = 180◦ C′ +c = 180◦ Suggestions for Proof: Extend (if necessary) arcsA′B′ andA′C′ till they meet the great circleBC in pointsD andE, respectively. Then arc DE is the measure of∠A′. Also ⁻BE = 90◦, and ⁻DC = 90◦. (Why?) But ⁻BE + ⁻DC = ⁻BC + ⁻ED =a +A′ = 180◦. 340. Corollary 1. If in two spherical triangles an angle of one is equal to an angle of the other, then the corresponding sides of their polar triangles are equal; and conversely, if in two spherical triangles a side of one is equal to a side of the other, then the corresponding angles of their polar triangles are equal. 341. Corollary 2. If two spherical triangles are equal or sym- metrical, their polar triangles are equal or symmetrical.

136 SOLID GEOMETRY: BOOK V sum of angles of a spherical triangle 342. Theorem XVI. The sum of the angles of a spher- ical triangle is less than six right angles and greater than two right angles. Given the spherical triangleABC. To prove that(1) ∠A + ∠B + ∠C <6 rt. angles. (2) ∠A + ∠B + ∠C >2 rt. angles. Proof: Construct the polar triangleA′B′C′, with sidesa′, b′, c′. (1) By § 339∠A + ∠B + ∠C +a′ +b′ +c′ = 6 rt. ∠s. Since a′ +b′ +c′ > 0, □ ∠ A + ∠B + ∠C <6 rt. ∠s. (2) Using § 322, show that∠A + ∠B + ∠C >2 rt. ∠s. 343. Corollary 1. State and prove the theorem on trihedral angles which corresponds to Theorem XVI. 344. Corollary 2. The sum of the angles of a spherical polygon of n sides is greater than2(n− 2) right angles and less than2n right angles. Proof: Divide the polygon inton− 2 triangles. Then by § 342 the sum of the angles> 2(n− 2) rt. ∠s. Since each angle is less than two right angles, it follows that the sum is less than2n right angles. 345. Corollary 3. State and prove the theorem on polyhedral angles which corresponds to Corollary 2.

THE SPHERE 137 equal and symmetrical spherical triangles 346. Theorem XVII. Two triangles on the same sphere, or on equal spheres, are equal or symmetrical if two angles and the included side of one are equal respectively to two angles and the included side of the other. Proof. By §§ 340 and 329 the polar triangles of the given trian- gles are equal or symmetrical. Hence, by § 341, the given triangles themselves are equal or symmetrical. 347. Corollary. State and prove the theorem on trihedral angles which corresponds to Theorem XVII. 348. Theorem XVIII. Two triangles on the same sphere, or on equal spheres, are equal or symmetrical if the angles of one are equal respectively to the angles of the other. Proof: By §§ 340 and 328 the polar triangles of the given trian- gles are equal or symmetrical. Hence, by § 341, the given triangles themselves are equal or symmetrical. 349. Corollary. State and prove the theorem on trihedral angles which corresponds to Theorem XVIII. Is there a theorem on plane triangles corresponding to that of § 348?

138 SOLID GEOMETRY: BOOK V construction of spherical triangles 350. Problem On a given sphere to construct a spherical triangle when its sides are given. Solution. Let O be the given sphere, anda, b, c the sides of the required triangle, and letAA′ be any diameter of the sphere. With A as a pole, construct circlesDBE and FCG, whose

polar distances from A are c and b respectively. With B as a pole, construct a circleHCK, whose polar distance from B is a. Then construct the three great circle arcs,AB, BC, CA. ABC is the required triangle. Why? sight work 1. What restrictions if any is it necessary to impose upon the three given sides of the triangle in § 350? (See §§ 321, 322.) 2. In plane geometry two equal triangles may be constructed upon the samebaseandonthesamesideofit. Isacorrespondingconstructionpossible on the sphere? 3. If in the above construction each of two sides of the required triangle is very great, that is, nearly a semicircle, show from the construction that the third side must be very small.

THE SPHERE 139 construction of spherical triangles 351. Problem To construct a spherical triangle when its three angles are given. Solution. Let the three given angles beA, B, C, and leta′, b′, c′ be arcs such thata′ + ∠A = 180◦,b′ + ∠B = 180◦, andc′ + ∠C = 180◦. Then the triangle whose arcs area′, b′, c′ will be the polar triangle of the required triangle. This latter triangleA′B′C′ may be constructed by the method of § 350. Then construct the polar triangle ofA′B′C′, which will be the required triangle. Give reasons in full for each step. 352. Problem To construct a trihedral angle when its face angles are given. Solution. Construct the corresponding spherical triangle by the method of § 350. Give the construction in full and prove each step. 353. Problem To construct a trihedral angle when its dihedral angles are given. Solution. Construct the corresponding spherical triangle by the method of § 351. Give reasons in full for each step. sight work 1. If two spherical triangles having angles respectively equal are con- structed on the same sphere, how are these triangles related? Prove. 2. If two trihedral angles with face angles respectively equal are con- structed as in § 352, how are they related? Prove. 3. If two trihedral angles with dihedral angles respectively equal are con- structed as in § 353, how are the trihedral angles related? Prove. 4. What restrictions if any must be placed upon the given anglesA, B, C in § 351? Compare Example 1, page 138.

140 SOLID GEOMETRY: BOOK V 5. What restrictions if any are needed in Examples 2 and 3? exercises 1. If two angles of a spherical triangle are un- equal, the sides opposite them are unequal, the greater side being opposite the greater angle. Suggestion. In the triangle ABC let ∠B be greater than ∠A. Draw ¯BD, making ∠ABD = ∠A. Then, ¯AD = ¯BD, and ¯BD + ¯DC > ¯BC. Hence, show that¯AC >¯BC. 2. State and prove a theorem on trihedral angles corresponding to the preceding. 3. If the sides of a spherical triangle are 60◦, 80◦, 120◦, find the angles of its polar triangle. 4. If the angles of a spherical triangle are72◦, 104◦, 88◦, find the sides of the polar triangle. 5. If a triangle is isosceles, prove that its polar triangle is isosceles. 6. If each side of a spherical triangle is a quadrant, describe its polar triangle. 7. Is it possible to construct a spherical triangle whose sides are50◦, 60◦, 120◦? 8. Is it possible to construct a spherical triangle whose sides are 100◦, 120◦, 150◦? Suggestion. Consider the polar triangle of such triangle. 9. Consider the questions on trihedral angles corresponding to the two preceding. 10. If the sides of a spherical triangle are75◦, 95◦, and 115◦ respec- tively, find the angles of each triangle formed by the polar construction.

THE SPHERE 141 11. If it is given that a spherical triangle is equilateral, can we infer from the theorems thus far proved that its polar triangle is equilateral? symmetrical triangles are equal in area 354. Equal Areas Defined. Two spherical polygons are said to have equal areas if they can be made to coincide, or if they can be divided into parts which can be made to coincide in pairs. Compare this with the definition of equal areas in Plane Geometry. 355. Theorem XIX. Two symmetrical spherical trian- gles are equal in area. Proof: Let ABC be one of the given triangles. Extend the radii AO,BO,CO to meet the sphere inA1,B1,C1, thus forming a triangle symmetrical to△ABC. § 327 LetP be a pole of the circle throughA,B,C. ExtendPO to meet the sphere inP1. Draw ¯PA, ¯PB, ¯PC, and ˙P1A1, ˙P1B1, and ˘P1C1. Suppose thatP lies within△ABC. Thesphericaltriangles PAB,PBC ,PCA,P1A1B1,P1B1C1,P1C1A1 are all isosceles. § 296 Now prove (1)△PAB =△P1A1B1; (2) △PBC =△P1B1C1; (3)△PCA =△P1C1A1. ∴ area△ABC = area△A1B1C1.

142 SOLID GEOMETRY: BOOK V But△A1B1C1 is equal to any other triangle which is symmetrical to△ABC. ∴ Two symmetrical spherical triangles are equal in area. birect angular spherical triangles 356. A birectangular spherical triangleis one having two right angles, as△PAB in the figure. If the third angle of a birectangular triangle is1◦, the triangle con- tains one of720 equal parts of the surface of the sphere. 357.SphericalDegree. Theareaofabirect- angular triangle having a vertex angle of one de- gree is called aspherical degreeand is used as a unit of measure of areas of spherical polygons. In a similar manner we define a spherical minute and aspherical second. 358. The Lune.A lune is a figure formed by two great semicircles having the same end-points. The angle between these semicircles is theangle of the lune. 359. Corollary 1. The area of a birectangular spherical triangle in terms of spherical degrees is equal to the number of degrees in the third angle of the triangle. 360. Corollary 2. The area of a lune in terms of spherical degrees is twice the number of degrees in the angle of the lune. 361. Spherical Excess.The number of degrees by which the sum of the angles of a spherical triangle exceeds180◦ is called thespherical excess of the triangle. The spherical excess of a spherical polygon is the sum of its angles less (n− 2)180◦, wheren is the number of sides of the polygon.

THE SPHERE 143 exercise Show that the spherical excess of a spherical polygon is less than four right angles. area of a spherical triangle 362. Theorem XX. The area of a spherical triangle in terms of spherical degrees is equal to its spherical excess.1 Proof: We are to show that area△ABC = ∠A + ∠B + ∠C− 180◦. By § 360, the areas of the lunesACDB,CAEB,BCFA in spherical degrees are as follows: △ABC +△BCD =ACDB = 2∠A. △ABC +△BAE =CAEB = 2∠C. △ABC +△CFA =BCFA = 2∠B. Hence, adding, 3△ABC +△sBCD, BAE, CFA = 2(∠A + ∠B + ∠C). Now△sBCD and AEF are symmetrical and equal in area. Hence, 2△ABC +△sABC, AEF, BAE, CFA = 2(∠A + ∠B + ∠C). But△sABC, AEF, BAE, CFA together constitute a hemisphere or 360 spherical degrees. 1This theorem was discovered by Cavalieri. See Frontispiece.

144 SOLID GEOMETRY: BOOK V Hence, 2△ABC + 360◦ = 2(∠A + ∠B + ∠C). Solving,△ABC = ∠A + ∠B + ∠C− 180◦. Q. E. D. Note.—The spherical degree differs fundamentally from the units of measure hitherto used. This unit is a certain fraction of the surface of the sphere and hence its actual size depends upon the size of the

sphere. area of a spherical pol ygon 363. Theorem XXI. The area of a spherical polygon in terms of spherical degrees is equal to its spherical excess. Proof: Join one vertex of the polygon to each non-adjacent vertex, thus formingn− 2 spherical triangles. Now since the sum of the spher- ical excesses of these triangles is the spherical excess of the polygon, the conclusion is evident. sight work 1. What is the area in spherical degrees of a birectangular triangle if one of its angles is54◦? if one angle is24◦? if one angle is36◦? 2. What is the spherical excess of a triangle the sum of whose angles is 285◦? 3. What is the spherical excess of a triangle whose angles are75◦, 110◦, and 150◦? 4. What is the spherical excess of a spherical hexagon the sum of whose angles is 1060◦? 5. What is the spherical excess of a spherical pentagon whose angles are 90◦, 120◦, 75◦, 108◦, and 165◦? 6. What is the angle of a lune whose area is160 spherical degrees? 7. The spherical excess of a triangle is120◦. Two of its angles are110◦ and 108◦ respectively. Find the third angle.

THE SPHERE 145 8. Between what limits is the sum of the angles of a spherical polygon of eight sides? 9. If the sum of the angles of a spherical polygon is11 right angles, what is known about the number of its sides? 10. If the sum of the angles of a spherical polygon is14 right angles, what is known about the number of its sides? 11. The sides of a spherical triangle are85◦, 95◦, 110◦. Find the area of each of the eight triangles formed by the polar construction from this triangle. 12. The area of a spherical triangle is74 spherical degrees. One angle is 105◦. Of the other two angles one is twice the other. Find all the angles of the triangle. la teral area of a frustum of a cone 364. Theorem XXII. The lateral area of a frustum of a right circular cone is equal to the altitude of the frustum multiplied by the circumference of a circle whose radius is the perpendicular distance from a point in the axis of the frustum to the middle point of an element.

146 SOLID GEOMETRY: BOOK V Given a frustum with an elementAA′ whose middle point is B, CC′ the axis of the frustum,BD⊥CC′ and EB⊥AA′. To prove that the lateral area is equal to2π·EB·CC′. Proof: By § 269, the lateral area is2π·BD·AA′. Hence, we must show that EB·CC′ =BD·AA′. To do this, draw A′F ⊥ AC and show that △AFA′∼△EDB. 365. Corollary. The lateral area of a right circular cone is equal to its altitude times the length of a circle whose radius is the perpendicular from a point in the axis to the middle point of an element. Note. The above theorem and corollary are needed in deducing the area of the surface of a sphere. We have already computed the area of a spherical triangle in terms ofspherical degrees, but we now wish to derive the area of the spherical surface in terms ofplane units of measure. circumscribed and inscribed cones and frustums 366. About a circle circumscribe a polygon as follows: Construct two diametersAB and CD at right angles to each other and divide each quad- rant into aneven number of parts by points, asE, F,G. At each alternate division point, beginning with the first pointE, draw a tangent. There results a regular polygon with two ver- tices on each of the diametersAB and CD ex- tended. Ifnowwe constructanotherpolygon inthe same mannerbydividing each quadrant intotwice as many arcs, it will likewise have two vertices on each of the diametersAB and CD extended. The vertices onAB will lie betweenM and N. This construction may be repeated at pleasure, thus obtaining a sequence of polygons each lying closer to the circle than the preceding.

THE SPHERE 147 Now inscribe a polygon similar to the first one of the set just circum- scribed, by joining the pointsC andF,F andA, and so on, repeating this process to form a sequence of inscribed polygons. If now the whole figure is made to revolve aboutAB as an axis, the circle generates a sphere, and the circumscribed and inscribed poly- gons generate sets of circumscribed and inscribed cones and frustums of cones. 367. Fundamental Assumption on the Area of a Sphere.We assume that A sphere has a definite area which is less than the surface of any circumscribed figure and greater than the surface of any inscribed convex figure. The student should note that while the statement just preceding is ob- viously true, it is not capable of proof by pure deduction. It is therefore necessarily in the nature of anassumption or axiom. formula for the area of a sphere 368. Theorem XXIII. The area of a sphere whose radius isr is 4πr2. Proof: (a) Denote byr the radius of the circle which generates the sphere. That is, in the figure, r =OG =OE =OH. By § 364 the lateral areas of the outside frustums whose axes areOL andOK are 2πr· OL and 2πr·OK, andby§365thelateralareas of the cones whose axes areLN and KM are respectively 2πr·LN and 2πr·KM. Hence, the total surface of the whole circumscribed figure is 2πr·MK + 2πr·KO + 2πr·OL + 2πr·LN, or 2πr(MK +KO +OL +LN) = 2πr·MN. If now a polygon of twice the number of sides is constructed, as described in § 366, we obtain another circumscribed figure whose area

148 SOLID GEOMETRY: BOOK V is 2πr·M′N′ where M′ and N′ are the two vertices on the lineAB extended, but lying betweenM andN, so that2πr·M′N′ < 2πr·MN. As this process goes on, the total surface generated decreases and may be made to approximateas nearly as we pleaseto 2πr×AB = 2πr× 2r = 4πr2. Therefore, the area of the sphere cannot be greater than4πr2. (b) Using the same figure, letOG′ be the apothem of the inscribed polygon. Then, as under (a), we find that the area of the figure developed by revolving the inscribed polygon aboutAB is 2π·AB·OG′ = 4πr·OG′. By continuing to double the number of sides,OG′ may be made to approach as nearly as we please toOG =r,OG′ being always less than OG. Hence, the total surface developed increases and approaches as nearly as we please to 4πr×OG = 4πr2. Therefore the area of the sphere cannot be less than4πr2. Since from (a) and (b) the area of the surface of the sphere can be neither greater than4πr2 nor less than4πr2, it follows that it is exactly equal to 4πr2. Q. E. D. sight work Use the valueπ = 3.1416. 1. Find the surface of a sphere whose radius is one inch. 2. Find the surface of a sphere whose radius is10 inches. 3. What is the relation between the surface of a sphere and the area of a circle of the same radius? 4. Find the area in square inches of a spherical degree on a sphere of radius 15 inches.

THE SPHERE 149 5. What fraction of the surface of a sphere is occupied by a birectangular triangle having one angle of65◦? by a lune with an angle of84◦? 6. Find the area in square inches of a birectangular spherical triangle with one angle equal to35◦, if the triangle is on a sphere of radius10 inches. 7. Find the area in square inches of a lune whose angle is42◦, if the lune is on a sphere of radius20 inches. exercises in arithmetic comput a tion Example. Find the area in square inches of a spherical triangle whose angles are80◦, 85◦, and97◦ if the radius of the sphere is6 inches. Solution. The spherical excess of the triangle is 80◦ + 85◦ + 97◦− 180◦ = 262◦− 180◦ = 82◦. The total area of the sphere is4π× 62 = 452.3904 sq. in., and one

spherical degree is 1 720 of 452.3904 = .62832 sq. in. Hence 82 spherical degrees is82×.62832 = 51.52224 sq. in. 1. Find the area in square inches of a spherical triangle whose angles are 70◦, 80◦, 90◦ if the radius of the sphere is10 inches. Suggestion. First find the spherical excess of the triangle. 2. Find the area in square inches of a spherical triangle whose angles are 95◦, 110◦, 75◦ if the radius of the sphere is15 inches. 3. Find the area of a spherical polygon whose angles are110◦, 120◦, 130◦, and 95◦ if the radius of the sphere is8 inches. 4. Find the area of a spherical polygon whose angles are130◦, 140◦, 110◦, 100◦, 160◦, and 150◦ if the radius of the sphere is20 inches.

150 SOLID GEOMETRY: BOOK V exercises in algebraic comput a tion 1. The area of a sphere iss square inches. Find the radius of the sphere in terms ofs. 2. The spherical excess of a spherical triangle ise and the radius of the sphere isr. Find the area of the triangle in terms ofe and r. 3. The spherical excess of a spherical polygon ise and its area is a square inches. Find the radius of the sphere in terms ofe and a. 4. Theareainsquareinchesofasphericaltriangleis aandtheradius of the sphere isr. Find the spherical excess in terms ofa and r. the volume of a sphere by inspection 369. The formula for the volume of a sphere may be inferred directly from the accompanying figure. The sphere is covered with a network of spherical quadrilaterals. If these are taken small enough, they may be regarded as approximatelyplane sur- faces. On this supposition we have a set of pyramids with a common al- titude r and the sum of their bases approximately equal to the area of the sphere. Hence, their combined volume is1 3r× (area of sphere) or 1 3r· 4πr2. That is, the volume is4 3πr3. Itisclearthat, bymakingthesequadrilateralssufficientlysmalleach one may be made to approach as nearly as we please to a plane surface. Hence, each pyramidal figure with vertex atO is made to approach the form of a true pyramid as nearly as we please. 370. Fundamental Assumption on the Volume and Surface of a Sphere. In the formal proof on page 151, we use polyhedrons circumscribed about the sphere and we assume that The surface and the volume of a sphere may be approximated as nearly as we please by taking the surfaces and the volumes of a series of circumscribed polyhedrons all of whose faces are made to decrease indefinitely.

THE SPHERE 151 Note. This assumption, when taken together with § 367, implies that the surface obtained by taking the surfaces of a series of polyhedrons is the same as the surface obtained by taking the surfaces of the circumscribed figures described in § 366. This is of course obvious at a glance, though a formal deductive proof is very difficult. exercise Using § 369 find the volume of a sphere whose radius is10 in.; also one whose diameter is6 ft. formula for the volume of a sphere 371. Theorem XXIV. The volume of a sphere whose radius isr is 4 3πr3. Proof: Consider a sphere of radiusr with a cube circumscribed about it. Connect its vertices with the center of the sphere. Then the cube is divided into six pyramids, each having an altituder. Since the volume of each pyramid equals the area of its base times one third its altitude, it follows that the sum of the volumes of these pyramids is equal to the sum of their bases times one third their common altitude. Consider now a polyhedron obtained from this cube by cutting off its eight vertices by planes tangent to the sphere. This new polyhe- dron is circumscribed about the sphere and approaches it more nearly than the cube. Continuing in this mannere we may obtain a series eSee Transcriber’s Notes.

152 SOLID GEOMETRY: BOOK V of circumscribed polyhedrons approaching the sphere as nearly as we please. By joining the vertices of each polyhedron to the center of the sphere we obtain a set of pyramids, each with altituder. Suppose the total surfaces of these successive polyhedrons ares1, s2,s3,s4,... . Then their volumes arer 3s1, r 3s2, r 3s3,... . But, by § 370, s1, s2, s3, ... approach the surface of the sphere, or4πr2, as nearly as we please. Hence the successive volumes approachr 3× 4πr2 = 4 3πr3 as nearly as we please. That is, Surface of a sphere= 4πr2, and Volume of a sphere= 4 3πr3. sight work 1. Givenasphereofradius 6inches, isthereanyupperlimittothevolume of its circumscribed polyhedrons? That is, can polyhedrons be circumscribed having a volume as large as we please? 2. With the same sphere is there any lower limit to the volume of its circumscribed polyhedrons? 3. Show that the areas of two spheres are in the same ratio as the squares of their radii or of their diameters. 4. Show that the volumes of two spheres are in the same ratio as the cubes of their radii or of their diameters. 5. Express the area of a sphere whose radius is8 in. in terms ofπ. 6. Express the volume of a sphere whose radius is10 ft. in terms ofπ. 7. The surface of a polyhedron circumscribed about a sphere of radius 4 in. is 420 sq. in. Find its volume. exercises in arithmetic comput a tion 1. The volume of a polyhedron circumscribed about a sphere of ra- dius 3.5 in. is 450 cu. in. Find its surface. 2. If the area of a sphere is227 sq. ft., find its radius. 3. If the volume of a sphere is335 cu. in., find its radius. 4. If the volumes of two spheres are27 cu. in. and729 cu. in., find the ratio of their radii.

THE SPHERE 153 exercises in algebraic comput a tion 1. The volume of a sphere isv cubic inches. Find its radius in terms of v and π. 2. The area of the surface of a sphere is s square inches. Find its volume in terms ofs and π. 3. The area of the surface of a sphere is numerically equal to its volume. Find the radius of the sphere if an inch is the unit of measure. 4. The difference between the volume of a cube and that of its in- scribed sphere isv. Find the radius of the sphere in terms ofv and π. spherical segments, cones, sectors 372. Zone, Segment. That part of a spherical surface included between two parallel planes cutting it is called azone. The perpendic- ular distance between the planes is thealtitude of the zone and of the corresponding segment. The portion of a sphere included between two parallel planes cutting it is called aspherical segment, and the two circular sections made by the parallel planes are itsbases. If one of the cutting planes is tangent to the sphere, then the spher- ical segment and the corresponding zone are said to have butone base. The altitude in this case is the perpendicular distance from the base to the tangent plane. 373. Spherical Cone, Sector.If one nappe of a convex conical surface has its vertex at the center of a sphere, the portion of the sphere cut out by this surface is called aspherical cone. If two spherical cones have the same axis, one lying within the other, the figure formed by their two lateral surfaces, together with the part of the sphere intercepted between them, is called aspher- ical sector. If the two cones are right circular cones, they in- tercept circles on the sphere, and the zone thus in- cluded is called thebase of the spherical sector.

154 SOLID GEOMETRY: BOOK V If the accompanying figure be revolved aboutLM as an axis, then any arc, asGD orMD, generates a zone, the former with two bases, the latter with one. The figureMDF ofFDGE generates a spherical segment, the

former with one base, the latter with two. The figure CAL or CBL generates a spherical cone,CL being the common axis. The figureCBA generates a spherical sector and arcAB generates the zone which is the base of the spherical sector. area of a zone. volume of a spherical cone 374. Area of a Zone. An argument precisely like that of § 368 shows that the area of a zone is s = 2πrh, where h is the altitude of the zone. That is, instead ofAB, the diameter in case of the sphere, we should have the sum of the altitudes of the frustums circumscribed about the zone equal toh, the altitude of the zone. 375. Volume of a Spherical Cone and a Spherical Sector. An argument precisely like that of § 371 shows that the volume of a spherical cone is v = r 3·s, wheres is the area of the zone of one base which is cut out of the sphere by the cone. Hence, ifh is the altitude of this zone, we have v = r 3· 2πrh = 2π 3 r2h. In like manner thevolume of a spherical sectoris v = 2π 3 r2h, whereh is the altitude of the zone of two bases which is cut out by the sector.

THE SPHERE 155 exercises 1. The radius of a sphere is6 in. and the altitude of a zone is5 in. Find the area of the sphere and of the zone. 2. The area of a zone is36π sq. ft. and its altitude4 ft. Find the radius of the sphere. 3. On a sphere of radius8 in. a spherical cone cuts out a zone whose altitude is 2 in. Find the volume of the cone. 4. Find the volume of the spherical sector cut out of a sphere of radius 9 in., if the altitude of the zone is2 in. volume of a spherical segment 376. Problem To find the volume of a spherical segment. Solution. Let r be the radius of the sphere, andr1 andr2 the radii of the bases of the segment, h the altitude of the segment, and let the segment be generated by revolving the figureACDB about AO as an axis. We have Vol. generated by ODB = 2π 3 r2h. § 375 Vol. generated by OAB = π 3r2 1(h +d). (Why?) Vol. generated by OCD = π 3r2 2d. (Why?) Hence, v = 2π 3 r2h + π 3r2 1(h +d)− π 3r2 2d = π 3 î 2r2h +r2 1h +d(r2 1−r2 2) ó . (1) From r2 =r2 2 +d2 and r2 =r2 1 + (h +d)2 we obtain d = r2 2−r2 1−h2 2h . (2)

156 SOLID GEOMETRY: BOOK V Substituting this value ofd in r2 =r2 2 +d2, we get r2 = r4 2 +r4 1 +h4− 2r2 1r2 2 + 2h2r2 2 + 2h2r2 1 4h2 . (3) Substituting (2) and (3) in (1) and reducing, we have v = πh 2 (r2 1 +r2 2) + π 6h3. Q. E. F. 377. Corollary. The volume of a spherical segment of one base is v =πh2 Ç r− h 3 å . Suggestion. In this case r2 1 = 0 and r2 2 = r2− (r−h)2 = 2rh−h2. Substituting in the formula above, we have the result. summar y of book v 1. Define sphere, diameter, radius. 2. Collect the theorems of Book V involving plane sections of the sphere. 3. Define axis and pole of a circle, great circle, polar distance. 4. Define tangent plane to a sphere, inscribed and circumscribed polyhedrons. 5. Arrange in parallel columns the corresponding theorems on tri- hedral angles and spherical triangles which are proved without the use of polar triangles. 6. Collect the definitions on polar triangles. 7. Collect the theorems on polar triangles. 8. Continue the lists begun in Example 5, adding the theorems proved by means of polar triangles. 9. Make a list of the definitions involving polyhedral angles and spherical polygons.

THE SPHERE 157 10. Collect the theorems involving polyhedral angles and spherical polygons. 11. Collect the theorems on the areas of spherical triangles and poly- gons. 12. Give the definitions and assumptions pertaining to the area and volume of the sphere. 13. State all the theorems pertaining to the area and volume of

the sphere. 14. Give the definitions and theorems pertaining to spherical figures, such as zones, cones, sectors, segments. 15. Collect all the mensuration formulas in this Book. 16. Collect all the mensuration formulas of solid geometry. 17. Describe some of the most important applications in this Book. Return to this question after studying the following sets of problems. problems on book v 1. What part of the earth's surface lies in the torrid zone? What part in the temperate zones? What part in the frigid zones? The parallels 23 1 2 ◦ north and south of the equator are the boundaries of the torrid zone, and the parallels66 1 2 ◦ north and south are the boundaries of the frigid zones. 2. Find to four places of decimals the area of a sphere circumscribed about a cube whose edge is6. No square root is to be approximated in the process, and the value ofπ is taken as3.1416. 3. Can the volume of the sphere in the preceding exercise be ap- proximated without finding a square root? Find the volume. 4. Find the area of a sphere circumscribed about a rectangular par- allelopiped whose sides area, b, andc. 5. Find the volume of the sphere in the preceding example.

158 SOLID GEOMETRY: BOOK V 6. A fixed sphere with centerO has its center on another sphere with centerO′. Show that the area of the part ofO′ which lies withinO is equal to the area of a great circle of the sphereO, provided the radius of the sphereO is not greater than the diameter ofO′. Suggestion. Let the figure represent a cross section through the centers of the two spheres. Connect O with A and B. Then OA2 = OB× OD. But OD is the altitude of the zone ofO′ whichlieswithin O, andOB isthediameterofthe sphere O′. Hence, the area of the zone isπOB× OD =πOA2. 7. Given a solid sphere of radius 12 inches. A cylindrical hole is bored through it so that the axis of the cylinder passes through the center of the sphere. What area of the sphere is removed if the diameter of the hole is4 inches? 8. Find the volume removed from the sphere by the process de- scribed in the preceding exercise. 9. A cylindrical post6 in. in diameter is surmounted by a part of a sphere 10 in. in diameter, as shown in the figure. Find the surface and the volume of the part of the sphere used. 10. A cylindrical post5 ft. long and4 in. in di- ameter is surmounted by a part of a sphere9 in. in diameter as shown in the figure. Find the volume of the whole post including the spherical part. 11. Find the volume of a spherical shell one inch thick if its outer diameter is8 inches. 12. Compare the volumes and areas of a sphere and the circumscribed cylinder. 13. In a sphere of radiusr a cylinder is inscribed whose altitude is equal to its diameter. Compare its volume and area with those of the sphere. 14. Find the diagonal of a cube in terms of its side, and also a side in terms of half the diagonal.

THE SPHERE 159 15. Express the volume of a cube inscribed in a sphere in terms of the radius of the sphere. 16. Three spheres each of radiusr are placed on a plane so that each is tangent to the other two. A fourth sphere of radiusr is placed on top of them. Find the distance from the plane to the top of the upper sphere. 17. Find the vertical distance from the floortothetopofatriangularpileofspher- ical cannon balls, each of radius5 inches, if there are3 layers in the pile. 18. Solve a problem like the preceding if there are16 layers in the pile, each shot of radiusr. formulas developed in books iii, iv, v 1. If V is the volume of a rectangular parallelopiped whose dimen- sions area, b, c, then V =abc. 2. IfV is the volume,b the area of the base, andh the altitude

of a prism or cylinder, then V =hb. 3. If S is the lateral surface,p the perimeter of a right section of a prism or cylinder, ande the lateral edge or element, then S =pe. 4. If V is the volume,b the area of base, andh the altitude of a pyramid or cone, then V = 1 3bh. 5. IfS is the lateral area,p the perimeter of the base, andl the slant height of a regular pyramid or cone, then S = 1 2pl.

160 SOLID GEOMETRY: BOOK V 6. IfV is the volume,b the lower base,b′ the upper base, andh the altitude of a frustum of pyramid or cone, then V = 1 3h(b +b′ + √ bb′). 7. If S is the area of the surface,V the volume, andr the radius of a sphere, then S = 4πr2 and V = 4 3πr3. 8. If a is the spherical excess in degrees of a spherical polygon,S the area in square units, andr the radius of a sphere, then S = a 720· 4πr2. 9. If S is the area,h the altitude, andr the radius of a lune, then S = 2πrh. 10. If V is the volume of spherical cone,h the altitude, andr the radius of the sphere, then V = 2π 3 r2h. 11. If V is the volume of a spherical segment,r the radius of the sphere, r1 and r2 the radii of the bases, and h the altitude of the segment, then V = πh 2 (r2 1 +r2 2) + π 6h3. supplement ar y exercises in comput a tion 1. Find the volume of a rectangular parallelopiped if (1) a = 6 b = 8 c = 7 2 3 (2) a = 7 8 b = 9 10 c = 3 (3) a = √ 3 b = √ 6 c = 4 (4) a = 3 √ 5 b = √ 15 c = 3 2. Find volume of a prism, or cylinder, if (1) b = 12 h = 16 (2) b = √ 3 h = √ 6 (3) b = 4 √ 2 h = √ 10 (4) b = 3 1 4 h = 4 √ 3

THE SPHERE 161 3. Find volume of a pyramid or cone if (1) b = 46 h = 12 (2) b = 47 1 2 h = 10 (3) b = 34π h = 8 (4) b = 42π h = 6 √ 2 4. Find the lateral surface of a regular pyramid or cone if (1) p = 8 l = 7 (2) p = 3 √ 7 l = √ 21 (3) p = √ 7 l = √ 14 (4) p = 3 √ 2 l = 4 √ 6 5. Find the volume of a frustum of a pyramid or cone if (1) b = 8 b′ = 6 h = 4 (2) b = 2 √ 2 b′ = √ 3 h = 6 (3) b = 16π b′ = 9π h = 3 (4) b = 3 √ 2 b′ = 6 √ 3 h = 8 6. Find the volume or surface of a sphere if r = 4, r = 8, r = 3 √ 2, r = 7 √ 3, r = 12 √ 6 7. Find the area of a spherical polygon if (1) a = 84◦ r = 12 (2) a = 112◦ 30′ r = 8 (3) a = 49◦ 25′ 17″ r = 14 miscellaneous review exercises A. LOCUS PROBLEMS 1. Find the locus of all points in space equally distant from two parallel lines and also from two parallel planes. Discuss. 2. Find the locus of all points in space equally distant from each of two intersecting straight lines and also from two intersecting planes. Discuss. 3. What is the locus of all points at a perpendicular distance of2feet from a given line and lying in a plane parallel to the line? Discuss. 4. Find the locus of all points equidistant from two given pointsA and B, and also equidistant from two planesM and N. Discuss. 5. What is the locus of all points on the floor of a room which are equally distant from two diagonally opposite corners of the room, one in the floor and one in the ceiling? 6. Find the locus of all points on a sphere where it is met by line- segments of equal length drawn from a fixed pointP outside the sphere. Discuss.

162 SOLID GEOMETRY: BOOK V 7. Find the locus of all points which are at the same fixed distance from each of two intersecting planes,M andN, and also equally distant from two planes,P and Q. 8. Given a planeM and a pointP not inM. Find the locus of a point which divides in a given ratio each segment connectingP with a point inM: (a) if the segments are divided internally; (b) if they are divided externally. 9. A segment AB of fixed length is free to move so that its end- points lie in two fixed parallel planes. Find the locus of a pointC on AB if AC is of fixed length. 10. Given two fixed points in space, through each of which passes a system of straight lines. If each

line of one system is perpendicular to a line of the other system, find the locus of the intersection points. B. PROBLEMS IN NUMERICAL COMPUTATION 1. A pedestal for a monument is in the shape of a frustum of a regular hexagonal pyramid, the radius of the upper base being4 ft., that of the lower base6 ft., and the altitude of the frustum8 ft. Find its volume, slant height, and lateral surface. 2. The area of the lower base of a frustum of a pyramid is42 sq. ft., its altitude 8 ft., and volume 200 cu. ft. Find the area of the upper base. 3. The area of the base of a pyramid is480 sq. ft. and its altitude 30 ft. Find the volume of the frustum remaining after a pyramid with altitude 10 ft. has been cut off by a plane parallel to the base. 4. The area of the base of a pyramid is250 sq. in. If a plane section of the pyramid parallel to the base and at a distance of5 in. from it has an area of175 sq. in., find the altitude of the pyramid. 5. The figure below represents a solid whose base is a rectangle50 ft. long and 40 ft. wide. Its height is12 ft. and its top a rectangle20 ft. by 10 ft. Find its volume.

THE SPHERE 163 6. A frustum of a right circular cone has an altitude one half that of the cone. If its slant height is8 ft. and lateral area64π sq. ft., find the diameters of its bases. 7. If the area of the base of a cone is16π sq. in. and its altitude 6 in., find the distance from the vertex to a plane, parallel to the base, which cuts out a section of area9π sq. in. C. PROBLEMS IN ALGEBRAIC COMPUTATION 1. Find the total area and the volume of a regular tetrahedron each of whose edges ise. 2. If the numerical values of the volume and of the total area of a regular tetrahedron are equal, what is the length of its edge? 3. Find the length of an edge of a regular tetrahedron if its volume is numerically equal to the square of the edge. 4. Cut a pyramid of altitudeh by means of a plane parallel to the base so that the perimeter of the section shall be one third that of the base.f 5. If the altitude of a pyramid ish, how far from the base must a plane parallel to it be drawn so that the area of its cross section shall be one third that of the base of the pyramid? 6. In a regular right pyramid a plane parallel to the base cuts it so as to make a section whose area is one third that of the base. Find the ratio between the lateral area of the pyramid and that of the small pyramid cut off by the plane. fThis is the declarative form of Question 5. —Trans.

164 SOLID GEOMETRY: BOOK V 7. If the diameter of a right circular cylinder is equal to its altitude, determine the diameter so that the total area of the cylinder shall be equal numerically to its volume. 8. Cut a right circular cone of altitudeh by a plane parallel to the base so that the area of the section shall be one third that of the base. Find the distance from the vertex to the plane. 9. Show that the lateral area of the small cone cut off in Example 8 is one third the lateral area of the original cone. 10. In a right circular cone, with altitude h, andr the radius of its base, a cylinder is inscribed as shown in the figure. Find the radiusOF of the cylinder if the area of the ring bounded by the circles OF and OA is equal to the lateral area of the small cone cut off at the top. D. PROBLEMS IN CONSTRUCTION 1. Construct a plane tangent to a given sphere and parallel to a given plane. How many such planes are there? Suggestion. From the center of the sphere pass a line perpendicular to the given plane and draw tangent planes at the points where this line meets the surface of the sphere. See §§ 306, 129. 2. Construct a plane tangent to a given sphere and perpendicular to a given line. How many such planes are there? 3. How many planes may be tangent to a sphere at a point on the sphere? How many lines? Show how to

construct them. 4. Throughagivenpointexteriortoaspherеconstructalinetangent to the sphere. 5. How many lines tangent to a sphere can be constructed from a point outside the sphere? 6. Through a given point exterior to a sphere construct a plane tangent to the sphere. How many planes can be passed through a given exterior point tangent to the sphere?

THE SPHERE 165 7. How many planes tangent to a sphere can be passed through two given pointsA and B outside a sphere? Discuss fully if the lineAB (1) meets the sphere in two points; (2) is tangent to the sphere; (3) does not meet the sphere. Show how to make the constructions. 8. Is it possible to construct a spherical triangle each of whose an- gles is a right angle? Show how to construct one. In a trirectangular spherical triangle what is the length of each side in terms of degrees? 9. Construct a spherical triangle such that its polar triangle is iden- tical with the given triangle. 10. Construct a spherical triangle whose sides are70◦, 80◦, and110◦, respectively, and find the sides of each of the eight triangles formed by its polar construction. E. THEOREMS TO BE PROVED 1. Prove that planes perpendicular to the faces of a trihedral angle and bisecting its face angles meet in a line. 2. Prove that if in two tetrahedrons three faces of one are equal respectively to three faces of the other and similarly placed about a vertex, the tetrahedrons are equal. 3. Prove that two tetrahedrons are equal if two faces and the in- cluded dihedral angle are equal and similarly placed. 4. In any regular tetrahedron, an altitude equals three times the perpendicular from its foot to any face; or an altitude equals the sum of the perpendiculars to the faces from any point within the tetrahedron. 5. In a cylinder of revolution the diameter of whose base equals the altitude, the volume equals one third the product of the total surface by the radius of the base. 6. If two intersecting planes are each tangent to a cylinder, show that their line of intersection is parallel to an element of the cylinder and also parallel to the plane containing the two elements of contact. 7. The volume of a frustum of a right circular cone equals the sum of a cylinder and a cone of the same altitude as the frustum, and with

166 SOLID GEOMETRY: BOOK V radii which are respectively the half sum and the half difference of the radii of the frustum. 8. Of circles on a sphere whose planes pass through a given point within the sphere, the smallest is that circle whose plane is perpendic- ular to the diameter through the given point. 9. Arighttriangularprismiscutbyaplanenot parallel to the base, but such that its intersection DE isparalleltothebasesegment AB. Showthat the volume of the part thus cut off is one third the product of the sum of the three vertical edges and the area of the base.

THE SPHERE 167 Adrien Marie Legendre(1752–1833) was a celebrated French math- ematician. He was one of three commissioners who introduced the met- ric system in France, having also been a member of the commission for determining the length of the meter. Besides many treatises on advanced subjects, he wrote a book on el- ementarygeometrywhichwasthemostsuccessfulofthemanyattempts to supersede Euclid as a text-book. It went through many editions in French and was translated into almost every other civilized language.

168 SOLID GEOMETRY: APPENDIX I appendix i: similar solids 378.SimilarCylinders. Tworightcircularcylindersare similar if they are generated by similar rectangles revolving about corresponding sides. 379. Theorem I. If in two similar cylinderss and s′ are the lateral areas,S andS′ the total areas,V andV′ the volumes, h andh′ the altitudes, andr andr′ the radii, then s s′ = S S′ = r2 r′2 = h2 h′2 and V V′ = r3 r′3 = h3 h′3. Suggestions for Proof: To find the ratioss s′ and S S′, make use of the following, giving reasons for each in detail. (1) s = 2πrh, (2) S = 2πr(r +h), (3) r r′ = h h′, and (4) r +h r′ +h′ = r r′ = h h′. To find the ratioV V′ make use ofV = πr2h, andV = πr′2h′, and r r′ = h h′.

SIMILAR SOLIDS 169 ra tios rela ting to similar cones 380. Similar Cones. Two right circular cones aresimilar if they are generated by two similar right triangles revolving about correspond- ing sides. 381. Theorem II. If in two similar coness and s′ are the lateral areas,S and S′ the total areas,V and V′ the volumes, h andh′ the altitudes, andr andr′ the radii, then s s′ = S S′ = r2 r′2 = h2 h′2 and V V′ = r3 r′3 = h3 h′3. Suggestions for Proof: See suggestions under § 379. exercises 1. The lateral area of a cone is36 square inches. What is the lateral area of a similar cone whose altitude is3 4 that of the given cone? 2. The total area of one of two similar cones is three times that of the other. Compare their altitudes and also their radii. 3. The sum of the total areas of two similar cones is 144 square inches. Find the area of each cone if one is1 3 4 times as high as the other. 4. The volume of one of two similar cones is5 times that of the other. Compare their altitudes.

170 SOLID GEOMETRY: APPENDIX I conditions for similarity of tetrahedrons 382. Similar Polyhedrons. Two polyhedrons aresimilar if they have the same number of faces similar each to each and similarly placed, and have their corresponding polyhedral angles equal. Any two parts which are similarly placed are calledcorresponding parts, as corresponding faces, edges, vertices. 383. Theorem III. Two tetrahedrons are similar if three faces of one are similar respectively to three faces of the other, and are similarly placed. Given the tetrahedrons P−ABC and P′−A′B′C′ having △APB~△A′P′B′,△APC~△A′P′C′, and△BPC ~△B′P′C′. To proveP−ABC~P′−A′B′C′. Proof: (1) Show that△ABC~△A′B′C′. (2) Show that trihedral∠sP and P′ are equal. Likewise ∠A = ∠A′, ∠B = ∠B′, ∠C = ∠C′. Hence, by definition, the polyhedrons are similar. sight work 1. If the two prisms in the figure are similar, name the pairs of corre- sponding parts. Likewise for two sim- ilar pyramids.

SIMILAR SOLIDS 171 tetrahedrons ha ving two equal trihedral angles 384. Theorem IV. The volumes of two tetrahedrons, having a trihedral angle of the one equal to a trihedral angle of the other, are proportional to the products of the edges which meet in the vertices of these angles. Given the tetrahedrons P− ABC and P′−A′B′C′ whose volumes areV and V′ and in which Tri.∠P = Tri. ∠P′. To prove thatV V′ = PA·PB·PC P′A′·P′B′·P′C′. Proof: PlaceP′−A′B′C′ so that Tri.∠P′ coincides with Tri.∠P. Let CM and C′M′ be the altitudes ofP−ABC and P−A′B′C′ from the verticesC and C′ upon the planePAB. Let AN and A′N′ be the altitudes of the△sPAB and PA′B′. Then V V′ = 1 3CM·area PAB 1 3C′M′·area PA′B′ = CM·PB·AN C′M′·PB′·A′N′. Now prove CM C′M′ = PC PC′ and AN A′N′ = PA PA′. Hence, we haveV V′ = PC·PB·PA P′C′·P′B′·P′A′.

Give all the steps and reasons in detail.

172 SOLID GEOMETRY: APPENDIX I volumes of similar tetrahedrons 385. Theorem V. The volumes of two similar tetra- hedrons are in the same ratio as the cubes of their corre- sponding edges. Given P−ABC∼P′−A′B′C′, with volumesV and V′. To prove thatV V′ = PA 3 P′A′3. Proof: We haveV V′ = PA·PB·PC P′A′·P′B′·P′C′. § 384 Now use the properties of similar triangles to complete the proof. Use the figure of § 384. exercises 1. Show that two tetrahedrons are similar if they have a dihedral angle of one equal to a dihedral angle of the other and the including faces similar each to each and similarly placed. 2. Show that the total areas of two similar tetrahedrons are in the same ratio as the squares of any two corresponding edges. 3. Show that if each of two polyhedrons is similar to a third they are similar to each other. 4. Two similar tetrahedral mounds have a pair of corresponding di- mensions 3 ft. and 4 ft. If one mound contains40 cu. ft. of earth, how much does the other contain? 5. The edges of a tetrahedron are3, 4, 5, 6, 7, and8. Find the edges of a similar tetrahedron containing64 times the volume. 6. Find what fraction of the altitude of a tetrahedron must be cut off by a plane parallel to the base, measuring from the vertex, in order that the new pyramid thus detached may have one fifth of the original volume.

SIMILAR SOLIDS 173 figures ha ving a center of similitude 386. Center of Similitude. Two figures are said to have a center of similitudeO, if for any two points,A andB, of the one figure the linesAO and BO meet the other in two points,A′ and B′, called corresponding points, such that OA :OA′ =OB :OB′. See figures under §§ 387–392. 387. Theorem VI. Any two figures which have a center of similitude are similar. Proof: (1) Two triangles. Given OA OA′ = OB OB′ = OC OC′. Let the student prove that△ABC∼△A′B′C′. In case the triangles do not lie in the same plane, use § 101 to show that the corresponding∠s are equal. (2) Two polygons. Given OA OA′ = OB OB′ = OC OC′, etc.

174 SOLID GEOMETRY: APPENDIX I Give the proof both for polygons in the same plane and not in the same plane. (3) Two tetrahedrons. With the same hypothesis as before, we must prove △PAB ~ △P′A′B′, △PBC ~ △P′B′C′, △PCA ~ △P′C′A′, and then use § 383. (4) Any two polyhedrons. (a) Prove corresponding polygonal faces similar. (b) Prove corresponding polyhedral angles equal. The last step requires not only equalface anglesabout the vertex, as in the case of the tetrahedron, but also equaldihedral angles. Note that two dihedral angles are equal if their faces are parallel right face to right face and left face to left face. (Why?) (5) Consider any two figures whatsoeverhaving a center of simili- tude. (a) Take any three pointsA, B, C, in one figure and the three corresponding points,A′, B′, C′, in the other. Then AB and A′B′, AC and A′C′, etc., are called corresponding linear dimensions, andthetriangles ABC andA′B′C′ arecorresponding triangles. (b) It is clear that any two corresponding linear dimensions have the same ratio as any other two, and that any two corresponding triangles are similar. In this sense any two figures having a center of similitude are thus proved to besimilar.

SIMILAR SOLIDS 175 similar triangles placed in similitude 388. Ratio of Similitude.The ratio of similitudeof two similar figures is the common ratio of their corresponding linear dimensions. This ratio is the same as thedistance ratioof

corresponding points from the center of similitude. 389. Theorem VII. Two similar triangles may be so placed as to have a center of similitude. Giventhesimilartriangles T andT′, inwhichA′B′ AB = A′C′ AC = B′C′ BC . To prove that they may be placed with a center of similitude. Proof: From any pointO drawOA, OB, OC. On these rays takeA1, B1, C1 so that OA1 OA = OB1 OB = OC1 OC = A′B′ AB . Now show the following: (1) △T1~△T, and hence△T1~△T′. (2) △T1 =△T′. For this show thatA1B1 =A′B′ by means of the equations A1B1 AB = OA1 OA and A′B′ AB = OA1 OA .

176 SOLID GEOMETRY: APPENDIX I LikewiseA1C1 =A′C′ and B1C1 =B′C′. (3) Finally, OM1 OM = OA1 OA , where M and M1 are any two corre- sponding points. Hence O is the required center of similitude. similar tetrahedrons placed in similitude 390. Theorem VIII. Two similar tetrahedrons may be so placed as to have a center of similitude. Given the similar tetrahedronsT and T′. To prove that they can be placed so as to have a center of similitude. Proof: WithO as a center of similitude, constructT1, making OA OA1 = OB OB1 = etc. = AB A′B′. Now show as in § 389 thatT1 =T′, and hence thatT′ can be placed in the positionT1 so as to have withT the center of similitudeO. Give all the steps in detail. 391. Corollary. Any two similar polyhedrons may be placed so as to have a center of similitude. Suggestion. The argument is precisely similar to that in § 390.

SIMILAR SOLIDS 177 center of similitude within the figure 392. In the proofs for the two preceding theorems the center of similitude was taken between the two figures oron the same side of them. The center may be taken equally wellwithin them, as in the following illustrations: In the case of similar convex polyhedrons with the center of simil- itude thus placed, the faces are the bases of pyramids whose vertices are all at the center of similitude. If, further, the polygonal faces be divided into triangles by drawing their diagonals, these triangles become the bases of tetrahedrons, all of whose vertices are at the center of similitude. Moreover, each inner tetrahedron is similar to its corresponding outer tetra- hedron. (Why?) The volumes of the two similar polyhe- drons are thus composed of the sums of sets of similar tetrahedrons. sight work 1. Give the proof of § 389, usingFig. 1 above, and extend the argument to two similar polygons, usingFig. 2. 2. Draw two similar tetrahedrons with their center of similitude within them, and give the proof of § 390 with the figure thus made. 3. Give the proof of § 390, usingFig. 3 above.

178 SOLID GEOMETRY: APPENDIX I ra tio of volumes of similar pol yhedrons 393. Theorem IX. The volumes of any two similar poly- hedrons are proportional to the cubes of their corresponding edges. Proof: Place the polyhedrons whose volumes areV and V′ so as to have their center of similitude within them as in the third figure of § 392. Call the volumes of the similar tetrahedronsT1,T2,T3,... , andT′ 1, T′ 2, T′ 3, ... , and letAB and A′B′ be two corresponding edges. Then we have AB 3 A′B′3 = T1 T′ 1 = T2 T′ 2 = T3 T′ 3 =⋯ . (Why?) And T1 +T2 +T3 +⋯ T′ 1 +T′ 2 +T′ 3 +⋯ = T1 T′ 1 = AB 3 A′B′3. (Why?) But T1 +T2 +T3⋯ =V and T′ 1 +T′ 2 +T′ 3⋯ =V′. Hence, V V′ = AB 3 A′B′3. 394. Corollary. The volumes of any two similar solids are proportional to the cubes of any two corresponding linear dimensions. This proposition may be rendered evident by noticing that any two similar three-dimensional figures may be built up to any degree of ap- proximation by means of pairs of similar tetrahedrons

similarly placed. The proposition then holds for any two corresponding figures used in the process of approximation. Note that the ratio of similitude of two similar figures may be ob- tained from the ratio of any pair of their corresponding linear dimen- sions.

SIMILAR SOLIDS 179 applica tions of similarity 395. The Pantograph. The theorem that any two figures which have a center of similitude are similar is the geometric basis of many mechanical contrivances for enlarging or reducing both plane and solid figures; that is, for constructing figures similar to given figures and having with them a given ratio of similitude. The essential property of all such contrivances is that one point O is kept fixed, while two pointsA and B are allowed to move so that O, A, and B always remain in a straight line, and so that the ratio OA : OB remains the same. See page 216 of Plane Geometry. In the first figure on this pageO is a fixed point. SegmentsOD, CB, and the sides of the parallelogramACED are of fixed length. Prove that ifB is once so taken on the lineEC as to be in the lineOA, the pointsO,A, andB will always remain collinear, and that OA :OB remains a fixed ratio. In the second figure is shown an ordinary pantograph used for copy- ing and at the same time for reduc- ing or enlarging maps, designs, etc. The lengths of the various segments are adjustable, as shown, thus ob- taining any desired scale. The same contrivance may be used for copying figures inspace, such as relief maps, and at the same time reducing or enlarging them. ra tios rela ting to any two similar solids 396. Corresponding Cross Sections. Now consider any two similar figures whatever so placed as to have a center of similitudeO. We have seen that if pointsA, B and A′, B′ are corresponding points of the two figures, then the ratio of thecorresponding linear dimensions AB andA′B′ is equal to the ratio of similitudem :n of the two figures.

180 SOLID GEOMETRY: APPENDIX I Also if A, B, C, D and A′, B′, C′, D′ are corresponding points, then△sABC and A′B′C′, and the tetrahedronsABCD and A′B′C′D′ are similar, and we have area ABC area A′B′C′ = m2 n2 and vol.ABCD vol.A′B′C′D′ = m3 n3. The points A, B, C and A′, B′, C′ determine two planes, each of which intercepts a certain plane figure in the solid figure to which the points belong. These two plane figures we callcorresponding cross sections. We assume without full argument: 397. Theorem X. (1) The ratio of the areas of any pair of corresponding cross sections or any pair of corresponding surfaces of similar figures is equal to the square of their ratio of similitude,and (2) The ratio of the volumes of any two similar figures is equal to the cube of their ratio of similitude. The fact that the ratio of the areas of corresponding surfaces of similar solids is equal to thesquare of their ratio of similitude, while the ratio of their volumes equals thecube of this ratio is one of the most important and far-reaching conclusions of geometry. Thus the ratio of the weights of two similar shells used in gunnery, or the ratio of the weights of two men of similar build, may be found when their ratio of similitude is known. problems and applica tions 1. If it is known that a steel wire of radiusr will carry a certain weightw, how great a weight will a wire of the same material carry if its radius is2r? Suggestion. The tensile strengths of wires are in the same ratio as their cross-section areas. 2. Find the ratio of the diameters of two wires of the same material if one carries twice the load of the other; three times the load.

SIMILAR SOLIDS 181 3. In a laboratory experiment a heavy iron ball is suspended by a steel wire. In suspending another ball of twice the diameter a wire of twice the radius of the first one is used. Is this perfectly safe if it is known that the first wire will just safely carry the ball suspended from it? Discuss fully. 4. In two schoolrooms of the same shape (similar figures) but of different size, the same proportion of the floor space is occupied by desks. Which contains the larger amount of air for each pupil? 5. If the shells used in guns are similar in shape, find the ratio of the total surface areas of an eight-inch and a twelve-inch shell. 6. Find the ratio of the weights of the shells in the preceding prob- lem, weights being in the same ratio as the volumes. 7. If a man5 ft. 9 in. tall weighs165 lb., what should be the weight of a man6 ft. 1 in. tall, if they are similar in shape? 8. What is the diameter of a gun which fires a shell weighing twice as much as a shell fired from an eight-inch gun, supposing the shells to be similar bodies? 9. Supposing two trees to be similar in shape, what is the diameter of a tree whose volume is three times that of one whose diameter is 2 feet? What is the diameter if the volume is five times that of the given tree? What if it isn times that of the given tree? 10. Assuming that the weights of schoolboys vary as the cubes of their heights, construct a graph representing the relation between their heights and weights, if a boy5 feet 9 inches tall weighs130 pounds. Suggestion.If w represents the number of pounds in weight andh the number of feet in height,w =kh3. Fromw = 130, whenh = 5 3 4, we havek =.684. For the purpose of the graph,k =.7 is accurate enough. 11. From the graph constructed in the preceding example find the weight of a boy 5 feet tall; one 5 feet 4 inches; one 5 feet 6 inches. Compare with the weights of boys in your class. 12. If a man6 feet tall weighs185 pounds, construct a graph repre- senting the weights of men of similar build and of various heights.

182 SOLID GEOMETRY: APPENDIX I 13. If steamships are of the same shape, their tonnages vary as the cubes of their lengths. The Mauretania is 790 feet long, with a net tonnage of 32, 500. Construct a graph representing the tonnage of steamships of the same shape, and of various lengths. Other ships which at one time or another have held ocean records are: the (former)Deutschland, length 686 ft. and tonnage16, 500; the (for- mer)Kaiser Wilhelm der Grosse, length648ft. and tonnage14, 300; the Lucania, length 625 ft. and tonnage13, 000 (nearly); and theEtruria, length 520ft. and tonnage8000. By means of this graph decide whether or not these boats have greater or less tonnage than theMauretania as compared with their lengths. 14. Raindrops as they start to fall are extremely small. In the course of their descent a great many are united to form larger and larger drops. If 1000 such drops unite into one, what is the ratio of the surface of the large drop to the sum of the surfaces of the small drops? 15. The strength of a muscle varies as its cross-section area, which in turn varies as the square of the height or length of an animal, while the weight of the animal varies as the cube of its height or length. Use these facts to explain the greater agility of small animals. For example, compare the rabbit and the elephant. 16. Assuming the velocities the same, the amounts of water flowing through pipes vary directly as their cross-section areas. How many pipes, each 4 in. in diameter, will carry as much water as one pipe 12 in. in diameter? 17. What must be the diameter of a cylindrical conduit which will carry enough water to

supply ten circular intakes each8 feet in diam- eter? 18. Awaterreservoir, includingitsfeedpipes, isreplacedbyanother, each of whose linear dimensions is twice the corresponding dimensions of the first. If the velocity of the water in the feed pipes of the new system is the same as that in the old, will it take more or less time to fill the new reservoir than it did the old? What is the ratio of the new time to the old? 19. If two engine plants are exactly similar in shape, but each linear dimension in one is three times the corresponding dimension of the

PROJECTION OF LINE-SEGMENTS 183 other, and if the steam in the feed pipes flows with the same velocity in both, compare the speeds of the engines. 20. If two men, one5 ft. 6 in. and the other6 ft. 2 in. in height, are similar in structure in every respect, how much faster must the blood flow in the larger person in order that the body tissues of both shall be supplied equally well? Suggestion. Note that the amount of tissue to be supplied varies as the cube of the height, while the cross-section area of the arteries varies as the square of the height. appendix ii: projection of line-segments 398. Length of Projection.The projection of a line-segment on a plane was shown in § 121 to be another line-segment. Thelength of this projectionwill now be computed in terms of the given line-segment. 399. Cosine of Projection Angle. The acute angle between a line-segment and a given line on which it is projected is called the projection angle. Ifl is the length of a line-segment andp the length of its projection, then the ratiop :l is called thecosine of the projection angle. E.g., in Fig. 1,p l = cosine ∠BAE. 400. Sine, Cosine, Tangent.In any right triangleABC (Fig. 2), either acute angle, as ∠A, is the projection angle between the hy- potenuse and the side adjacent to the angle. Hence thecosine of an acute angle of a right triangle is theratio of the adjacent side to the hypotenuse.

184 SOLID GEOMETRY: APPENDIX II Likewise we define thesine of an acute angle of a right triangle as the ratio of the opposite side to the hypotenuse, and thetangent of an acute angle of a right triangle as theratio of the opposite side to the adjacent side. Using the common abbreviations, sin, cos, and tan, we have in Fig. 2, sinA = a c, cosA = b c, tanA = a b. exercises on sines, cosines, t angents The sine, cosine, and tangent are of great importance in many com- putations. By careful measurement (and in other ways) their values may be computed for any acute angle, and a table formed, like that on page 186. E.g., if ∠A = 35◦ (measured with a protrac- tor), and if we measureAC, AB, and BC, and then compute the ratios a c, b c, and a b, we shall find the values ofsin 35◦, cos 35◦, tan 35◦. With an ordinary ruler it will not usually be possible to make these measurements with sufficient accuracy to obtain more than one decimal place. exercises 1. Using a protractor, construct angles of10◦, 30◦, 50◦, 70◦, and by measurement determine the sine, cosine, and tangent of each. 2. Prove that the cosine of any given angle is the same, no matter what point is taken in either side from which to let fall the perpen- dicular to the other side. Prove the same for the tangent. 3. Show that if the hypotenuse be takenone decimeter in length, then the length of the side adjacent, measured in decimeters, is the cosine of the angle, and the length of the side opposite is thesine of the angle.

PROJECTION OF LINE-SEGMENTS 185 4. Show that if the side adjacent be taken one decimeter in length, the length of the side opposite, measured in decimeters, is

thetangent of the angle. 5. Without any direct measurement, show how to compute the three ratios for each of the angles30◦, 45◦, 60◦. Suggestion. Make use of the fact that if one acute angle in a right triangle is 30◦, the side opposite it is one half the hypotenuse.

186 SOLID GEOMETRY: APPENDIX II t able of sines, cosines, and t angents Angle Sin Cos Tan Angle Sin Cos Tan 0◦ 0 1.000 0 46◦ .719 .695 1.04 1◦ .017 1.000 .017 47◦ .731 .682 1.07 2◦ .035 .999 .035 48◦ .743 .669 1.11 3◦ .052 .999 .052 49◦ .755 .656 1.15 4◦ .070 .998 .070 50◦ .766 .643 1.19 5◦ .087 .996 .087 51◦ .777 .629 1.23 6◦ .105 .995 .105 52◦ .788 .616 1.28 7◦ .122 .993 .123 53◦ .799 .602 1.33 8◦ .139 .990 .141 54◦ .809 .588 1.38 9◦ .156 .988 .158 55◦ .819 .574 1.43 10◦ .174 .985 .176 56◦ .829 .559 1.48 11◦ .191 .982 .194 57◦ .839 .545 1.54 12◦ .208 .978 .213 58◦ .848 .530 1.60 13◦ .225 .974 .231 59◦ .857 .515 1.66 14◦ .242 .970 .249 60◦ .866 .500 1.73 15◦ .259 .966 .268 61◦ .875 .485 1.80 16◦ .276 .961 .287 62◦ .883 .469 1.88 17◦ .292 .956 .306 63◦ .891 .454 1.96 18◦ .309 .951 .325 64◦ .899 .438 2.05 19◦ .326 .946 .344 65◦ .906 .423 2.14 20◦ .342 .940 .364 66◦ .914 .407 2.25 21◦ .358 .934 .384 67◦ .921 .391 2.36 22◦ .375 .927 .404 68◦ .927 .375 2.48 23◦ .391 .921 .424 69◦ .934 .358 2.61 24◦ .407 .914 .445 70◦ .940 .342 2.75 25◦ .423 .906 .466 71◦ .946 .326 2.90 26◦ .438 .899 .488 72◦ .951 .309 3.08 27◦ .454 .891 .510 73◦ .956 .292 3.27 28◦ .469 .883 .532 74◦ .961 .276 3.49 29◦ .485 .875 .554 75◦ .966 .259 3.73 30◦ .500 .866 .577 76◦ .970 .242 4.01 31◦ .515 .857 .601 77◦ .974 .225 4.33 32◦ .530 .848 .625 78◦ .978 .208 4.70 33◦ .545 .839 .649 79◦ .982 .191 5.14 34◦ .559 .829 .675 80◦ .985 .174 5.67 35◦ .574 .819 .700 81◦ .988 .156 6.31 36◦ .588 .809 .727 82◦ .990 .139 7.12 37◦ .602 .799 .754 83◦ .993 .122 8.14 38◦ .616 .788 .781 84◦ .995 .105 9.51 39◦ .629 .777 .810 85◦ .996 .087 11.43 40◦ .643 .766 .839 86◦ .998 .070 14.30 41◦ .656 .755 .869 87◦ .999 .052 19.08 42◦ .669 .743 .900 88◦ .999 .035 28.64 43◦ .682 .731 .933 89◦ 1.000 .017 57.29 44◦ .695 .719 .966 90◦ 1.000 0 45◦ .707 .707 1.00

PROJECTION OF LINE-SEGMENTS 187 length of projection of a line-segment 401. Theorem I. The length of the projection of a line-segment upon a given line is equal to the length of the line-segment multiplied by the cosine of the projection angle. Given the projectionp of the line- segment l on the lineCD, with the projection angleA. To prove thatp =l cosA. Proof: By definition we havep l = cosA. Hence, p =l cosA. exercises 1. Find the cosines of the angles35◦ 30′, 54◦ 15′, 15◦ 45′. Suggestion.The cosine of35◦ 30′ lies betweencos 35◦ and cos 36◦. We assume that it lies halfway between these numbers. This assumption, while not quite correct, is very nearly so for small differences of angles, as in this case, where the total difference is only one degree. From the tablecos 35◦ =.819, cos 36◦ =.809. The number midway between these is.814, which we take as the cosine of35◦ 30′. This process is calledinterpolation. A similar process is used for sines and tangents. 2. Find the tangents of the angles25◦ 20′, 47◦ 45′, 63◦ 40′. 3. Find the angle whose tangent is1.74. Solution. Fromthetablewehave tan 60◦ = 1.73and tan 61◦ = 1.80. Hence, the required angle must lie between60◦ and 61◦. Moreover, the number1.74 is one seventh the way from1.73 to 1.80. Hence, we assume the angle to lie one seventh the way from60◦ to 61◦, which gives 60◦ + 1 7× 1◦ = 60◦ + 9′ nearly. The required angle is60◦ 9′. 4. Find the angles whose sines are.276; .674; .437. 5. Find the angles whose

cosines are.940; .094; .435.

188 SOLID GEOMETRY: APPENDIX II 6. Find the angles whose tangents are.781; 1.41; 3.64. 7. At what angle with the horizontal must the base of a right circular cylinder be tilted to make it just topple over if its diameter is6 ft. and its altitude 8 ft.? Suggestion. The center of gravity is at the middle point C of the axis of the cylinder. The base must be tilted so that the lineAC becomes vertical. The required angle is∠ACB. 8. The Leaning Tower of Pisa is179 feet high and 31 feet in diameter. It now leans so that a plumb line from the top on the lower side reaches the ground 14 feet from the base. At what an- gle is its side now inclined from the vertical? At what angle would its side have to incline from the vertical before it would topple over? 9. A four-inch hole is cut in a board, and a ball 8 in. in diameter is made to rest on it. At what angle must the board be held so that the ball will just roll out of the hole? Suggestion. The board must be held so that the line OA becomes vertical. 10. Using a ball 8 inches in diameter, what must be the radius of the hole in the board of the preceding problem so that the ball shall just roll out when the board is inclined at an angle of 45◦ to the horizontal? 11. If the figureABCD−H is a cube, find each of the following angles:∠ECA, ∠AEC. Check by using the fact that the sum of the angles of a triangle is180◦.

PROJECTION OF LINE-SEGMENTS 189 al titude of oblique prism or cylinder 402. Theorem II. The altitude of an oblique prism or cylinder is equal to an element multiplied by the cosine of the angle between the plane of the base and that of a right section. Given an oblique prism or cylinder with baseb and right section c, and letBE be a perpendicular between the planes of the bases. Consider the plane determined byBE and the elementAB. This plane is⊥ to the plane ofb and also to the plane ofc. (Why?) Hence, it is⊥ to the line of intersection of the planes ofb and c. (Why?) Let this plane cut the planes ofb andc inGD andFD respectively. Then ∠D is the measure of the dihedral angle between the planes of b and c. (Why?) To prove that BE =AB· cosD. Proof: We have BE =AB· cos ∠ABE. Why? But ∠D = ∠ABE. Why? Hence, BE =AB· cosD. 403. Corollary. The dihedral angle between the planes of the base and a right section of an oblique cylinder or prism is equal to the angle between an element and the altitude.

190 SOLID GEOMETRY: APPENDIX II exercises 1. Given a line-segment10 inches long. Find the length of its pro- jection on a plane if the projection angle is20◦. If the angle is30◦, 45◦, 60◦, 90◦, 0◦. 2. A kite string forms an angle of40◦ with the ground. The distance from the end of the string to a point directly beneath the kite is200 ft. Find the length of the string and the perpendicular height of the kite. 3. The altitude of an oblique prism is15 inches. Find the length of an element if it makes an angle of45◦ with the perpendicular between the bases. 4. A right section of a cylinder makes an angle of20◦ with the plane of the lower base. Find the ratio between the altitude and an element. 5. Prove that by joining the middle points of six edges of a cube, as shown in the figure, a regular hexagon is formed. 6. Prove that in the preceding exam- ple the plane of the regular hexagon, KLMNOP , is perpendicular to the diag- onal DF of the cube. 7. How large a cube will be required from which to cut a stopper for a hexagonal spout, each of whose sides is4 inches? 8. In the figure find

the angleKQH. Suggestion. Let a be a side of the cube. ComputeKH, KQ, andHQ in terms a. Note that ∠QKH = rt. ∠. 9. Find the area of the projection of the hexagonKLMNOP on the faceBCGF . Note that this projection equals the whole square less △NCO +△KEL. See § 404. Find the area of the hexagon in terms of the sidea of the cube.

PROJECTION OF LINE-SEGMENTS 191 area of the projection of a plane-segment 404. Projection of a Plane-Segment. If from each point in the boundary of a plane-segment a perpendicular is drawn to a given plane, the locus of the feet of these perpendiculars will bound a portion of the plane, which is called theprojection of the plane-segmenton the given plane. E.g., the plane-segmentA′B′C′ in the plane N is the projection of the plane-segment ABC from the planeM upon N. The angle of projection is the angle be- tween the planesM and N. 405. Theorem III. The area of the projection of a plane- segment on a plane is equal to the area of the plane-segment multiplied by the cosine of the projection angle. Proof: Let the boundary of the given plane-segmentb be any con- vex polygon or closed curve. Using a line perpendicular to the given plane as agenerator, develop aprismaticorcylindricalsurfaceofwhich bisasection. Thegivenplane will cut this surface in a right section whose area we denote byc. Now cut the surface by a plane parallel tob, forming the upper base b′ of a prism or cylinder whose altitude ish, edgee, and volumeV.

192 SOLID GEOMETRY: APPENDIX II Then c is the projection ofb upon the given plane, and∠1 = ∠2 is the projection angle. We are to show that c =b cos ∠1. We know that V =ce =bh. Why? But h =e cos ∠2. Hence, ce =be cos ∠2. That is, c =b cos ∠2 = b cos ∠1. Note. The foregoing theorem may be proveddi- rectly in case the plane-segment is a rectangle with one side parallel to the line of intersection of the two planes. In the figure letS be the given rectangle and S′ its projection, withAB∥ to the line of intersection of the planes in whichS andS′ lie, and ∠1 the angle between them. Then S =AB·BC and S′ =A′B′·B′C′. But AB =A′B′ and BE =B′C′. Why? But BE =BC· cos ∠1 Why? and S′ =A′B′·B′C′ =AB·BC cos ∠1. That is, S′ =S cos ∠1. In the case of any plane-segment, rectangles may be inscribed in it in this position and their number increased indefinitely, so that their sum will approach more and more nearly to the area of the plane-segment, and in this way it may be shown to any desired degree of approximation, that the projection of a plane-segment equals the given plane-segment multiplied by the cosine of the projection angle. 406. Ellipse. An important special case of the theorem of § 405 is the area of the figure obtained by projecting a circle upon a plane not parallel to its own plane, nor at right angles to it. This figure is called an ellipse. On page 193 the principle developed above is used to find the area of the ellipse.

PROJECTION OF LINE-SEGMENTS 193 the area of an ellipse 407. Projection of a Circle. In the figure two planes,M and M′, meet in a linePQ. The circleO in M has a diameterAB∥ PQ and a diameterCD⊥PQ. In projecting the whole figure upon the planeM′ the diameterAB projectsintoitsequal A′B′, whileCD projectsinto C′D′ sothat C′D′ = CD cos ∠1. By theorem § 405 the area of the ellipseA′C′B′D′ equals the area of the circleACBD multiplied by cos ∠1. Hence, πr2· cos ∠1 is the area of the ellipse.

But r cos ∠1 = O′C′ and r =O′B′. (§ 401) Hence, the area of the ellipse isπ·O′C′×O′B′. The segmentsA′B′ and C′D′ are called respectively the major and minor axes of the ellipse, andO′B′ and O′C′ the semimajor and the semiminor axes. These latter are usually denoted bya and b. Hence, the area of the ellipse is πab. Note that whena and b are equal, the ellipse becomes a circle, and this formula reduces toπa2 as it should.

194 SOLID GEOMETRY: APPENDIX II problems and applica tions 1. Given a right circular cylinder the radius of whose base is6inches. Find the area of an oblique cross section inclined at an angle of45◦ to the plane of the base. 2. Given an oblique circular cylinder the radius of whose right sec- tion is 10 inches. Find the area of the base if it is inclined at an angle of 60◦ to the right section. 3. If an oblique circular cylinder has an altitudeh, an elemente, radius of right sectionr, and ∠A the inclination of the base to the right section, express the volume in two ways and show that these are equivalent. 4. A six-inch stovepipe has a45◦ elbow angle, that is, it turns at right angles. (The angleCAB is called the elbow angle.) Find the area of the cross section ofAB. Likewise if it has a60◦ elbow angle. 5. At what angle must the damper in a circular stovepipe be turned in order to obstruct just half the right cross sectional area of the pipe? Suggestion.The damper must be turned so that the projection of the damper upon a right cross section is equal to half that cross section. 6. The comparatively low temperature of the earth's surface near the pole, even in summer, when the sun does not set for months, is due partly to the obliqueness with which the sun's rays strike the earth. That is, a given amount of sunlight is spread over a larger area than in lower latitudes. Thus, if in the figureD′C is a horizontal line, andD′D the direction of the sun's rays, then a beam of light whose right cross section isABCD is spread over the rectangle A′BCD′. In other words, a patch of groundA′BCD′ receives onlyasmuchsunlightasapatchthesizeof ABCD receives when the sun's rays strike it vertically.ABCD =A′BCD′ cos ∠1.

VARIABLES. LIMITS 195 Hence, each unit of area inA′BCD′ receives cos ∠1 times as much light as a unit inABCD. Hence, to compare the heat-producing powers of sunlight in any latitude with that at the place where the sun's rays fall vertically, we need to know how the projection angle,∠1, is related to the difference in latitude of the two places. 7. If ∠1 = 30 ◦, compare the amount of heat received by a unit of area inABCD and A′BCD′. 8. What must ∠1 be in order that a unit of area inA′BCD′ shall receive only 1 5 as much light as a unit inABCD? 9. The figure represents a cross section of the earth with an indica- tion of the direction of the rays of light as they strike it at the summer solstice when they are vertical atA, the tropic of Cancer.B represents the latitude of Chicago,C the polar circle, andP the north pole. The angles PDE, CGF, BKH represent the projection angle,∠1, for the various latitudes. Prove: ∠PDE = ∠POK, ∠CGF = ∠COK, ∠BKH = ∠BOK. That is, ∠1 for each place is the latitude of that place mi- nus the latitude of the place where the sun's rays are ver- tical. 10. Find the relative amount of sunlight received by a unit of area at the tropic of Cancer and at the north pole at the time of the summer solstice. 11. Find the ratio between the amount of light received by a unit of the earth's area at Chicago and at the tropic of Cancer at the time of the summer solstice. 12. Find the same ratio for the equator and Chicago at the winter solstice when the sun is vertical at latitude23 1 2 ◦ south.

196 SOLID GEOMETRY: APPENDIX III appendix iii: v ariables. limits 408. Variables and Functions. It is often useful to think of a geometric figure asvarying in size and shape. E.g., if a rectangle has a fixed base, say10 inches long, but an altitude which varies from3 inches to 5 inches, then the area varies from3· 10 = 30 to 5· 10 = 50 square inches. We may even think of the altitude as starting at zero inches and increasing, in which case the area starts at zero and increases continuously. The altitude which we think of as varying at our pleasure is called the independent variable, while the area, being dependent upon the altitude, is called thedependent variable. The dependent variable is sometimes called afunction of the in- dependent variable, meaning that the two are connected by a definite relation such that, for any definite value of the independent variable, the dependent variable also has a definite value. Thus, in the formula for the area of a rectangle,a = bh, if b is fixed and h varies, thena is a function ofh, since for every value ofh there is determined a definite value ofa. 409. Illustrations of Limits. If a regular polygon is inscribed in a circle of fixed radius, and if the number of sides of the polygon be continually increased, for instance by repeatedly doubling the number, then the apothem, perimeter, and area are all variables depending upon the number of sides. That is, each of these is a function of the number of sides. Now the greater the number of sides the more nearly does the apothem equal the radius in length. Indeed, it is evident that the difference between the apothem and the radius will ultimately become less than any fixed number, however small. Hence we say that the apothem approaches the radius as a limitas the number of sides in- creases indefinitely. Similarly the perimeters of the polygons may be made as nearly equal to the circumference as we please by making the number of sides sufficiently great.

VARIABLES. LIMITS 197 Hence we may define the circumference of a circle as thelimit of the perimeter of a regular inscribed polygon as the number of sides increases indefinitely. The circumference of a circle may also be defined as the limit of the perimeter of a circumscribed polygon as the number of sides is increased indefinitely. Likewise we may define the area of a circle as thelimit of the area of the inscribed or the circumscribed polygon as the number of sides is increased indefinitely. The notion of a limit may be used to define the length of a line- segment which is incommensurable with a given unit segment. Thus, the diagonald of a square whose side is unity isd = √ 2. Hence d may be defined as thelimit of the variable line-segmentwhose successive lengths are 1, 1.4, 1.41, 1.414, ...g In like manner, the length of any line-segment, whether commen- surable or incommensurable with the unit segment, may be defined in terms of a limit. Thus, if a variable segment is increased by successively adding to it one half the length previously added, then the segment will approach a limit. If the initial length is1, and if the successive additions are1 2, 1 4, 1 8, 1 16, 1 32, etc., then the successive lengths are1, 1 1 2, 1 3 4, 1 7 8, 1 15 16, 1 31 32, etc. Evidently this segment approaches the limit2. Hence 2 may be defined as the limit of the variable segment whose suc- cessive lengths are1, 1 1 2, 1 3 4, 1 7 8, etc., as the number of successive additions is increased indefinitely. The idea of a functional relation between variables and the idea of a limit, as illustrated above, are two of the most important concepts in all mathematics. definition of a limit 410. Constant. A quantity which remains fixed in value through- out a discussion is

called aconstant. E.g., the base of the rectangle mentioned in § 408 is a constant. gSee Transcriber's Notes.

198 SOLID GEOMETRY: APPENDIX III 411. Variable. A quantity which continuously changes in value, or which takes on a succession of different values, is called avariable. E.g., the altitude and the area mentioned in § 408 are variables. 412. Limit of a Variable. If a variable may be made to ap- proach a certain constant quantity in such a way that the difference between the constant and the variable becomes and remains less than any assignable value, however small, then the constant is called the limit of the variable. E.g., the fixed circumference of the circle is the limit of the variable perimeters of the polygons mentioned in § 409. sight work 1. Findthelimitofavariableline-segmentwhoseinitiallengthis 6inches, and which varies by successive increments each equal to one half the preced- ing, the first increment being2 inches. 2. Construct a right triangle whose sides are1 and 2. By approximating a square root, find five successive lengths of a segment which approaches the length of the hypotenuse as a limit. 3. If one tangent to a circle is fixed and another is made to move so that their intersection point approaches the circle, what is the limiting position of the moving tangent? What is the limit of the measure of the angle formed by these tangents? 4. The arcAB of 74◦ is the greater of the two arcs intercepted between two secants meeting atC outside the circle. The pointsA and B remain fixed whileC moves up to the circle. What angle is the limit of the variable angle formed by the varying secants? What is the limit of the measure of this angle? 5. If in the preceding the secants meet within the circle, what is the limit of their angle and also of the measure of this angle as the intersection point moves up to the center of the circle?

VARIABLES. LIMITS 199 the incommensurable cases 413. The Incommensurable Cases.We have seen that there are segments which areincommensurable; that is, which have no common unit of measure,—for instance, the side and the diagonal of a square. For practical purposes the lengths of such segments are approxi- matedtoanydesireddegreeofaccuracy, andtheirratiosareunderstood to be the ratios of these approximate numerical measures. But fortheoretical purposesit is important to consider these incom- mensurable cases further; just as in algebra we not onlyapproximate such roots as $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, etc., but we also deal with these surds as exact numbers. Instances of this kind occur in such operations as $(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2}) = 3-2 = 1$. Whilethelengthofthediagonalofaunitsquarecannotbeexpressed as an integer or as a rational fraction, that is, as the quotient of two integers, we nevertheless think of such a segment as having adefinite length, or what is the same thing, adefinite ratiowith the unit segment forming the side of the square. We now fix our attention on theincommensurable ratios themselves, and the method of determining them, rather than on theprocess of approach and the practical computation based on it. 414. Representation by Irrational Numbers.In general, the ratio between any two incommensurable geometric magnitudes of the same kind may be represented by what is called anirrational number; that is, a number which isneither an integer nor a quotient of two integers. Examples of irrational numbers are the surds, such as $\sqrt{2}$, $3\sqrt{5}$, etc., and the numberπ. numbers defined by sequences 415. Numbers Defined by Sequences. The following is a method for

determining any number, whether rational or irrational. For simplicity it is applied first to the integer1.

200 SOLID GEOMETRY: APPENDIX III Throughout this discussion the expressions “point on a line” and “number” will be used interchangeably. 1 2 3 4 7 80 1 On a straight line mark a certain point 0 (zero), and one unit to the right of it mark another point1. Also lay off points such that their distances from0 are 1 2, 3 4, 7 8, 15 16, ... If this sequence of points is carried ever so far, it will never reach the point 1. If, however, we select a point to the left of1, no matter how near to it, we may always go far enough along this sequence to reach points between this point and 1. That is, 1 is the limit of the terms of this sequence. See § 412. The point 1 has two definite relations to this sequence: (a) Every point of the sequence is to the left of1. (b)For any fixed point to the left of1 there are points of the sequence between it and1. We see that1 is theonly point on the whole line such that both (a) and (b) are true of it. For every point to the right of1 (a) is true, but (b) is not. For every point to the left of1 (b) is true, but (a) is not. It follows therefore that, while the points of the sequence merely approach the point 1 as a limit, the sequence,taken as a whole, serves to distinguish that point from all other points on the line. In the above sequence, the terms continuallyincrease. The number 1 may be determined equally well by adecreasing sequence. For instance, the sequence 1 1 2, 1 1 4, 1 1 8, 1 1 16, ... has the following properties which distinguish the point 1 from all other points on the line. (a) Every point of the sequence is to the right of1. (b) For any fixed point to the right of 1 there are points of the sequence between it and1. limit of an infinite sequence 416. Infinite Sequences. An endless sequence of either kind just described is called aninfinite sequence. Not every infinite sequence serves to single out a definite point in the manner shown above. Thus the sequence1, 2, 3, 4, ... fails to do so, because its terms grow large beyond all bound. Such sequences are

VARIABLES. LIMITS 201 said to beunbounded, while the sequences1 2, 3 4, 7 8, ... and 1 1 2, 1 1 4, 1 1 8, ... are bounded. Again, the sequence1, 2, 1, 2, 1, 2, ... fails to single out a definite point. This sequence is said to beoscillating, since its terms increase, then decrease, then increase, etc., while1 2, 3 4, 7 8,... and 1 1 2, 1 1 4, 1 1 8,... are non-oscillating. 417. Limit of a Sequence.The number 1 is said to be theleast upper bound of the sequence 1 2, 3 4, 7 8, ... That is, 1 is the smallest number beyond which the sequence does not go.1 is also said to be the limit of the sequence. See § 412. Similarly, 1 is thegreatest lower boundor thelimit of the sequence 1 1 2, 1 1 4, 1 1 8, ... ; that is, the greatest number such that the sequence contains no number less than it. In the manner described above any integer may be determined by an increasing or a decreasing infinite sequence. E.g., the number2 is the limit of the increasing sequence1, 1 1 2, 1 3 4, 1 7 8, 1 15 16, ... , or of the decreasing sequence2 1 2, 2 1 4, 2 1 8, 2 1 16, ... Two different sequences, both increasing or both decreasing, may also define the same number. E.g., each of the increasing sequences1 2, 3 4, 7 8, 15 16,... and 2 3, 4 5, 6 7, 8 9,... determines the number1. We notice, however, that no matter what definite number we select in either of these sequences, there is a number in the other greater than it; that is, neither sequence contains a number greater than every number in the other. See § 420. It is apparent that any number may be defined as the limit of a sequence, and we shall assume the

converse, namely, that any bounded increasing or decreasing sequence has a limit. fundament al principles of limits 418. Axiom We now assume that: Every bounded increasing se- quence has a least upper bound, and every bounded decreasing sequence has a greatest lower bound. This axiom may also be stated:

202 SOLID GEOMETRY: APPENDIX III Every bounded increasing or decreasing sequence has a limit. Thisaxiomsimplymeansthateverysuchsequencesinglesoutadefi- nitenumber, rationalorirrational, inthemannerdiscussedonpage201. Thus, if we attempt to approximate the square root of 2, we obtain a sequence, 1, 1.4, 1.41, 1.414, 1.4142, ... , having for its limit adefinite number represented by √ 2, which corresponds to the length of the diagonal of a square whose side is unity. Again, if we attempt to approximate the value ofπ, we obtain a sequence 3, 3.1, 3.14, 3.141, 3.1415, ... , having for its limit the definite number rep- resented byπ, which is the ratio of the circumference to the diameter of a circle. 419. Theorem I. Two bounded sequences have the same limit if they are equal term by term. Proof: If the sequences area1,a2,a3,... andb1,b2,b3,... , and if a1 =b1,a2 =b2,a3 =b3,... , then they are one and the same sequence and hence have the same limit by § 418. Historical Note. The rigorous treatment of limits has been developed during the last fifty years. The foundation on which it rests is the theory of irrational numbers, which itself was placed on a firm basis in the early seventies. The Greeks dealt with the incommensurable cases in an interesting man- ner. Thus, to prove that two incommensurable ratios are equal they showed that neither can be less than the other. However, they failed to make explicit definitions of incommensurables and they did not explicitly state the funda- mental axiom of § 418 or its equivalent. Their treatment would be complete and rigorous if supplemented by the proper definitions and axioms. fundament al principles of limits 420. Theorem II. Two increasing bounded sequences have the same limit if neither sequence contains a number greater than every number of the other. Let a1, a2, a3, ... and b1, b2, b3, ... denote two infinite sequences with limitsa and b, such that no term of the first sequence is greater

VARIABLES. LIMITS 203 than every term of the second, and no term of the second is greater than every term of the first. To prove thata =b. Proof: Suppose that a is not equal tob but is less thanb. Then there must be numbers of the sequenceb1, b2, b3, ... greater thana, since by § 415 there are numbersb1, b2, b3, greater than any fixed number whatever which is less thanb. This contradicts the hypothesis of the theorem, and hencea cannot be less thanb. In like manner we show thatb is not less thana. Hencea =b. See § 20. 421. Theorem III. Two decreasing bounded sequences have the same limit if neither sequence contains a number less than every number of the other. Proof: The proof is exactly similar to that of Theorem II. exercises 1. Show that the sequences 1 2, 3 4, 7 8, ... and 2 3, 4 5, 6 7, ... have the same limits. 2. Show that the sequences 1 2, 1 4, 1 8, ... and 1 3, 1 5, 1 7, ... have the same limits. The applications of the theory of limits to geometry consist chiefly in showing that two numbers are equal because they are the limits of the same sequence, or of sequences having the property stated in the theorem of § 420 or of § 421. applica tion of limits to plane geometr y 422. Problem On a given segmentAB to lay off a sequence of pointsB1, B2, B3, ... , of whichB is the limit, such that each of the

segmentsAB1,AB2,AB3,... is com- mensurable with a given segmentCD.

204 SOLID GEOMETRY: APPENDIX III Solution: Using m1, an exact divisor ofCD, as a unit of measure, lay off onAB a segmentAB1, an exact multiple ofm1, such that the remainder B1B is less thanm1. ThenCD and AB1 are commensurable. Using a unitm2, likewise a divisor ofCD, and less thanB1B, lay off AB2 such thatB2B is less thanm2. Then AB2 > AB1, andCD and AB2 are commensurable. Continuing in this manner, using as units of measure segmentsm3, m4, ... each an exact divisor ofCD and each less thanB2B, B3B, ... respec- tively, we obtain a sequence of segmentsAB1, AB2, AB3, ... , each greater than the preceding and each commensurable withCD. If the unitsm1, m2, m3, ... are so selected that they approach zero as a limit, it follows thatB is the limit point of the sequenceB1, B2, B3, ... If a different sequence of divisors ofCD, asm′ 1,m′ 2,m′ 3,... , is used, we obtain a sequence of pointsB′ 1,B′ 2,B′ 3,... , likewise satisfying the conditions of the problem. Any two such sequences of pointsB1, B2, B3, ... and B′ 1, B′ 2, B′ 3, ... determined as above, are bothincreasing and each is such thatno point of it is to the right ofevery point of the other. Hence, by § 420 any two such sequences have the same limit pointB. 423. Limit of a Sequence of Segments.If B1, B2, B3, ... is a sequence of points on the segmentAB having the limitB, then the segment AB is said tobe the limit of the sequence of segmentsAB1, AB2, AB3, ... incommensurable ra tios 424. Ratio of Two Incommensurable Segments. The ratio of two incommensurable segments has not been explicitly defined, but this may now be done in terms of thelimit of a sequence. Consider two incommensurable segmentsAB and CD. Let a1, a2, a3,... be the lengths of segments each commensurable withCD, form- ing a sequence whose limit is the segmentAB, and letb be the length of the segmentCD.

VARIABLES. LIMITS 205 Then a1 b , a2 b , a3 b , ... is an increasing bounded sequence having a limit which we callR. If a′ 1, a′ 2, a′ 3, ... are the lengths of segments forming any other sequence whose limit isAB, the sequence a′ 1 b , a′ 2 b , a′ 3 b , ... is another increasing bounded sequence with limitR′. By § 420 we now know thatR =R′. This numberR is defined asthe ratio of the incommensurable seg- ments AB and CD. 425. Ratio of Two Incommensurable Arcs.The considerations of §§ 422, 423, 424 apply to two arcs of equal circles. Thus, in the figure, the ratio of the two incommensurable arcsAB and CD is the limit of the sequenceAB1 CD , AB2 CD , AB3 CD , ... , and the ratio of the incommensurable angles AOB and COD is the limit of the sequence ∠AOB1 ∠COD , ∠AOB2 ∠COD , ∠AOB3 ∠COD , ... applica tions of limits to plane geometr y 426. Theorem IV. A line parallel to the base of a triangle, and meeting the other two sides, divides them in the same ratio.

206 SOLID GEOMETRY: APPENDIX III Given the△ABC withDE∥BC and cut- ting AB and AC. To prove thatAD AB = AE AC. Proof: Consider the case whenAD and AB are incommensurable. LetD1,D2,D3,... be a sequence of points on AB whose limit isD. Through these points draw parallels toBC, meetingAC in E1, E2, E3, ... Then E is the limit of the sequence of pointsE1, E2, E3, ... For suppose it is not, and that there is a pointK on AE such that there is no point ofE1, E2, E3, ... between K and E. Draw a line parallel toBC through K, meetingAB in H. But there are points of the sequenceD1,D2,D3, betweenH andD, and hence there must be points of the

sequenceE1,E2,E3,... between K and E, which shows thatE is the limit of the sequenceE1, E2, E3, ... Then the sequence (1)AD1 AB , AD2 AB , AD3 AB , ... has a limitR which, by definition, is the ratio of the segmentsAD and AB. § 424 Similarly, the sequence (2) AE1 AC , AE2 AC , AE3 AC , ... has a limit R′ which is the ratio ofAE to AC. Now AD1 AB = AE1 AC , AD2 AB = AE2 AC , AD3 AB = AE3 AC , ... § 321, P. G. Hence, the two sequences, (1) and (2), are identical and have the same limit (§ 419). HenceR =R′. Q. E. D. applica tions to plane geometr y 427. Theorem V. In the same circle or in equal circles the ratio of two central angles is the same as the ratio of their intercepted arcs.

VARIABLES. LIMITS 207 Proof: In case the arcs are commensurable the proof is obvious. For, in that case, the common measure of the arcs divides each into a number of equal parts, saym andn respectively, and these arcs subtend the same number of equal angles at the center (§ 244, Plane Geometry.) Hence the ratio of the arcs ism : n and the ratio of the angles is also m :n. If the arcs AB and CD are not commensurable, let AB1, AB2, AB3, ... be a sequence of arcs whose limit is AB, each arc being commensurable with the arcCD. Then the sequenceAB1 CD , AB2 CD , AB3 CD , ... has a limitR = AB CD. Similarly the sequence ∠AOB1 ∠COD , ∠AOB2 ∠COD , ∠AOB3 ∠COD , ... has a limit R′ = ∠AOB ∠COD. But AB1 CD = ∠AOB1 ∠COD , AB2 CD = ∠AOB2 ∠COD ,... , as shown above for the commensurable case. Hence, these two sequences are identical and have the same limit (§ 419). Therefore it follows thatR =R′. Q. E. D. Note.—The student should note the successive steps in the proofs of §§ 426, 427: (a) Definitions of the incommensurable ratios by means of infi- nite sequences; (b) Proof that these sequences are identical; (c) Conclusion that the limiting ratios are equal. 428. Products of Irrational Numbers. If a1, a2, a3, ... is a sequence with limita, thenka1,ka2,ka3,... is a sequence whose limit is defined as ka where k is any number. If a1, a2, a3, ... and b1, b2, b3, ... are two sequences with limitsa and b respectively, thena1b1, a2b2, a3b3, ... is a sequence whose limit is defined as ab.

208 SOLID GEOMETRY: APPENDIX III Similarly if a1, a2, a3, ... , b1, b2, b3, ... and c1, c2, c3, ... are sequences with limits a, b, c, then abc is defined as the limit of the sequence a1b1c1, a2b2c2, a3b3c3, ... For a complete treatment it would be necessary to show that these definitions of multiplication of irrational numbers are consistent with the rest of arithmetic and also that these new sequences are such as to determine definite limits. This is possible but is beyond the scope of this book. 429. Area of a Rectangle. If the sides of a rectangle are in- commensurable with the unit segment, wedefine its area as follows: Let a1, a2, a3, ... be a sequence of rational numbers whose limit is the altitudea; and letb1, b2, b3, ... be a sequence whose limit is the base b. Then the area of the rectangle is the limit of the sequencea1b1,a2b2, a3b3, ... But by definition the limit ofa1b1,a2b2,a3b3,... is the productab. Hence we have the theorem: 430. Theorem VI. The area of a rectangle is the product of its base and altitude.

VARIABLES. LIMITS 209 applica tions to plane geometr y 431. Incommensurable Ratios Related to Two Circles.In a circle inscribe a sequenceP1, P2, P3, ... of regular polygons, each having twice the number of sides of the one preceding it. Let the

perimeters of these polygons bep1, p2, p3, ... and their areas A1, A2, A3, ... Then the lengthc of the circle is defined to be the limit of the sequencep1,p2,p3,... and its areaA is defined to be the limit ofA1,A2,A3, ... The sequence of polygons thus inscribed is called anapproximating sequence of polygons. That these sequences areincreasing and bounded is obvious from the figure. 432. Theorem VII. The lengths of two circles are in the same ratio as their radii, and their areas are in the same ratio as the squares of their radii. Proof: Let the radii of the two circles ber andr′. Denote the ratio r′ :r byk. Thenr′ =kr. Inscribe in one circle an approximating sequence of polygons with perimeters p1, p2, p3, ... and areas A1, A2, A3, ... In the other circle inscribe a sequence of similar polygons. By §§ 476, 477 (Plane Geometry), the perimeters of the latter arekp1,kp2,kp3,... and their areas arek2A1, k2A2, k2A3, ... By § 428, if the limit ofp1, p2, p3, ... is c and the limit ofA1, A2, A3,... isA, then the limits ofkp1,kp2,kp3,... , andk2A1,k2A2,k2A3, ... are kc and k2A, respectively. That is, the ratio of the lengths of the circles iskc c = k = r′ r and the ratio of their areas isk2A A =k2 = r′2 r2.

 433. Incommensurable Solids. The areas of curved surfaces, the volumes of cones, cylinders, and spheres, and even of rectangular solids involve incommensurable quantities. We have already treated these cases informally and have devised means for computing them approximately. We now give a formal treatment for logical completeness. 434. Rectangular Parallelopiped.If the three concurrent edges a, b, c of a rectangular parallelopiped are incommensurable with the unit segment, the volume inclosed is defined as follows: Let a1, a2, a3, ... be a sequence of rational numbers whose limit is the sidea. Let b1, b2, b3, ... and c1, c2, c3, ... be similar sequences whose limits are respectively the dimensionsb and c. Then the volume is the limit of the se- quence a1b1c1, a2b2c2, a3b3c3, ... But the limit of this sequence is by definition the product of the limits of the three sequences a1, a2, a3, ... , b1, b2, b3, ... , c1, c2, c3, ... , or abc. Hence, we have the theorem: 435. Theorem VIII. The volume of a rectangular par- allelopiped is equal to the product of its three concurrent edges. exercises 1. Give in decimals four terms of a sequence which approximates the volume of a rectangular parallelopiped whose dimensions are2, 4, √ 5. 2. Give in decimals four terms of a sequence which approximates the circumference of a circle whose radius is10 in.; also of a sequence which approximates the area.

 436. Triangular Pyramid. In a triangular pyramid inscribe a set of prisms in a manner similar to that shown in the figure of § 243. Call V1 the sum of the volumes of these inscribed prisms. Using as altitudes one half the altitudes of the first set of prisms, inscribe a second set the sum of whose volumes isV2. Continuing in this manner, we obtain a sequence of sets of prisms with volumesV1, V2, V3, ... The limitV of this sequence we define asthe volume of the pyramid. Circumscribed prisms may also be used for defining the volume of a pyramid, in which, case we get adecreasingsequence of volumesU1,U2,U3, ... , with limitU. That U = V follows from the fact that the sum of the volumes of the circumscribed prisms exceeds that of the inscribed prisms by the volume of the lowest circumscribed prism, and this may be made as small as we

please. 437. Theorem IX. Two pyramids with the same alti- tudes and equivalent bases have equal volumes. Proof: Call a sequence of inscribed prisms in one pyramidV1, V2, V3, ... , and in the otherV′ 1, V′ 2, V′ 3, ... Since corresponding sets of these prisms have equal volumes (§ 200), that is,V1 = V′ 1, V2 = V′ 2, V3 =V′ 3, ... , the theorem follows from § 419. 438. Convex Closed Curves.We assume that in aconvex closed curve (§ 202) it is possible to inscribe a sequence of polygonsP1, P2, P3,... , having perimetersp1,p2,p3,... and areasA1,A2,A3,... with limitsp andA respectively, and to circumscribe a sequence of polygons P′ 1,P′ 2,P′ 3,... , having perimetersp′ 1,p′ 2,p′ 3,... and areasA′ 1,A′ 2,A′ 34, ... with limitsp′ and A′ respectively, such thatp =p′ and A =A′. These limitsp and A we now define asthe perimeter and the area respectively of the curve. applica tion of limits to the cylinder 439. The Cylinder.Given any cylinder with a convex right cross- section and an elemente. In this cross-section inscribe a sequenceP1, P2,P3,... of polygons, as in § 438, with perimetersp1,p2,p3,... and

 areas A1, A2, A3, ... , thus defining the perimeterp and the areaA of the cross-section. Consider a set of prisms inscribed in this cylinder, of whichP1, P2, P3, ... are right cross-sections. Then the areas and the volumes of these prisms are respectivelyp1e, p2e, p3e, ... and A1e, A2e, A3e, ... The lateral area and the volume of the cylinder are now defined as the limits of these sequences. But by § 428 these limits are equal respectively tope and Ae. Hence, we have the theorem: 440. Theorem X. The lateral area of a cylinder is the product of an element, and the perimeter of a right section and its volume is the product of an element and the area of a right section. exercises 1. Prove as above that the volume of any convex cylinder is equal to the product of its altitude and the area of its base. 2. Prove that the lateral area of a right circular cone is equal to half the product of the slant height and the perimeter of its base. 3. Prove that the volume of any convex cone is equal to one third the product of its altitude and the area of its base. Suggestion. The treatment required in these exercises is a very close para- phrase of the definitions and proof given in § 439. Observe that we cannot begin to make a proof until we havedefined the subject matter of the the- orem. That is, we must first definethe areas and volumes in question .

 applica tion of limits to the sphere 441. The Sphere. Through two pointsP andQ of a sphere pass a great circle forming a hemisphere with centerO. DivideOA, the radius perpendicular to the plane of the great circle, into the equal partsOC, CB, andBA. ThroughC and B pass planes parallel to the plane ofPOQ, meeting the sphere in pointsD and E, respectively. Construct right circular cylinders with axesOC and CB and radii CD and BE. Denote byV1 the sum of the volumes of these cylinders. Now divide the radiusOA into six equal parts and construct five cylinders in the same manner as above. Let the sum of these volumes be V2. Continuing in this manner, each time dividingOA into twice as many equal parts as in the preceding, we obtain a sequence of sets of cylinders and a corresponding sequenceV1, V2, V3, ... of volumes. We now define the volume of the hemisphere as the limit of the sequence V1, V2, V3, ... 442. Construct a right circular cylinder with its base in the plane of POQ and with radius and altitude both equal toOA. Denote byF the figure formed by the lower base of the cylinder, its lateral surface and the lateral

surface of the cone whose base is the upper base of the cylinder and whose vertex is atO′. (See the right- hand figure.) Draw segmentsO′M and O′N. Let the planes through C and B cut O′A′ in C′ and B′ and O′M in H and K. Now form the cylinderO′C′H whose axis isO′C′ and whose radius is C′H. Likewise form the cylinderC′B′K.

214 SOLID GEOMETRY: APPENDIX III LetV′ 1 denote the sum of the volumes ofO′C′D′ andC′B′E′ minus the sum of the volumes ofO′C′H and C′B′K. In a similar manner, using the planes which divideOA and hence O′A′ into six equal parts, we form another set of five cylinders, the sum of whose volumes minus that of the smaller inside cylinders we denote byV′ 2. Continuing in this manner, we obtain a sequence of volumesV′ 1, V′ 2, V′ 3, ... whose limit V′ we define as the volume of the given figure F. We now prove thatV1 =V′ 1, V2 =V′ 2, ... Denote OA byr, and note thatO′B′ =B′K. (1) Vol.C′B′E′−Vol.C′B′K =πC′B′(B′E′2 −B′K 2) =πC′B′(r2−O′B′2). (2) Vol.CBE =πCB·BE 2 =πCB (r2−OB 2), since BE 2 =r2−OB 2 . ButOB =O′B′ and CB =C′B′. Hence, Vol.CBE =πC′B′(r2−O′B′2). Hence, Vol.CBE = Vol.C′B′E′−Vol.C′B′K. Similarly we show that Vol.OCD = Vol.O′C′D′−Vol.O′C′H. Hence, V1 =V′ 1. In like mannerV2 =V′ 2, V3 =V′ 3, ... Hence, V =V′, since they are the limits of the same sequences. But the volume of the cylinderO′A′M isπr3 and of the cone whose volume was subtracted, 1 3πr3. That is, the volume ofF is 2 3πr3, and hence that of the hemisphere is2 3πr3. Hence, we have the theorem: 443. Theorem XI. The volume of the sphere is4 3πr3. Note that the above proof consists essentially in showing that the area of the circleBE is equal to that of the ring between the circles B′E′ and B′K, and that the area of the circleCD is equal to that of the ring betweenC′D′ and C′H and so on. Indeed, this theorem and also that of § 437 are special cases of what is known asCavalieri's Theorem.

VARIABLES. LIMITS 215 444. Theorem XII. If two solid figures are regarded as resting on the same planeb, and if in every plane parallel to b the sections of the two figures have equal areas, the figures have equal volumes. The proof of this general theorem is more difficult than any thus far given, inasmuch as it involves sequences whichoscillate; that is, which are neither constantly increasing nor constantly decreasing. 445. The Area of the Sphere. About a sphere of radiusr construct a sequence of circumscribed polyhedrons such that the largest face in each polyhedron becomes as small as we please when we proceed along the sequence. Let s1, s2, s3, ... be the total surfaces of these polyhedrons. This forms a decreasing sequence with limitS which we define asthe surface of the sphere. The volumes of these polyhedrons will be 1 3rs1, 1 3rs2, 1 3rs3, ... Then the volume V of the sphere is defined2 as the limit of this sequence of volumes. Hence, by § 428,V = 1 3rS. But by § 443,V = 4 3πr3. Then S = 3 r· 4 3πr3 = 4πr2. Hence, we have the theorem: 446. Theorem XIII. The area of the sphere is4πr2. exercises on limits 1. In addition to those which are found in the text give other exam- ples of infinite sequences which do not determine definite numbers. 2. Give two increasing sequences which determine the number 3. Show that the theorem of § 420 applies and proves that these sequences determine the same number. 3. Give two decreasing sequences each of which determines the num- ber 5. Apply § 421 to show that these sequences determine the same number. 2This definition can be shown to be consistent with that of § 441.

216 SOLID GEOMETRY: APPENDIX III 4. State fully the relation between a bounded increasing sequence and the number determined by it. State also the relations between a bounded decreasing sequence and the number determined by it. 5. State fully what is meant by “a limit of a sequence” both for increasing and decreasing sequences. 6. Given two incommensurable segmentsAB and CD. Lay off on the lineAB a decreasing sequence of segments, each of which is com- mensurable with CD, such that the limit of the sequence is the seg- mentAB. 7. Ifa1,a2,a3,... is an increasing sequence defining the number4, prove that 3a1, 3a2, 3a3, ... defines the number3× 4 = 12. 8. If a1, a2, a3, andb1, b2, b3, ... are increasing sequences defining the numbers3and 5, show that the sequencea1b1,a2b2,a3b3,... defines the number 15. 9. Describe in more detail the meaning of the ratio of two incommen- surable arcs as indicated in § 425. Show that this ratio is independent of the sequence of units of measurement used, so long as the limit of this sequence is zero. 10. Treat the ratio of two incommensurable angles in a manner sim- ilar to the treatment of arcs in the preceding exercise.

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TRANSCRIBER’S NOTES transcriber ’s notes The spellings inclose and parallelopiped have been retained. Mi- nor typographical corrections and regularizations of spelling have been made without comment. Page 12In the third (rightmost) figure after Theorem 73, the la- bels □1 and □2 are exchanged relative to their use in the subsequent exposition. The book contains at least five potentially ambiguous assertions. Some are complicated to resolve completely. Page 53(How many vertices and edges has a hexahedron?) Com- binatorially there are multiple types of hexahedron, including but not limited to (i) a parallelepiped (8 vertices and 12 edges), (ii) a square-basedpyramidwithonebasecornershavedoff(7verticesand11edges), (iii) a pentagonal pyramid (6 vertices and 10 edges), and (iv) a com- pound of two regular tetrahedra placed base-to-base (5 vertices and 9 edges). Page 75(A closed plane curve isconvexif it can be cut by a straight line in not more than two points.) Presumably this was to be inter- preted as “...if it cannot be cut by a straight line inmore than two points.” However, even the latter is inconsistent with usage throughout the book, since some polygons are convex while no polygon satisfies this condition. Page79: Approximatingacurvedsurfacebyplanarpatchesismore subtle than the analogous approximation of a curve by an inscribed sequence of segments. For example, bounded polyhedra inscribed in a right circular cylinder may have arbitrarily large area. Page 151: The construction of a sequence of polyhedra circum- scribed about a sphere depends on the points of tangency chosen at each stage, and these points cannot be chosen arbitrarily. Page 197: The “definition” of √ 2 as the limit of its sequence of finite decimal truncations is circular since it requires the firstn decimal places to be known in order to generaten terms of the sequence.

TRANSCRIBER’S NOTES

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